Lower Bounds for Cycle Lengths in the Juggler Map

Philippe Cochin
philippe@cochin.fr

11 September 2026

Abstract

The Juggler map sends an even positive integer to the integer part of its square root and an odd positive integer to the integer part of its three-halves power. We obtain restrictions on hypothetical nontrivial cycles. For a cycle with minimum n, length L, and o odd steps, we prove the cycle-financing inequality

$$n \log n \,(3^o - 2^L) \leq L\, 3^o.$$

It bounds the formal expansion that accumulated floor losses can offset. A refinement transports these losses to a reduced base and bounds the resulting exponent-walk charge using an irrational rotation, Denjoy–Koksma estimates, and finite Ostrowski decompositions. Combined with the verified descent inputs described in the paper, the inequalities give period lower bounds of 25781, 176251, 478245, and 780239 at floors 10^6, 26254995, 162849448, and 350000000, respectively. Separately, finite-word exclusions give at least four even steps without a descent-floor input; an exact computational classification strengthens this to eight even steps and period at least twenty-two. For cycles whose maximum is below the cube of their minimum, we also determine the complete rank order and mechanical itinerary, derive a uniform log-log grid bound, and prove the further height restriction $M < m^3 - m^{15/8}$ for $m \geq 7$, where m, M are the minimum and maximum. Successive genuine return-gap contractions sharpen this to $M < m^3 - (1/2)m^{253/128}$ for $m \geq 2^{24}$, and to $M < m^3 - (1/2)m^{127/64}$ for $m \geq 2^{128}$, with a slightly stronger exponent in the latter case. A terminal word factorization identifies the expansion still uncontrolled by these contractions. Exact short-return cells identify the rounding information discarded by several relaxed no-cycle criteria. The core inequalities and selected classifications are formalized in Lean 4. The descent computations, per-length numerical comparisons, and remaining analytic identifications are distinguished from those formal proofs. A limitation result applies to charges retaining a positive contribution at a fixed floor. Neither the exclusion of all nontrivial cycles nor universal termination is established.

2020 Mathematics Subject Classification. Primary 11B83; secondary 37P99, 11Y55.

Keywords. Juggler map, Juggler sequence, floor-power map, cycle financing, integer dynamics.

1. Introduction

The Juggler sequence was introduced by Pickover [1,2] as an interesting variation of the Collatz problem. Pickover's later exposition is Chapter 45 of [2], pp. 102–106. We study $J : \mathbb{N} \to \mathbb{N}$,

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

The *Juggler sequence* starting at n is the trajectory of iterates n, $J(n)$, $J^2(n), \ldots$. The On-Line Encyclopedia of Integer Sequences records the one-step values of J as A094683 [3] and the number of steps to reach 1 (when that occurs) as A007320 [4]; those catalogue entries are not theorems of this paper.

The trajectory of 3 is $3, 5, 11, 36, 6, 2, 1$. The trajectory of 37 already peaks at 24906114455136. Universal arrival at 1 remains an open conjecture. This paper does not prove that conjecture. It proves period lower bounds for a *hypothetical* nontrivial cycle, once a verified descent floor is given.

Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$. Write J^k for the k-fold iterate, and write $\lfloor x \rfloor$ for the integer part of a real $x \geq 0$: discard the fractional part. Thus $\lfloor 5.196\ldots \rfloor = 5$ and $\lfloor 6 \rfloor = 6$. The floor is applied after every branch of J, not once at the end of a walk.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor,$$

so $J(n) = E(n)$ when n is even and $J(n) = O(n)$ when n is odd.

Trajectory. The *trajectory* of $n \in \mathbb{N}$ is the sequence of values $n_0 = n$, $n_{i+1} = J(n_i)$ (also called an orbit in some dynamics texts).

Itinerary. The *itinerary* of the first k steps is the length-k string $w \in \{O, E\}^k$ with $w_i = O$ if n_i is odd and $w_i = E$ if n_i is even. Synonyms: itinerary = parity sequence; trajectory = orbit. The itinerary is the list of branch labels, not the list of values. The integer k is any finite prefix length; the definition does not assume that the trajectory reaches 1.

Ideal exponent. An itinerary of length k with o odd letters has ideal exponent $3^o/2^k$. Ignoring floors, those letters would send a start n to $n^{3^o/2^k}$; equivalently, the ratio multiplies $\log n$. Applying floors can only decrease the image.

Realized itinerary. An itinerary w is *realized* at n when the first $|w|$ parities of the trajectory of n are exactly the letters of w. Equivalently: a formal string over $\{O, E\}$ is an itinerary only when some start actually follows those branches. That is what is meant by: an itinerary is available only when the trajectory realizes those parities. Floors are applied after every letter of a realized itinerary.

One-step preimage. The map J is not invertible. For $m \in \mathbb{N}$, the *one-step preimage* of m is the set

$$J^{-1}(m) = \{k \in \mathbb{N} : J(k) = m\}.$$

For each target m, its even parents are the even integers k with $m^2 \leq k < (m+1)^2$. Its odd parents satisfy $m^2 \leq k^3 < (m+1)^2$, and there is at most one such odd integer (Lemma 3.1). These are statements about the parent's branch, independent of the target's parity.

Lemma 1.1 (three fates). Let $n \in \mathbb{N}$. The trajectory of n does exactly one of the following: (i) some iterate equals 1; (ii) some iterate $m \geq 2$ returns, and the trajectory is eventually periodic through a cycle containing m; (iii) the trajectory is unbounded. In particular, a bounded infinite trajectory is eventually periodic.

Proof. The map J is a well-defined function $\mathbb{N} \to \mathbb{N}$, so the forward trajectory is an infinite sequence in $\mathbb{N}$. If some term equals 1, we are in (i); note $J(1) = 1$. If the trajectory is unbounded, we are in (iii). If the trajectory is infinite and bounded, the pigeonhole principle supplies a repeated value m; uniqueness of the successor $J(m)$ then forces a cycle, which is (ii) unless $m = 1$, already covered by (i). □

The lemma lists the only logical possibilities. It does not assert which fate occurs for a given start, and it is not a termination theorem.

Object	Meaning
J	the one-step map
E, O	even and odd branches
trajectory	the sequence of values
itinerary	a finite string in $\{O, E\}$ of parities
realized itinerary	the trajectory actually follows those letters
$3^o/2^k$	ideal exponent of an itinerary of length k with o odd letters
$J^{-1}(m)$	the one-step preimage of m
N_0	a verified descent floor (a computational input)
CycleMin	a minimum-based rotation of a cycle itinerary

The mechanism is the interaction of three elementary facts: exact integer one-step preimages, a logarithmic defect, and cycle minimality. Exact one-step preimages give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. That is Theorem 4.4:

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

The inequality $\log(1+u) \leq u$ is the only analytic input; it is not the content of the theorem.

Write N_0 for a *verified descent floor*: every start $2 \leq n \leq N_0$ reaches 1. A floor is an input, not the result. Four instances are used. The base instance is $N_0 = 10^6$, reported by Weisstein [5] and recomputed here by exact first-passage. Combined with Theorem 4.4 it gives

$$\text{known verification through } 10^6 \ + \ \text{this inequality} \quad \Rightarrow \quad L \geq 25781.$$

The laboratory instance is $N_0 = 26254995$, certified by the same exact first-passage method (Proposition 5.1); the same table then gives $L \geq 50508$ (Theorem 5.2), and the walk-charge envelope of Section 5 amplifies it to $L \geq 176251$ (Theorem 5.9). Corollary 5.10 evaluates the same kill criterion at the second certified floor $N_0 = 162849448$, every survivor lying inside the $[50508, 16785921)$ window of Theorem 5.8, whose envelope therefore covers their charge, and gives $L \geq 478245$. The main numerical result is Corollary 5.11: at the third certified floor $N_0 = 350000000$ the same comparison gives $L \geq 780239$. The census-free window theorem covers $[50508, 16785921)$ — every nonendpoint member $L_0, \dots, L_{54}$ of the semiconvergent fan of Proposition 5.12; the endpoint $L_{55} = 16785921$ is excluded by the half-open interval — so the charge side of the later-floor kills needs no per-length dynamic program. The kill comparison itself stays per-length, because its left-hand side $\theta(L)$ is a Diophantine quantity the envelope does not control. The architecture is

$$\text{envelope} \to \text{cycle minimum} \to \text{finance} \to n_{\max}(L) \to N_0 \to L \geq 25781 \text{ at } 10^6,$$

extended at the laboratory floor by

$$\begin{aligned} &\text{transport} \to \text{hug adversary} \to \text{itinerary identity} \\ &\qquad \to \text{Denjoy–Koksma} \to \text{window} \to L \geq 176251. \end{aligned}$$

The computation supplies the endpoint N_0. The mathematics amplifies that floor to the period bound.

Roadmap. Section 2 records the power envelope and the exact defect identity that explains it. Section 3 classifies minimum-based cycle itineraries and proves that every nontrivial cycle has at least four even letters (Theorem 3.22); Theorem 3.31 raises that to eight even letters, and period at least twenty-two, once the cycle minimum is at least 300. Section 4 records the excursion necklace of a minimum-based itinerary, unrolls the one-step-preimage logarithm around that minimum, obtains the finance inequality, and applies it at the known floor 10^6. The necklace is the geometry of the unroll, not a fourth main theorem. A run-type packing of the same identity is a supporting refinement; the arithmetic of the leftover lengths is secondary. Section 5 certifies the laboratory floor 26254995, replaces the length-only charge by a coupled exponent-walk charge, identifies its extremal word as a rotation itinerary, and proves a census-free envelope for every length in the window $[50508, 16785921)$; the resulting kill table gives the period bound 176251, a certified evaluation of the same kill criterion at a second certified floor raises it to 478245 (Corollary 5.10), and a third certified floor raises it to 780239 (Corollary 5.11). Section 6 records limitations.

A nonempty realized itinerary w with $J^{|w|}(n) = n$ is a *cycle itinerary*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \geq 2$. A cycle word at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. Every cycle itinerary has a minimum-based rotation.

All exceptional sets below are *finance-survivor* sets: lengths that a stated form of the finance inequality does not exclude at the verified descent floor. They are not candidate cycle sets. A survivor count "through X" is inclusive: it counts every surviving length $L \leq X$.

1.0 Main results

Contribution 1 — cycle-financing inequality. For a minimum-based cycle of length L with o odd steps,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o$$

(Theorem 4.4). Floor defects become a quantitative bound on the cycle minimum.

Contribution 2 — structural itinerary obstruction. Every nontrivial cycle itinerary has at least four even letters, and hence period at least eleven (Theorem 3.22 and

Corollary 3.23). That is the Lean statement, for every $n \geq 2$, and it uses no descent floor. The argument classifies the minimum-based itinerary geometry; it is not a raw census of itineraries of length at most ten. Theorem 3.31 is a computational strengthening of the same two envelopes: once the cycle minimum is at least 300 (every $n \leq 299$ reaches 1), no cycle itinerary has fewer than eight even letters, so a nontrivial cycle has period at least twenty-two. That enumeration is not Lean.

Contribution 3 — explicit conditional consequence for hypothetical cycles. Combined with the independently verified descent floor $N_0 = 10^6$, there is no nontrivial Juggler cycle of length at most 25780. Equivalently, any nontrivial cycle has period at least 25781 (Theorem 4.6(A)). At the laboratory-certified floor $N_0 = 26254995$ the same table gives period at least 50508 (Theorem 5.2).

Contribution 4 — walk-charge envelope and the main period bound. On a minimum-based cycle every state is coupled through one closed exponent walk. Transport of the floor losses to a reduced base (Theorem 5.3), identification of the extremal walk as the rotation (hug) itinerary (Theorem 5.4, Lemma 5.6), and a Denjoy–Koksma bound over certified Ostrowski blocks (Theorem 5.7) give a uniform envelope for every length in the window $[50508, 16785921)$ (Theorem 5.8) — census-free on that window, which covers $L_0, \ldots, L_{54}$ but not its endpoint $L_{55} = 16785921$. At the laboratory floor the resulting kill table leaves a single finance survivor below $2 \cdot 10^5$: any nontrivial cycle has period at least 176251 (Theorem 5.9), the first laboratory instance. The second-floor evaluation is Corollary 5.10: at $N_0 = 162849448$ the same kill criterion, evaluated on the additional survivors — all inside the window, so all with a census-free charge bound — leaves only the semiconvergent fan member 478245. The main numerical result is Corollary 5.11: at the third certified floor $N_0 = 350000000$ that comparison leaves only the next fan member 780239: any nontrivial cycle has period at least 780239.

Contribution 5 — floor-free gap transfer and the short-cycle reduction. The finance inequality ties the relative surplus to the linear form $\Lambda = o \log 3 - L \log 2$ without any floor: $n \log n \cdot \min(\Lambda, 1) \leq 2L$ (Theorem 4.10, Lean `cycleMin_gap_transfer`). With Rhin's effective measure this excludes every cycle with $L^{14.3} \leq n \log n/915$ (Corollary 4.11), for all $n \geq 2$. The statement is a reduction, not a kill: it is weaker than the finance table at every certified floor, and it shows that the no-cycle problem is exactly the exclusion of long cycles, $L > (n \log n/915)^{1/14.3}$, where the finance survivors ($L \approx n^{0.59}$) live.

Contribution 6 — complete cycle order under a height hypothesis. For a primitive cycle with $m > 1$ and $M < m^3$, the sorted states rotate by the even count, the period and odd count are coprime, and the itinerary from its minimum is exactly the ceiling mechanical word (Theorem 3.33). The associated threshold maps have a common period and interlacing cycles at each fixed threshold (Theorem 3.35); Proposition 3.36 controls the entire sorted cycle in log-log coordinates. Propositions 3.37–3.38 show precisely why a one-unit successor allowance or exact within-branch differences alone cannot prove no-cycle. They concern different maps and are not counterexamples to the Juggler conjecture.

Contribution 7 — a periodic height gap and exact short-return guards. For $m \geq 7$, Theorem 3.39 strengthens the hypothesis $M < m^3$ to $M < m^3 - m^{15/8}$. It uses the ordered return images of the minimum and maximum retained states, followed by exact parity faces of the floor cells. Appendix E gives exact carry recovery, a finite bound on consecutive transitions in a polynomial growth family, and counterfamilies delimiting two proposed guard summaries. The family obstructions concern prescribed blocks; the height gap uses actual periodicity. None increases the numerical period floor.

Contribution 8 — successive return gaps and the terminal passage. Theorem 3.40 strengthens the height strip through two and then three genuine changes of the return boundary, at the stated distinct minimum thresholds. It retains every floor error and actual intermediate guard. Proposition 3.41 factors the terminal words around a mixed OE/EO passage, whose gap contracts, and identifies the remaining uncontrolled prefix. Appendix F gives the full error, rank and height proofs, and the precise obstruction to a uniform one-sided error certificate.

Section 6.3 complements the height results with an upper-cell charge bound. At the exact counts $(780239, 492276, 287963)$, it restricts an actual cubic-cycle minimum to $350000000 < m < 520000000$. The local odd-to-odd upper-square gap is Lean verified; the charge proof and its interval evaluation have separate evidence boundaries. Appendix E.7 rules out full eventually periodic and single-polynomial domains for an invariant all-OOE construction, while leaving sparse trajectories unexcluded.

These statements are not interchangeable. Theorem 4.4 is the conceptual sharp inequality (constant 1). Corollary 4.5 is the convenient length-only statewise bound that turns a descent floor into a per-length exclusion. Theorem 4.6 is the numerical certification of that bound, and it uses a conservative coefficient 6/5 only as a uniform majorant; the headline cutoff 25781 is not an artifact of that majorant. Theorem 4.7 is a run-type refinement of the same defect sum. The leftover lengths through 10^5 are supporting material, not a second main theorem.

1.1 Related work

Pickover's later exposition is Chapter 45 of [2], already cited above. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6. That verified descent floor is the computational input to Theorem 4.6; the first-passage run of Appendix B recomputes it. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to exact cycles. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form $n(2^K - 3^p) = C$. A check of Pickover [1,2], Weisstein [5], the OEIS records [3,4,6], the Prasad–Prasad estimates [7], and the standard Collatz cycle-bound sources [8–13] found no published explicit lower bound on the period of a nontrivial cycle for this exact floor-power Juggler map. Informal near-miss notes and later unverified webpage floors are not used.

Goodstein's theorem is a superficially similar statement about natural numbers — every Goodstein sequence terminates at 0 — which cannot be proved in Peano arithmetic [14]. No such independence claim is made for the Juggler conjecture, and this paper does not study Goodstein sequences.

Companion manuscripts. Two later manuscripts by the author build on this one and are cited where they bear on the cycle problem. Paper B [16] proves parity equidistribution of the nested floor powers along the itineraries of J, with power savings, completely through depth four for odd-rooted itineraries and for the two length-five contractors, giving the certified-descent densities 13/16 and 7/8; its only input from this paper is the contraction criterion of Theorem 2.2. Paper C [17] proves that every nonempty backward-closed set — in particular the basin of any nontrivial cycle, and the set of divergent starts — has logarithmic count $\gg (\log x)^{\lambda}$ for $\lambda < \lambda^{**} = 0.4926$, and reduces the Juggler conjecture to a Tao-type almost-all statement whose bounded target is the certified floor of Section 5 and whose descent step is the power envelope of Theorem 2.2 (`power_bound_word`). Section 6.1 states precisely what those results add to the cycle problem, and what they leave untouched.

The layers of the argument are as follows.

1. *Classical ideas, not claimed as new:* cycle financing (Simons–de Weger [12]); logarithmic and continued-fraction approximation of $\log 2/\log 3$, including Rhin's effective measure for $o\log 3 - L\log 2$ [15], used only in Corollary 4.11; cycle-itinerary restrictions and leftover packaging (Eliahou [13]; Lagarias [8,9]); the Denjoy–Koksma inequality and Ostrowski numeration, used as known tools in Section 5.
2. *New object:* the exact one-step floor-power preimages of J.
3. *New theorem:* the Juggler-specific cycle-minimum finance inequality of Theorem 4.4.
4. *New consequence:* $L \geq 25781$ at the verified descent floor 10^6.
5. *Supporting organization:* the run-packing refinement of the same defect sum (Theorem 4.7).
6. *New theorem:* the reduced-base transport and the uniform window envelope of Section 5 (Theorems 5.3 and 5.8).
7. *New consequence:* period at least 176251 at the laboratory floor 26254995 (Theorem 5.9), 478245 at the second certified floor 162849448 (Corollary 5.10), and 780239 at the third certified floor 350000000 (Corollary 5.11), every survivor inside the window of Theorem 5.8.
8. *Reduction:* the floor-free gap transfer (Theorem 4.10) and the short-cycle exclusion $L^{14.3} > n\log n/915$ (Corollary 4.11), which locate the open problem in the long regime without claiming anything there.

The argument below is elementary and independent of the Diophantine tools of [12]: floor-power defects are relatively $O(1/x)$ in logarithms, so a uniform logarithmic floor-error bound, valid above the verified floor, excludes every length that is not a finance-survivor for Theorem 4.4. To the best of our knowledge, the Juggler-specific inequality and the explicit period bound it produces are new.

Novelty statement. For this nonlinear floor-power map, exact floor defects convert into a cycle-financing inequality that forces the minimum of a hypothetical cycle below the independently verified descent region unless the period is at least 25781. Coupling the states through the closed exponent walk and bounding the extremal rotation itinerary by Denjoy–Koksma over certified Ostrowski blocks raises that period bound to 176251 at the laboratory floor, census-free on an explicit window of lengths, and — by certified evaluation of the same kill criterion on the surviving lengths, all of which lie inside that window, the comparison against $\theta(L)$ remaining per-length — to 478245 at the second certified floor and 780239 at the third.

1.2 Verification

The arguments of Sections 2, 3, and 4 may be read without machine assistance. The core mathematical lemmas are mechanized in Lean 4; selected finite classifications and numerical tables are independently certified computations. Appendix A records the Lean names. The finite tables used by Lemmas 3.5, 3.7, 3.11 and Theorems 3.12–3.20 are `decide +kernel` evaluations in the modules named there: the reduction is performed by the Lean kernel, so these tables add no trust assumption beyond it. That holds throughout the Juggler layer with one exception, `window_digit_scan`: a scan of a quarter of a million window lengths in Section 5's Ostrowski certification, which uses `native_decide` and so also trusts the Lean compiler and runtime. `#print axioms` on any theorem displays whether it depends on it.

The structural digit identity `greedy_eq_ostro` identifies the fold and recursive algorithms for all lengths; `greedy_eq_ostro_below_window` is its bounded corollary. Likewise `greedy_reconstruct_all` is an exact reconstruction theorem. The remaining native scan sharpens the digit cap from 47 to 37 on its stated sub-window. The bound `window_digit_cap` inherits that scan's runtime dependency, as recorded by the axiom audit.

Section 3.10's finite-order statements, exact rounding results, and quantitative log-log grid inequalities are also formalized. The grid proof starts from integer square-cell inequalities and the actual successor map; its geometric conclusion is not an assumption. Appendix A identifies the public declarations. The illustrative asymptotic comparison following Proposition 3.36 and the open absolute-cell question remain written analysis. Section 3.11's actual-cycle seam and integer height gap are formalized from periodic-orbit hypotheses. Appendix E's formal coverage includes the short OE/OOE cells, subtractive rank returns, exact family-chain bound, and retained-remainder recovery. Its larger analytic constructions and full tower discussions remain written where Appendix A indicates. The new proofs use kernel verification and add no native scan. Section 3.12's two height restrictions are formalized from actual periodic-set hypotheses. The section, forced batches, odd endpoint gaps and genuine transfers are constructed internally. Appendix F's exact floor-loss bounds and finite certificate threshold are also formalized. Section 3.13's primitive terminal construction is also formalized from an ordinary actual periodic orbit, with its minimum and attained maximum. The proof retains original-set adjacency and all guards, identifies the absolute odd/even cut, and proves the strict mixed-gap inequality. Common-prefix amplification remains uncontrolled. No general no-cycle theorem is asserted. The real projection description and the higher-difference consequence are explanatory deductions from the listed results. The grid endpoints use the namespace `Problems.Juggler.CubicGrid`.

Reproducibility has three layers. The repository command `research.juggler_sequence.paper_a_audit` recomputes selected parity thresholds, finite fan identities, and constants, and checks the stored walk-comparison records. The standalone script `tools/check_paper_a_numeric.py` independently screens every length below the four headline cutoffs using rational logarithm bounds and interval arithmetic; it assumes the archived descent floors. Replaying those floors requires the separate exact-integer first-passage computation. Appendix B identifies its coverage and records. Neither an archive hash nor the arithmetic audit replaces that replay.

Near a convergent, double precision is insufficient to identify an exact integer crossing without a guard: the 25781 threshold can shift by one integer when $L \log 2 - o \log 3$ is evaluated naively. The independent checker uses outward interval comparisons and rational screening. The historical GPU and CPU tables remain separately identified in Appendix B.

The build command `python tools/build_paper_a.py` produces the canonical PDF and synchronized distribution copies from this source. `python tools/build_paper_a.py --check` checks their provenance and hashes. The manuscript source, formalization map, and reviewer packet in `docs/theory/` are the editorial inputs; files in `juggler_review/` and the companion's paper directories are generated exports. No external submission or universal termination proof is implied by a successful build.

Proposition 1.3 (certified computational input). An exact-integer first-passage computation checks that every odd start $3 \le n \le 10^6$ reaches a value strictly below its start. Even starts descend in one step, so strong induction gives arrival at 1 for every $2 \le n \le 10^6$. The archived files retain chunk summaries, counts, maximal first-passage lengths, and exceptional-seed resolutions; they do not retain a realized word for every start. Reproducing the descent assertion requires the exact computation, not only checking a summary's hash. The first four chunks of the 26254995-floor run cover the odd starts through 10^6; their maximal first-passage lengths are $253, 213, 188, 188$. Thus the maximum is 253, at seed 78901. Weisstein [5] reports the same coverage; the archived run is a separate computation. Artifact locations and commands appear in Appendix B.

Roles. Independently proved: the finance inequality (Theorem 4.4). Computational input: every $2 \le n \le 10^6$ reaches 1 (this proposition), every $2 \le n \le 26254995$ reaches 1 (Proposition 5.1, the laboratory instance), every $2 \le n \le 162849448$ reaches 1 (Corollary 5.10, the second laboratory instance), and every $2 \le n \le 350000000$ reaches 1 (Corollary 5.11, the third laboratory instance). Independently recomputed: the exact first-passage runs of Appendix B. Not proved: global termination.

Theorem 4.6 applies Corollary 4.5 to this input and certifies the table with the conservative coefficient $6/5$. The whole comparison it tests is Lean: `cycleMin_defect_threeTerm` (`FinanceTransfer.lean`) is the certified identity `cycleMin_defect_finance` composed with the three-class charge `cycleMin_threeTerm`, so what stays verified computation is the per-length arithmetic and the descent floor, not the inequality. Theorem 4.7 carries two hypotheses — the itinerary is primitive and contains no `EE` — without which its displayed counts fail; §4 prices them below. Given them the display is Lean (`cycleMin_sixTerm`), so they are sufficient as well as necessary. Theorem 4.8 reuses the gap table under that packing, and 18 of its 42 exclusions inherit the second hypothesis. Proposition 4.9 is integer arithmetic in Lean. In Section 5's Denjoy–Koksma core is Lean end to end relative to the circle integral. Its observable's variation is `observable_window_variation_lt_two`, proved for `periodicObservable` itself on every unit window; `denjoy_koksma_rotation` is Lean from positivity, coprimality and $|\theta - p/q| \le 1/q^2$, and `denjoy_koksma_blocks` composes the blocks uniformly in their starting phase. The one-statement display is `hugCharge_sub_circleMean_le`. Two bridges in the printed Theorem 5.7 stay human: it quantifies over any decomposition into actual convergents while that named theorem fixes the certified Ostrowski assembly, and it writes the circle integral as the explicit C_* of Proposition 5.5. Lemma 5.6 is Lean on both halves: the itinerary identity (`budgetedWord_eq_hugWord`) and the rotation identification (`hugOdds_eq_sub_floor`, `hugWalk_eq_fract`, `HugRotation.lean`). The Laplace bound of Proposition 5.5 is Lean (`rotation_average_le`, `rotationAverage_gap`), and `denjoy_koksma_blocks` proves convergence of the finite hug averages to the circle integral for the bounded-variation observable. The elementary change of variables identifying that circle integral with the displayed `rotationAverage (log n')` remains prose; the transport inequality of Theorem 5.3 (`cycleMin_transport`), the defect-to-hug-charge consequence of §5.2 (`cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge`), the charge-maximisation and strict-uniqueness halves of Theorem 5.4 (`hug_charge_maximal`, `hug_charge_unique`), the general digit machinery and the instantiated cap below $q_{13} = 301994$ (`ostroDigit_le`, `theta_digitSum_le`, `window_digit_scan`), and the kill template of Theorem 5.9 (`cycleMin_hug_kill_criterion`) are Lean-verified; Theorems 5.2 and 5.9 are independently certified computations at the laboratory floor, with the same trust boundary as Theorem 4.6 (exact integer arithmetic plus guarded float comparisons). This is not a claim that the paper as a whole is formally verified.

```
Repository:  https://github.com/sneakyweasel/btlab
Commit:      7802f78bec58c68cec92a2efb3db4a502f916277
Lean:        leanprover/lean4:v4.33.1
Mathlib:     v4.33.1 (lake-manifest rev 0df444a360eaa60ab8c11dca51a86af692955474)
Build:       lake build Problems.JugglerPaper   (from formal/)
Computation: python -m research.juggler_sequence.cycle_finance
             (parity table); run-type table in budget_opt.json
SHA-256:     Appendix B
```

The commit is the repository state that produced the finance tables. A later editorial commit of this text does not change those hashes.

1.3 Notation

The table records the core notation used in the envelope, itinerary exclusions, finance and walk estimates. Sections 3.10–3.13, 6.3 and Appendices E–F introduce additional local notation, defined where it is used. In the core estimates, G_a denotes the exponent in Lemma 3.3's constant, h the per-step exponent, and $T(u)$ the backward exponent of a suffix. These are distinct from the even count e, the step loss ε_i, and the digit sum $s(L)$.

symbol	meaning	first used
J	the Juggler map	§1
n	a start; on a cycle, its minimum	throughout
N_0	a verified descent floor: every start $\leq N_0$ reaches 1	§1
x_k	the state after k steps	§4
w, v, u	an itinerary word, a prefix of it, a suffix of it	§1, §3.9
O, E	the odd and the even letter	§1
L	the length of w; on a cycle, the period	§3
o, e	the odd and even letter counts, $e = L - o$	§3

symbol	meaning	first used
a, b, c	the odd-run lengths of the canonical form $O^aEO^bEO^cE$	§3
a_i, a_k	the i-th odd run; the odd count after k steps	§4, §5
r	the length of a trailing even run	§3.9
h	the per-step exponent: 1 on an even letter, 3 on an odd one	§4
G_a	$2(3^a - 2^a)$, the exponent of 2 in Lemma 3.3's constant	§3
X_a, Y_a	$3(3^a - 2^a)$ and $2 \cdot 3^a - 3 \cdot 2^a$, the sharp envelope exponents	§3.9
$B(u)$	the backward envelope of a suffix (Lemma 3.25)	§3.9
$T(u)$	$2^{\lvert u \rvert}/3^{\#O(u)}$, the backward exponent of a suffix	§3.9
θ	$1 - 2^L/3^o$, the expansion margin	§1, §4
Λ	$o \log 3 - L \log 2 = -\log(1-\theta)$	§3, §5
μ	$\log_2(3/2)$	§5
u_k	$(1+\mu)a_k - k$, the exponent walk	§5
w_k	$2^{u_k} = 3^{a_k}/2^k$, the walk weight	§5
n'	$n\, e^{-D}$, the reduced base, with D the deficit of §5.1	§5
C_L, C_*	the charge per letter of a word, and its limit	§5
g	the charge weight $1/(e^{W\nu} W\nu)$	§5
p_j/q_j	the continued-fraction convergents of $\log_2 3$	§5
b_j, $s(L)$	the Ostrowski digits of $L = \sum_j b_j q_j$, and their sum	§5
$n_{\max}(L)$	the largest cycle minimum the kill criterion admits at length L	§1, §5

In these core estimates, e is also Euler's number in the numerical displays of §3 and §5, where it appears as e^x. s is an integration variable inside the two displays of §5.4 that evaluate C_*; the digit sum always carries its argument, $s(L)$. These conventions govern the core estimates; local symbols are reset explicitly. In particular, I_b is the threshold interval and m, M are extrema in §3.10; $Q(s)$ is an odd-integer projection in §3.11; U, V, P, Q are words in §3.13; and $e_W(x)$ is the absolute word loss in Appendix F. Local d symbols in Appendices E–F denote quantities defined in their displays, including divisors, carries and gaps.

2. Envelope

Let $\mathcal{B} = \{E, O\}$. A finite itinerary $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the trajectory of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$-fold iterate.

Theorem 2.1 (fixed-itinerary monotonicity). If $n \leq m$ and both realize w, then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w. The empty itinerary is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. □

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-itinerary power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. The empty itinerary is the equality $n \leq n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \leq n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \leq x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting itinerary.

2.4 Exact floor defect

Theorem 2.2 is the inequality form of an exact identity

$$n^{3^{\#O(w)}} = J^{|w|}(n)^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \geq 0.$$

The identity, its vanishing law, and the two-term composition are Theorems 2.4–2.6 and Corollary 2.7 in Appendix C. A mixed realized word has $\Delta_w(n) > 0$. This explains why the envelope of Theorem 2.2 is true, and why a mixed cycle itinerary is formally expanding. Theorem 4.4 uses only the nonnegativity $\Delta_w(n) \geq 0$.

No uniform local tax exists: the relative slack of a single letter tends to 0 with the state. Finance must therefore be a global comparison, not a fixed cost per odd or even step. A quantitative itinerary-dependent lower bound on Δ_w, or a tighter upper bound on $\sum 1/(x_i \log x_i)$, would improve the period cutoff; neither is proved here.

3. Structural restrictions on cycle itineraries

The role of this section in the paper is structural, not numerical. The minimum geometry established here — the cycle minimum is odd, prefixes to even states are superquadratic (Theorem 3.2), a minimum-based itinerary has the canonical run form of Lemma 3.21b, and the last even letter lands in an explicit one-step preimage (Lemma 3.4(iv)) — is exactly what the finance unroll of Section 4 and the transport of Section 5 consume. The even-count exclusion (Theorem 3.22: every nontrivial cycle itinerary has at least four even letters, hence period at least eleven) is the section's own headline, but as a period bound it is superseded the moment financing appears; it is retained because it is floor-free. The small-period censuses (Theorems 3.6 and 3.8) are supporting. Main-text proofs are kept to the short structural lemmas; the longer case analyses — Lemmas 3.5 and 3.7, the censuses of Theorems 3.6 and 3.8, and the family exclusions of Theorems 3.12–3.21 — are Appendix D. After the one-step preimages, a minimum-based itinerary has a canonical run form. The three regimes $1 \le e \le 3$ use $O^{a_1}E \cdots O^{a_e}E$, with exactly e even letters. These forms are eliminated by a next-square obstruction, by long odd-run growth against a last-even one-step preimage, or by a finite exceptional window. The family calculations are Appendix D.

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd one-step preimages are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd one-step preimage indexed by m. Then $(a+1)^3 \le b^3 < (m+1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 < 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. □

Theorem 3.2 (cycle restrictions). Let w be a cycle itinerary at $n \ge 2$.

(i) The itinerary is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting itinerary cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

(iii) A realized itinerary v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix OOE is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$. Alternatively: a mixed cycle itinerary has $\Delta_w(n) > 0$ by Theorem 2.5 in Appendix C, so the envelope is strict; a monochrome tower cannot return for $n \geq 2$.

Every state on such a cycle is at least 2: once a trajectory reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return one-step preimage would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized itinerary v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. □

Lemma 3.21b (canonical run form). Let the cycle contain a state at least 2. After rotation to a minimum-based orientation, the itinerary begins with OO and ends with E; hence every itinerary with $1 \leq e \leq 3$ even letters has the canonical run decomposition $O^{a_1}E \cdots O^{a_e}E$, with $a_1 \geq 2$ and $a_i \geq 0$ for $i > 1$. The case $e = 0$ is all-odd and is already forbidden by the last-letter restriction. No other cyclic rotation needs a separate case.

Proof. Theorem 3.2: the minimum is odd, so the itinerary cannot begin with E or OE, and it cannot end with an odd letter. Thus a minimum-based itinerary starts OO and ends E, so $e \geq 1$. The remaining letters are odd runs separated by the e even letters, so the itinerary is $O^{a_1}E \cdots O^{a_e}E$ with $a_1 \geq 2$. The forms for $e = 1, 2, 3$ are respectively O^aE, O^aEO^cE, and $O^aEO^bEO^cE$. Every nontrivial cycle itinerary has a minimum-based rotation of the same even-count, and that orientation is already in this form. □

Thus it is enough to exclude the three regimes $e = 1, 2, 3$. The rest of the section does that. Theorems 3.6 and 3.8 record the short-period consequences; Theorem 3.22 is the structural statement.

Lemma 3.3 (coarse lower envelope). For $n \geq 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \leq n < (q+1)^2$. For $q \geq 1$ one has $(q+1)^2 \leq 4q^2$, and the second inequality is strict for $q \geq 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \leq 4\, J(n)^2$ and an odd step satisfies $n^3 \leq 4\, J(n)^2$. Composing along a realized itinerary v of length k with odd count o gives

$$n^{3^o} \leq C_v\, J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. □

Lemma 3.4 (next-square thresholds). (i) If $q \geq 5$ realizes OO, then $J^2(q) \geq (q+1)^2$. (ii) If $q \geq 3$ realizes OOO, then $J^3(q) \geq (q+1)^2$. (iii) If a realized itinerary v satisfies $J^{|v|}(q) \geq (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \geq (q+1)^2$. (iv) If vE is a cycle itinerary at n, then $J^{|v|}(n) < (n+1)^2$. (v) If $a \geq 3$, then O^aE is not a cycle itinerary at any $n \geq 2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \geq (q+1)^2$ follows from $m^3 \geq (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \geq (q+1)^2$. If $q = 5$, then $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \geq 6^4 = 1296$. Since q realizes OO, it is odd, so the only remaining case is $q \geq 7$. Then $m \geq q^{3/2} - 1$, so it is enough that $(q^{3/2} - 1)^3 \geq (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to $q^{9/2} - 3q^3 \geq (q+1)^4$, or equivalently $\sqrt{q} - 3/q \geq (1 + 1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the trajectory $3 \to 5 \to 11 \to 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO, then it realizes OO, so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$.

For (v), suppose O^aE is a cycle itinerary at $n \geq 2$. The start is odd, hence at least 3, and realizes O^a. Parts (ii) and (iii) give $J^a(n) \geq (n+1)^2$, contradicting (iv). □

Lemma 3.5 (two length-six exclusions). Neither $OOOEOE$ nor $OOOOEE$ is a cycle itinerary at any $n \geq 2$.

Proof. Appendix D.

Theorem 3.6 (small-cycle census). No itinerary of length at most six is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Appendix D. The reduction used there — an all-odd itinerary cannot return, and every mixed cycle itinerary rotates to an even-terminating cycle itinerary based at a cycle state $m \geq 2$ — is reused by Theorem 3.8 and Theorem 3.22.

Lemma 3.4(v) excludes every odd-run-then-even word $O^a E$ with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle itinerary at any $n \geq 2$.

Proof. Appendix D.

Theorem 3.8 (small-cycle census through length seven). No itinerary of length at most seven is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. Appendix D.

Theorems 3.6 and 3.8 are supporting period statements. The structural claim is the even-count exclusion of Theorem 3.22. The same one-step preimages organise leftover itineraries by even-count. Write

$$G_a = 2\big(3^a - 2^a\big)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle itinerary based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle itinerary at n. Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z)+1)^2 \leq \big((n+1)^{2^r}\big)^2 = (n+1)^{2^{r+1}}$. ☐

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a, then

$$n^{3^a} \leq 2^{G_a}\, J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized itinerary, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{G_a}$ with $G_0 = 0$ yields the recurrence $G_{a+1} = 3G_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $G_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. ☐

Lemma 3.11 (seven-odd window). No integer n with $2 \leq n < 256$ realizes the itinerary O^7.

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `decide +kernel` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A). ☐

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle itinerary at n.

Proof. Appendix D.

A cycle itinerary w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. The remainder after a proper prefix of a cycle itinerary need not itself be a cycle itinerary at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-itinerary exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover one-step preimage is measured against a larger start and need not contradict the tail at y. In particular, Theorem 3.13 does not assert that those itineraries fail to be cycle itineraries at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The itinerary O^aEEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEOEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOOEE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOOEE$ is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEEOE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOEOE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOEOE$ is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.21 (gapped leftovers as cycle itineraries). Let $n \geq 2$. No cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

Theorems 3.12–3.21 assemble into an even-count exclusion: no cycle itinerary has fewer than four even letters, so a nontrivial cycle has period at least eleven (Theorem 3.22). Section 4 excludes later periods by financing.

Every integer $2 \leq n < 12$ reaches 1, so a cycle minimum is at least 12. Write e for the number of even letters. Lemma 3.21b is the canonical run form.

Lemma 3.21a (classification). Every nontrivial minimum-based cycle itinerary with at most three even letters belongs to one of the families excluded by Lemma 3.4(v) and Theorems 3.12–3.21.

Proof. Lemma 3.21b gives $O^{a_1}E \cdots O^{a_e}E$ with $a_1 \geq 2$; write the last odd-run length as c when $e \geq 2$. If $e = 0$, the itinerary is all-odd. If $e = 1$, it is O^aE. If $e = 2$, a last run $c \geq 2$

is the internal-even bootstrap of Lemma 3.4, and the remaining shapes are the two-even families of Theorem 3.12. If $e = 3$, again $c \geq 2$ is the bootstrap, while $c \in \{0, 1\}$ is either a gapped leftover (Theorem 3.21) or one of the seven bunched families (Theorems 3.14–3.20). $\square$

e	remaining minimum-based forms	elimination
0	all-odd	cannot return (Theorem 3.6)
1	O^aE	next-square; Lemma 3.4(v)
2	O^aEE, O^aEOE, last run ≥ 2	Theorem 3.12; Lemma 3.4
3	bunched families and gapped leftovers	Theorems 3.14–3.21
≥ 4	not excluded by even-count	finance (Section 4)

Theorem 3.22 (even-count). No itinerary with fewer than four even letters is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial cycle itinerary has at least four even letters.

Proof. Every cycle itinerary has a minimum-based rotation, with the same even-count. Lemma 3.21b puts that orientation in canonical run form; Lemma 3.21a names the families. In detail:

If $e = 0$, the itinerary is all-odd and cannot return, as in Theorem 3.6.

If $e = 1$, the itinerary is O^aE with $a \geq 2$. The case $a = 2$ is OOE, excluded in Theorem 3.6. The cases $a \geq 3$ are Lemma 3.4(v).

If $e = 2$, the itinerary is O^aEO^cE with $a \geq 2$. A last odd-run $c \geq 2$ is an internal even letter followed by OO or OOO. Lemma 3.4 at the cycle minimum ($n \geq 12$) contradicts the last-even one-step preimage. The remaining shapes are O^aEE and O^aEOE. Expansion forces $a \geq 4$ and $a \geq 3$ respectively, so both are Theorem 3.12.

If $e = 3$, the itinerary is $O^aEO^bEO^cE$ with $a \geq 2$. Again $c \geq 2$ is the internal-even bootstrap. The remaining last runs are $c = 0$ and $c = 1$. For $c = 0$ and $b \geq 4$ the itinerary is a gapped leftover O^aEO^bEE, excluded by Theorem 3.21. For $c = 0$ and $b \leq 3$ the word is one of O^aEEE, O^aEOEE, O^aEOOEE, $O^aEOOOEE$, excluded by Theorems 3.14–3.17 once expansion supplies the stated lower bounds on a. For $c = 1$ and $b \geq 3$ the itinerary is a gapped leftover O^aEO^bEOE, excluded by Theorem 3.21. For $c = 1$ and $b \leq 2$ the itinerary is one of O^aEEOE, O^aEOEOE, $O^aEOOEOE$, excluded by Theorems 3.18–3.20.

Thus $e \leq 3$ is impossible. $\square$

Corollary 3.23. A nontrivial cycle, if one exists, has period at least eleven.

Proof. Theorem 3.22 gives four even letters, hence $o \leq L - 4$. Formal expansion is $2^L < 3^o \leq 3^{L-4}$. The comparison $2^L < 3^{L-4}$ first holds at $L = 11$. $\square$

In particular there is no cycle of length eight, nine, or ten.

3.9 One inequality behind the eleven exclusions

The earlier exclusions use two envelopes: the odd-run lower envelope of Lemma 3.10 pushes the state up, while the one-step preimage of the trailing letters bounds it from above. A single comparison explains the ten suffixes collected in Corollary 3.27.

Lemma 3.24 (odd-run envelope in closed form). If $n \geq 1$ realizes O^a, then

$$J^a(n) \;\geq\; 4\left(\frac{n}{4}\right)^{(3/2)^a}.$$

Proof. Lemma 3.10 is $n^{3^a} \leq 2^{G_a} J^a(n)^{2^a}$ with $G_a = 2(3^a - 2^a)$. Taking 2^a-th roots gives $J^a(n) \geq n^{(3/2)^a}\, 2^{-G_a/2^a}$, and $G_a/2^a = 2\big((3/2)^a - 1\big)$, so the factor is $4^{\,1-(3/2)^a}$. □

One constant and one exponent. The constant 4 is the whole cost of Lemma 3.3's coarse step bound, paid once rather than a times, and the exponent $(3/2)^a$ is the formal expansion rate. The next lemma is the matching upper envelope, and it is Lemma 3.9 with the hypothesis that the run is even deleted from the induction.

Lemma 3.25 (backward envelope). Let w be a cycle itinerary at n. For each suffix u of w, define an integer $B(u)$ by

$$B(\varepsilon) = n+1, \qquad B(Eu) = B(u)^2, \qquad B(Ou) = \min\{c \in \mathbb{Z} : c^3 \geq B(u)^2\}.$$

Then the state entering u is strictly less than $B(u)$.

Proof. Induction on $|u|$, from the empty suffix. The state entering ε is the return $n < n+1$. Let x be the state entering Eu or Ou and $y = J(x)$ the state entering u, so $y < B(u)$ and, both being integers, $y+1 \leq B(u)$. If x is even, the even one-step preimage gives $x < (y{+}1)^2 \leq B(u)^2$. If x is odd, the odd one-step preimage gives $x^3 < (y{+}1)^2 \leq B(u)^2$, so x is below the least integer whose cube reaches $B(u)^2$. □

Lemma 3.9 is the case $u = E^r$, where no cube root is taken and $B(E^r) = (n+1)^{2^r}$ exactly. In general, write

$$T(u) \;=\; \frac{2^{|u|}}{3^{\#O(u)}}$$

for the *backward exponent* of a suffix — one factor 2 per even letter, one factor $\frac{2}{3}$ per odd letter, read from the return backwards. Then

$$B(u) \;\geq\; (n+1)^{T(u)},$$

with the excess coming only from rounding cube roots up. That excess is additive in the state and therefore invisible in the exponent at scale: for the ten suffixes below it is at most $7 \cdot 10^{-5}$ in $\log_{n+1} B(u)$ at $n = 10^3$, and below 10^{-15} by $n = 10^{12}$.

Theorem 3.26 (run–suffix law). Let w be a minimum-based cycle itinerary at $n \geq 4$, and write $w = v\,O^a\,u$ where O^a is a maximal odd run, $a \geq 1$, so that u is empty or begins with an even letter. If u is nonempty, then

$$4\left(\frac{n}{4}\right)^{(3/2)^a} \;<\; B(u).$$

Proof. Let $m = J^{|v|}(n)$ be the state at the start of the run. The itinerary is minimum-based, so $m \geq n \geq 4$, and m realizes O^a. Lemma 3.24 at m, with the right side increasing in m, gives $J^a(m) \geq 4(n/4)^{(3/2)^a}$. That state is the one entering u, which Lemma 3.25 puts strictly below $B(u)$. $\square$

For a fixed suffix u, the leading exponent of the backward envelope is $T(u)$. Thus $(3/2)^a > T(u)$ identifies shapes that can be excluded once n is sufficiently large. At a specified finite n, however, the test is the exact displayed comparison with $B(u)$; the constants and the $n+1$ terms cannot be dropped. This does not assert formal non-expansion of every proper tail. The next table evaluates the finite-envelope comparison for the stated suffixes.

Corollary 3.27 (eleven statements, ten suffixes). Each row below is Theorem 3.26 for one suffix. The column *least* a is the least odd-run length with $(3/2)^a > T(u)$; the column n_u is the least cycle minimum at which the law fires, computed against the exact envelope $B(u)$ of Lemma 3.25.

suffix u	$T(u)$	least a	n_u	printed in
E	2	2	1032	Lemma 3.4(v), at $a \geq 3$
EE	4	4	205	Theorems 3.12, 3.21
EOE	8/3	3	109	Theorems 3.12, 3.21
EEE	8	6	73	Theorem 3.14
$EOEE$	16/3	5	60	Theorem 3.15
$EOOEE$	32/9	4	45	Theorem 3.16
$EOOOEE$	64/27	3	30	Theorem 3.17
$EEOE$	16/3	5	60	Theorem 3.18
$EOEOE$	32/9	4	45	Theorem 3.19
$EOOEOE$	64/27	3	30	Theorem 3.20

Nine of the ten least odd-run lengths match the earlier statements. The tenth is a strengthening: for $u = E$ the law excludes $a \geq 2$, so the word OOE needs no appeal to the census of Theorem 3.6. Where a theorem prints a threshold on n, the law's is smaller — 205 against the 256 of Theorem 3.12, 73 against the 128 of Theorem 3.14, 45 against the 256 of Theorem 3.16 — because the backward exponent is carried exactly as $T(u)$ rather than rounded up to the next power of two. Every threshold in the table is below the certified descent floor by four orders of magnitude or more, so above that floor Theorem 3.22 is Theorem 3.26 evaluated ten times.

Two caveats keep the table from replacing Appendix D outright. The theorems it reproduces are stated for every $n \geq 2$, and each closes its window below the threshold with a `decide +kernel` evaluation; the law says nothing there. And Theorem 3.21 upgrades Theorem 3.13 from minimum-based itineraries to all cycle itineraries, which the law, being a statement about a minimum-based orientation, does not do on its own. Above the certified floor neither caveat is active.

Lemma 3.28 (sharp odd-run envelope). Let O^a be an odd run inside a minimum-based cycle itinerary at n, starting at the state m, and put

$$X_a = 3\big(3^a - 2^a\big), \qquad Y_a = 2 \cdot 3^a - 3 \cdot 2^a.$$

Then

$$n^{X_a} \;<\; (n+1)^{Y_a}\bigl(J^a(m)+1\bigr)^{2^a}.$$

Proof. Induction on a. For $a=1$ the odd one-step preimage gives $m^3 < (J(m)+1)^2$, and $m \geq n$, which is the claim with $X_1 = 3$, $Y_1 = 0$. Suppose the display holds after k letters of the run, with the state $x_k \geq n$. Cubing gives

$$n^{3X_k} < (n+1)^{3Y_k}(x_k+1)^{3\cdot 2^k}.$$

Because $n \leq x_k$ one has $n(x_k+1) \leq (n+1)x_k$, hence $(x_k+1)^e n^e \leq (n+1)^e x_k^e$ for $e = 3\cdot 2^k$; multiplying through by n^e and substituting turns the display into

$$n^{3X_k+e} < (n+1)^{3Y_k+e}\, x_k^e.$$

Finally $x_k^{3\cdot 2^k} = (x_k^3)^{2^k} < \bigl((x_{k+1}+1)^2\bigr)^{2^k}$. So $X_{k+1} = 3X_k + 3\cdot 2^k$ and $Y_{k+1} = 3Y_k + 3\cdot 2^k$, whose solutions from $X_1 = 3,\ Y_1 = 0$ are the stated closed forms. $\square$

The difference from Lemma 3.24 is where minimum-basedness is used. Lemma 3.24 uses it once, at the start of the run, and then pays Lemma 3.3's factor 4 at every step. Lemma 3.28 uses it at *every* step: the intermediate state is itself above the cycle minimum, so the successor's $+1$ can be traded for a factor $(n+1)/n$ rather than for a factor 4. This is the chain already formalized at $a=7$ in `Problems/Juggler/O7EEEEGap.lean`, whose constants 6177 and 3990 are X_7 and Y_7; `absorb_odd_step` there is the inductive step above.

Theorem 3.29 (run–suffix law, sharp form). In the situation of Theorem 3.26,

$$n^{X_a} \;<\; (n+1)^{Y_a}\, B(u)^{2^a}.$$

Proof. Lemma 3.28 bounds $J^a(m)+1$ from below and Lemma 3.25 bounds $J^a(m)+1 \leq B(u)$ from above. $\square$

Since $X_a - Y_a = 3^a$, the leading-order reading is unchanged: with $B(u) \approx (n+1)^{T(u)}$ the criterion is again the crossing $(3/2)^a > T(u)$. What changes is the price. Theorem 3.26 has to buy the crossing against a factor $4^{(3/2)^a}$, which costs a margin $\log 4/\log n$; Theorem 3.29 buys it against $(1+1/n)^{Y_a}$, which costs a margin of order $1/(n\log n)$. The crude form needs a floor exponential in the reciprocal margin, the sharp form one merely linear in it.

Corollary 3.30 (the same table, sharply). For the ten suffixes of Corollary 3.27, the least cycle minimum at which Theorem 3.29 fires is

$$7,\ 6,\ 6,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5,\ 5$$

in the order printed there, against $1032, 205, 109, 73, 60, 45, 30, 60, 45, 30$ for Theorem 3.26 and 256 or 128 in the proofs of Appendix D. Every $n \leq 299$ reaches 1, so a cycle minimum is at least 300 and all ten are unconditional.

Theorem 3.31 (even count at least eight). Every cycle itinerary based at $n \geq 2$ has at least eight even letters. Consequently, every nontrivial cycle has period at least twenty-two.

Proof. Let a minimum-based cycle itinerary have $e \leq 7$ even letters and canonical run form $O^{a_0}E \cdots O^{a_{e-1}}E$. Apply Theorem 3.29 at run i. Its suffix has $e-i$ even letters and

$\sum_{j>i} a_j$ odd ones, so $T(u) \le 2^{e-i}$, with equality exactly when the later runs are empty; and the criterion is monotone in $B(u)$, so that empty case is the worst. Hence

$$a_i \;\le\; \left\lfloor (e-i)\,\tfrac{\log 2}{\log(3/2)} \right\rfloor,$$

and the thresholds of those bounding applications — rows $r \le 7$ of the table in Remark 3.32 — are all at most 55. The run bounds are $(5,3,1)$, $(6,5,3,1)$, $(8,6,5,3,1)$, $(10,8,6,5,3,1)$ and $(11,10,8,6,5,3,1)$ for $e = 3,\dots,7$, leaving 16, 186, 2037, 25353 and 325452 canonical forms once formal expansion $3^o > 2^{o+e}$ is imposed. Theorem 3.29 closes every one of them, the first four at $n \ge 16$ and the last at $n \ge 64$. The cases $e \le 2$ are Theorem 3.22. A cycle minimum is at least 300, so $n \ge 64$ is free. Given $e \ge 8$, formal expansion $2^L < 3^{L-8}$ first holds at $L = 22$. $\square$

This replaces $e \ge 4$ and period ≥ 11 in Theorem 3.22 and Corollary 3.23 — it doubles the floor-free period bound — which remain the statements proved for every $n \ge 2$ without any descent input. Numerically the gain is of no consequence — Theorem 4.6 already gives period ≥ 25781 at a floor of 10^6 — but Corollary 3.23 is the paper's only bound that needs no floor at all, and this is how far the same two envelopes carry it.

Remark 3.32 (where the law stops, and why Section 4 exists). The reading $B(u) \approx (n+1)^{T(u)}$ is good only while the backward exponent stays comfortably above 1, because the envelope of Lemma 3.25 rounds a cube root up at every odd letter. For short suffixes that rounding is invisible: over the ten suffixes of Corollary 3.27 it moves $\log_{n+1} B(u)$ by at most $7 \cdot 10^{-5}$ at $n = 10^3$. For a cycle candidate read as one long word it is the whole story. There the exponent of the full word is $2^L/3^o = 1-\theta$, just *below* one, so the envelope hovers at n itself and the $+1$ rounded into each of the L backward steps is no longer negligible; accumulated, it is exactly the factor L that Theorem 4.4 carries on its right-hand side. The run–suffix law is sharp on short tails and degenerates to the finance inequality on the whole cycle. That is the division of labour between this section and the next.

The margins the law has to pay for are the same objects Section 5 fights. For the pure trailing block $O^{a_r}E^r$, writing a_r for the least a with $(3/2)^a > 2^r$, the margin is θ at $(o,L) = (a_r, a_r + r)$:

r	(o,L)	L/o	θ_r	n threshold
1	$(2,3)$	$3/2$	0.1111	7
2	$(4,6)$	$3/2$	0.2099	6
3	$(6,9)$	$3/2$	0.2977	5
4	$(7,11)$	$11/7$	0.0636	16
5	$(9,14)$	$14/9$	0.1676	8
6	$(11,17)$	$17/11$	0.2601	6
7	$(12,19)$	$19/12$	0.0135	55
11	$(19,30)$	$30/19$	0.0762	15
14	$(24,38)$	$19/12$	0.0267	32

The spikes sit at $11/7$ and $19/12$, the semiconvergent and the convergent of $\log_2 3$ at smallest denominator, and the thresholds they carry grow like $1/\theta_r$ rather than like $\exp(1/\theta_r)$

— which is why they stay in the tens. What limits Theorem 3.31 is therefore not any threshold but the enumeration: the number of canonical forms with e even letters grows faster than $e!$. The obstruction that stops this section and the obstruction that leaves the semiconvergent fan of Section 5 standing are the same arithmetic; only the denominators differ.

What each further even letter is worth. Expansion at even count e needs $L\log(3/2) > e\log 3$, so each further even letter buys $\log 3/\log(3/2) \in (2.70, 2.71)$ in period: the thresholds are $11, 14, 17, 19, 22$ at $e = 4, 5, 6, 7, 8$. Both ends of that constant are integer certificates rather than numerical bounds on logarithms: $(3/2)^{27/10} < 3$ is $3^{17} < 2^{27}$, and $3 < (3/2)^{271/100}$ is $2^{271} < 3^{171}$, the latter tight to four significant figures (`expansion_rate_lower`, `expansion_rate_upper`, `expansion_e4`, `expansion_e5`, `expansion_e6`, `FanLaw.lean`).

An earlier draft of this section priced a fifth even letter as a family program in the style of Theorems 3.14–3.20: two further gapped-leftover theorems and some twenty-five bunched families, for three units of period. Theorem 3.31 makes that program unnecessary. The route it takes is not a longer case analysis but a cheaper envelope — Lemma 3.28 in place of Lemma 3.24 — and the case analysis it then needs is mechanical rather than hand-written. What the earlier accounting got right is that raising the even count is the only route to a stronger floor-free statement; what it got wrong is the assumption that the eleven exclusions had to be generalised one at a time.

3.10 Cycle order below the cubic height

The preceding restrictions concern the itinerary or its counts. A height hypothesis gives additional information about the order of all states in the closed orbit. Throughout this subsection a primitive cycle means its distinct states in one traversal; m and M denote its minimum and maximum. The bounds obtained here do not exclude every cycle or improve the numerical period bounds in Section 5.

Theorem 3.33 (cubic-band order). Suppose C is a primitive nontrivial Juggler cycle of length L, minimum $m > 1$, maximum $M < m^3$, and odd-state count o. Put $e = L - o$ and list its states as $c_0 = m < c_1 < \cdots < c_{L-1} = M$. Then

$$\begin{aligned} &c_i \text{ is odd} \iff i < o \iff c_i < m^2, \\ &J(c_i) = c_{(i+e) \bmod L}, \qquad \gcd(L, o) = 1. \end{aligned} \tag{CB1}$$

If a_k is the number of odd steps in the first k steps from the minimum, then, for $0 \le k \le L$,

$$a_k = \left\lceil \frac{ko}{L} \right\rceil. \tag{CB2}$$

Thus the letter at position k, starting with position zero, is O precisely when $\lceil (k+1)o/L \rceil - \lceil ko/L \rceil = 1$. This specifies the word, rather than only its letter counts.

In particular, a primitive nontrivial cycle whose minimum-based word is not (CB2), or whose counts satisfy $\gcd(L, o) > 1$, must satisfy $M \ge m^3$. Since m is odd and M even, the integer consequence is $M \ge m^3 + 1$. This is a conditional height bound, not a global cycle exclusion.

Proof. A cycle with minimum greater than 1 has odd minimum $m \geq 3$: an even minimum would map below itself. Its maximum is even, since $\lfloor\sqrt{x^3}\rfloor > x$ for every integer $x \geq 3$. Thus $o, e > 0$.

If an even state $x \in C$ satisfied $x < m^2$, then $J(x) < m$, impossible. If an odd state $x \in C$ satisfied $x \geq m^2$, then $J(x) \geq \lfloor\sqrt{(m^2)^3}\rfloor = m^3 > M$, also impossible. Therefore the first o sorted states are exactly the odd states.

For any even $x \in C$ and odd $y \in C$, monotonicity gives

$$J(x) \leq \lfloor\sqrt{M}\rfloor \leq \lfloor\sqrt{m^3}\rfloor \leq J(y).$$

The possible equality from taking floors must be handled: J permutes the states of a primitive cycle, so it is injective on C. Equality of these images would force $x = y$, impossible for opposite parities. Hence every even image is strictly below every odd image. Each branch is nondecreasing, and injectivity on C makes its restriction strictly increasing.

The e even images are consequently $c_0, \dots, c_{e-1}$, in that order, and the o odd images are $c_e, \dots, c_{L-1}$, in that order. This proves $J(c_i) = c_{(i+e) \bmod L}$. Adding e modulo L has an orbit of size $L/\gcd(e, L)$. Primitivity requires that size to be L, so $\gcd(e, L) = \gcd(o, L) = 1$.

From rank zero the rank at time k is $ke \bmod L$. An even step occurs exactly when adding e wraps past L: the starting rank is then at least $L - e = o$. The number of wraps in the first k additions is $\lfloor ke/L\rfloor$. The odd count is therefore $k - \lfloor ke/L\rfloor = \lceil ko/L\rceil$, proving (CB2). No exact logarithmic closure is used.

Proposition 3.34 (threshold comparison map). For every integer $b \geq 3$, define a different map on $I_b = \{b, b+1, \dots, b^3 - 1\}$:

$$S_b(x) = \begin{cases} \lfloor\sqrt{x^3}\rfloor, & x < b^2, \\ \lfloor\sqrt{x}\rfloor, & x \geq b^2. \end{cases} \tag{CB3}$$

It maps I_b into itself, has no fixed point, and therefore has a cycle of length at least two. All its primitive cycles have the rank-rotation and mechanical-word properties in (CB1)–(CB2), counting branch labels instead of actual odd integers. Their minima are at least b, so this construction is unbounded in scale.

Map distinction: (CB3) chooses its branch by size. It is not the Juggler map. A cycle of (CB3) is a Juggler cycle if and only if every state satisfies

$$x \text{ odd} \iff x < b^2. \tag{CB4}$$

The construction does not assert that (CB4) ever holds on a full cycle.

Proof. For $b \leq x < b^2$, $b \leq x < \lfloor\sqrt{x^3}\rfloor < b^3$. For $b^2 \leq x < b^3$, $b \leq \lfloor\sqrt{x}\rfloor < b^2 \leq x$. Both conclusions follow directly from the square/cube inequalities; the strict growth uses $x \geq 3$. Thus (CB3) preserves the nonempty finite set I_b, and neither branch has a fixed point there. Every orbit eventually repeats, yielding a primitive cycle of length at least two.

On any such cycle the states using the odd branch are its lower part $x < b^2$. The even-branch images are at most $\lfloor\sqrt{b^3 - 1}\rfloor$, and the odd-branch images are at least $\lfloor\sqrt{b^3}\rfloor$. Once again these weak inequalities become strict between cycle images by injectivity. The same sorting and modular-rank argument proves (CB1)–(CB2) with branch counts. Since the least state is at least b, these closed orbits exist beyond any fixed lower bound.

Their formal exponent is also strictly expanding. Composing $S_b(x) \le x^{3/2}$ or $S_b(x) \le x^{1/2}$ around a cycle with minimum $r > 1$ gives $r \le r^{3^o/2^L}$, hence $3^o \ge 2^L$. Equality is impossible for positive o, L, since a positive power of 3 is odd and a positive power of 2 is even. Consequently $3^o > 2^L$. This sign, exact floors, distinctness, closed-orbit order and unbounded minimum coexist in the relaxed map. They do not force the missing parity rule (CB4).

Theorem 3.35 (common period and interlacing). For a fixed $b \ge 3$, all primitive cycles of S_b have the same period, lower/upper counts, and minimum-based mechanical word. If there are t cycles of common period L, they interlace: in the sorted list of all tL periodic points, each cycle occupies exactly one residue class of ranks modulo t.

Proof. Let P be the union of all periodic points. It is a nonempty finite invariant set, and S_b permutes it. Put $N = |P|$, and let E count its upper-branch points. As in Theorem 3.33, branch monotonicity and separated image ranges, with injectivity removing ties, make the rank permutation addition by E modulo N. Write $t = \gcd(N, E)$. This permutation has exactly t orbits, each of length $L = N/t$, namely the rank residue classes modulo t. The upper states occupy the final E ranks. Since $t \mid E$, every residue class contains exactly $e = E/t$ of them. All cycles therefore have counts $(o, e) = ((N - E)/t, E/t)$, and Theorem 3.33's wrap count gives their identical minimum-based word. This does not assert uniqueness.

Proposition 3.36 (uniform sorted grid). Let C be a primitive threshold cycle, or an actual Juggler cycle with $m > 1$ and $M < m^3$. Retain its sorted states c_i and counts (L, o, e), and put

$$T = \log 3, \quad \alpha = \log(3/2), \quad \Lambda = o \log 3 - L \log 2 > 0, \quad v_i = \log \frac{\log c_i}{\log m}.$$

Extend $v_{i+L} = v_i + T$. With $w_i = v_i - iT/L$,

$$\operatorname{osc}(w) \le \left(1 - \frac{1}{L}\right)\Lambda, \qquad \left|v_i - \frac{iT}{L}\right| \le \left(1 - \frac{1}{L}\right)\Lambda. \tag{CB5}$$

The lifted adjacent gaps $h_i = v_{i+1} - v_i > 0$, including the gap at the seam, satisfy

$$\max h_i - \min h_i \le \Lambda, \qquad \left|h_i - \frac{T}{L}\right| \le \left(1 - \frac{1}{L}\right)\Lambda. \tag{CB6}$$

These are uniform bounds on every rank, without a factor L in the error.

Proof. Let $p_i = 3/2$ for $i < o$ and $p_i = 1/2$ otherwise, and let y_i be the successor of c_i. Define

$$\delta_i = \log \frac{p_i \log c_i}{\log y_i} \ge 0.$$

All states exceed 1. Exact floors give $y_i \le c_i^{p_i}$, which justifies the sign. The rank rule and the periodic lift give

$$v_{i+e} = v_i + \alpha - \delta_i. \tag{CB7}$$

For an upper step the lift adds T, converting $-\log 2$ to α. Summing over all ranks gives

$$\sum_i \delta_i = L\alpha - eT = \Lambda, \qquad w_{i+e} - w_i = \frac{\Lambda}{L} - \delta_i. \tag{CB8}$$

In particular $\Lambda \geq 0$. Equality would give $3^o = 2^L$, impossible for positive branch counts, so $\Lambda > 0$. For distinct ranks i, j, choose $1 \leq k \leq L-1$ with $j \equiv i + ke \pmod{L}$. The k indices traversed are distinct because $\gcd(e, L) = 1$. Thus

$$w_j - w_i = \frac{k\Lambda}{L} - \sum_{r=0}^{k-1} \delta_{i+re}.$$

The partial sum is between 0 and Λ, proving the oscillation bound. Since $w_0 = 0$, it also proves the absolute bound in (CB5). Subtracting (CB7) at neighboring ranks gives $h_{i+e} - h_i = \delta_i - \delta_{i+1}$. Iteration expresses any difference of two gaps as the difference of two partial sums of the δ_i; each lies in $[0, \Lambda]$. Hence the gap range is at most Λ. Its mean is T/L, and the maximum distance from the mean of L values of range at most Λ is at most $(1 - 1/L)\Lambda$, proving (CB6). The seam gap is positive because $M < m^3$, irrespective of whether (CB6) supplies a positive numerical lower bound.

Parity-scale limitation. Actual lower states are distinct odd integers, so $c_i \geq m + 2i$ for $i < o$. A necessary consequence of (CB5) is

$$\log \frac{\log(m + 2i)}{\log m} \leq \frac{i \log 3}{L} + \Lambda.$$

If $\Lambda \leq C/L$, the case $i = 1$ gives

$$L \leq \frac{\log 3 + C}{\log(\log(m+2)/\log m)} \sim \frac{\log 3 + C}{2} m \log m.$$

This does not conflict with a minimum of order L^2. More specifically, for an unanchored rank $i > 0$, the interval for a state permitted by (CB5) has width of order $c_i \log(c_i)\Lambda$ in the local regime $\Lambda \log(c_i) \ll 1$. In the illustrative regime Λ comparable to $1/L$ and m comparable to $L^2/(\log L)^2$, that permitted width is already of order $L/\log L$ at low unanchored ranks and therefore does not determine parity. The minimum $c_0 = m$ is fixed exactly and has no such uncertainty. This is a statement about the precision of (CB5), not a lower bound on the actual error or an assertion that a cycle realizes this scale. Proposition 6.3b supplies an additional upper-cell restriction, including a constraint on the leading constant.

Proposition 3.37 (one-unit successor allowance). For every odd $b \geq 3$, define

$$D_b = \{x \text{ odd} : b \leq x \leq b^2\} \cup \{x \text{ even} : b^2 + 1 \leq x \leq b^3 - 1\}.$$

For $b \leq z \leq b^3$, let $P_b(z) = \max\{d \in D_b : d \leq z\}$, and set

$$R_b(x) = \begin{cases} P_b(x^{3/2}), & x \text{ odd}, \\ P_b(\sqrt{x}), & x \text{ even}. \end{cases}$$

This finite map has a nontrivial cycle and every edge satisfies

$$R_b(x) \in \{J(x), J(x) - 1\}. \tag{CB9}$$

The exponent agrees with the actual source parity at every state. All its primitive cycles have $M < m^3$, the rank-rotation rule, coprime counts, and the mechanical word. Therefore correct source parity, this global order, and one-sided power-rounding loss below 2 admit closed orbits at arbitrarily large minima.

Proof and boundary check. Adjacent members of D_b have gap 2, except the gap 1 from b^2 to b^2+1. Therefore $0 \le z - P_b(z) < 2$, including $z = b^3$, whose image is $b^3 - 1$. Odd source images lie between $b^{3/2}$ and b^3; even source images lie strictly between b and $b^{3/2}$. The projection is defined in each case and preserves D_b. Odd steps strictly increase, since $x^{3/2} - x \ge 3\sqrt{3} - 3 > 2$ for $x \ge 3$; even steps strictly decrease. Finiteness gives a nontrivial cycle. Its minimum is odd and at least b, its maximum is even and below b^3, so $M < m^3$. The common nondecreasing projection preserves the weak separation of the two image ranges; injectivity on a cycle makes it strict. The rank proof applies again. Finally an integer output at one-sided distance less than 2 from the real power differs from its floor by either 0 or 1, proving (CB9).

The inclusion of b^2 is essential: ending the lower odd range at $b^2 - 2$ would create a gap 3 and would not prove (CB9). The exact-floor characterization "odd iff $x < m^2$" is **not** imported into this altered map; an odd source b^2 is allowed. The rank and mechanical-word conclusions, which do not need that characterization, are the ones retained.

No assertion is made that every R_b cycle contains an altered edge. Proving that assertion uniformly would itself exclude the cubic-band Juggler cycles: a cycle with all corrections zero is an actual Juggler cycle, and any actual cubic-band cycle lies in D_m and is unchanged by R_m. For the explicit cycle in Proposition 3.38, each odd edge is altered.

Equal rotation fractions do not detect an alteration. At $b = 3$, S_b has $3 \to 5 \to 11 \to 3$, while R_b has $3 \to 5 \to 10 \to 3$. They have identical period and counts, but different cycles. Thus replacing the map by a monotone downward projection need not strictly change its common rotation fraction.

Proposition 3.38 (branch offsets are invisible to differences). Define G on positive integers by $G(1) = 1$, $G(x) = J(x) - 1$ for odd $x \ge 3$, and $G(x) = J(x)$ for even x. It has the primitive cycle

$$13, 45, 300, 17, 69, 572, 23, 109, 1136, 33, 188, 13. \tag{CB10}$$

Its minimum is 13, its maximum is $1136 < 13^3$, and its counts are $(L, o, e) = (11, 7, 4)$. It is also a cycle of R_{11}. Every source uses its actual parity, and the same rank and mechanical-word conditions hold.

Nevertheless, for every pair of odd sources $x, x' \ge 3$, or every pair of even positive sources,

$$G(x') - G(x) = J(x') - J(x). \tag{CB11}$$

This is a global identity, not a numerical approximation on (CB10). On sources greater than 1, all within-branch higher finite differences erase the constant shift. Thus global rank order, correct source parity, and even the complete within-branch image-difference data of J restricted to sources greater than 1 do not by themselves exclude cycles. Differences involving the fixed point 1 are deliberately excluded: since $G(1) = J(1)$, they could anchor the odd-branch constant.

Exact witness certificate. The entries in the middle column are certified by $J(x)^2 \le x^h < (J(x) + 1)^2$, with $h = 3$ on odd sources and $h = 1$ on even sources. Subtracting one on each odd row yields the successive states of (CB10).

x	$J(x)$	$G(x)$
13	46	45
45	301	300
300	17	17
17	70	69
69	573	572
572	23	23
23	110	109
109	1137	1136
1136	33	33
33	189	188
188	13	13

All eleven states are distinct. Each row can be checked directly by the displayed square-cell inequalities. Formula (CB11) follows immediately by subtracting the same branch constant on both sides.

Why the strict nearest-even gap test also fails. Suppose two sources $x < x'$ on the same branch have same-parity successors in (CB10). Their image gap $d' = G(x') - G(x)$ is even. For the relevant real branch f, (CB11) gives

$$\left|d' - \big(f(x') - f(x)\big)\right| < 1,$$

because the difference of the two fractional parts lies strictly between -1 and 1. Thus d' is the unique even integer within distance less than 1 of the smooth increment. This strict gap test survives every changed odd edge of this example.

The lost information is the absolute position of each branch image inside its unit floor cell. Subtracting neighboring equations removes an additive constant for each branch. A proof using gaps must restore those constants through absolute values or a relation between the branches. This counterexample concerns a different map; it neither proves nor refutes the no-cycle statement for J.

The remaining absolute floor-cell question. Define the wrong-parity set

$$B_b = \{x \in I_b : x < b^2,\ x \text{ even}\} \cup \{x \in I_b : x \geq b^2,\ x \text{ odd}\}.$$

Must every cycle of S_b meet B_b, for every integer $b \geq 3$? An affirmative answer is equivalent to excluding all actual Juggler cycles with $m > 1$ and $M < m^3$. Indeed, a threshold cycle avoiding B_b is an actual cycle, and an actual cycle in this height range is an S_m cycle by Theorem 3.33. The regime $M \geq m^3$ is separate.

Proposition 3.38 rules out a contradiction based only on the within-branch differences on sources greater than 1: subtraction removes a branch constant. The exact conditions retain more information. With successor y, they are

$$y^2 \leq x^3 < (y+1)^2 \quad (x \text{ odd}), \qquad y^2 \leq x < (y+1)^2 \quad (x \text{ even}).$$

The uniform wrong-parity conclusion from these absolute cells remains open in this paper. Finite checks of the threshold or altered maps do not settle it.

3.11. A periodic return boundary and a sharper height ceiling

The absolute-cell question has a partial answer near the top of the cubic height band. This restriction uses the return map of the complete periodic set. Isolated admissible blocks do not supply its ordering condition.

Write $O(x) = \mathrm{isqrt}(x^3)$ and $E(x) = \mathrm{isqrt}(x)$. These are prescribed branches; a word in O, E need not agree with the parities of its intermediate states. Composition is chronological: $F_{OOE} = E \circ O \circ O$. Let $Q(s)$ denote the greatest odd integer not exceeding s.

Theorem 3.39 (periodic return height restriction). Suppose that an actual Juggler cycle has minimum $m \geq 5$ and maximum $M < m^3$. Put

$$q = O(m), \qquad t = E(M), \qquad z = F_{OOE}(m) = Q(m^{9/8}).$$

Then m, q, t, z are odd and

$$t^3 + 2 \leq [z(z-2)]^2, \qquad M + 2 \leq (t+1)^2. \tag{RC1}$$

Consequently, if T is the greatest positive odd integer satisfying $T^3 + 2 \leq [z(z-2)]^2$, then

$$M \leq (T+1)^2 - 2. \tag{RC2}$$

For $m \geq 7$, a simpler consequence is

$$\boxed{M < m^3 - m^{15/8}.} \tag{RC3}$$

Equivalently, the positive integer gap satisfies $(m^3 - M)^8 > m^{15}$.

Proof. We first record two branch estimates, valid without source-parity assumptions:

$$O(O(x)) \geq x^2 \quad (x \geq 3), \qquad F_{OOE}(x) > x \quad (x \geq 5). \tag{RC4}$$

For the first, take $k = \mathrm{isqrt}(x)$. If $x \geq 4$, then $k \geq 2$, $x \leq k^2 + 2k \leq k^3$, and $O(x) \geq kx$. Thus $O(x)^3 \geq x^4$, which implies the claim. At $x = 3$, $O(O(3)) = 11 \geq 9$. For the second, if $x \geq 9$, then $k \geq 3$ and $k^3 \geq x + 12$. Hence

$$O(x)^3 \geq (x+12)x^3 > (x+1)^4.$$

The last inequality follows from $8x^3 - 6x^2 - 4x - 1 > 0$. For $x = 5, 6, 7, 8$, the first images are $11, 14, 18, 22$, whose cubes are respectively at least $6^4, 7^4, 8^4, 9^4$. The excluded values $3, 4$ satisfy $F_{OOE}(x) = x$.

Let C be the cycle set. Its actual parities make it an S_m cycle: an even source has square root at least m, and so is at least m^2; an odd source at least m^2 would have image at least m^3, contrary to the height assumption. In particular, the minimum is an odd source and the maximum is an even source. Every even-branch image is below m^2, and two successive odd branches reach at least m^2 by (RC4). Therefore the successive returns after even steps have only the words OE and OOE.

The correct section, including its strict upper endpoint, is

$$Y = C \cap [m, q), \qquad q = O(m).$$

All odd-branch images are at least q. All even-branch images are at most $\mathrm{isqrt}(m^3-1) \le q$. Equality with q is impossible: q is already the image of m, and the cycle map is injective on C. Thus Y is exactly the set of even-branch images and $t = E(M) = \max Y$. Every point of Y is odd. The return word at $x \in Y$ is OOE when $x^3 < m^4$, and OE otherwise.

The minimum m uses OOE, since $F_{OE}(m) = \lfloor m^{3/4} \rfloor < m$. The maximum t uses OE, since an OOE return would increase it by (RC4). Set

$$z = F_{OOE}(m), \qquad w = F_{OE}(t).$$

Monotonicity and $t < q$ give

$$w \le F_{OE}(q) = z.$$

These images are distinct. Indeed, the first-return map is a permutation of Y, and $m < t$, since the return of m is larger than m. Hence $w < z$. Since both are odd,

$$w \le z - 2. \tag{RC5}$$

The short-word identity in Appendix E.1 gives $z = Q(m^{9/8})$; here its odd-output hypothesis is supplied by the actual cycle.

Write $p = O(t)$. The states p, M are even, whereas t, w, z are odd. The exact cells and their parity faces imply

$$p + 2 \le (w+1)^2 \le (z-1)^2, \qquad t^3 + 2 \le (p+1)^2 \le [z(z-2)]^2.$$

The cell for $M \mapsto t$ similarly gives $M + 2 \le (t+1)^2$. This proves (RC1) and its integer ceiling (RC2).

For the smooth bound (RC3), even the weaker consequence

$$t^3 < (z-1)^4, \qquad z^8 \le m^9$$

is sufficient. Set $r = m^{3/8}$. Since $m \ge 7$, $r \ge 2$. We have $z \le r^3$, and

$$(r^3-1)^4 \le (r^4-r)^3,$$

because the difference of the right and left sides is $(r^3-1)^3 \ge 0$. Thus $t < r^4 - r$, and

$$M < (r^4 - r + 1)^2 - 2 < r^8 - r^5.$$

The last comparison follows from the exact identity

$$r^8 - r^5 - \big((r^4 - r + 1)^2 - 2\big) = (r-2)(r^4 - r) + 1 > 0.$$

Since $r^8 = m^3$ and $r^5 = m^{15/8}$, this proves (RC3). $\square$

A further smooth consequence of the integer seam. The sharper first inequality in (RC1) also gives

$$t < m^{3/2} - \frac{4}{3} m^{3/8}.$$

Indeed $g(z) = z^{4/3} - \frac{4}{3} z^{1/3}$ is positive for $z \ge 3$, is increasing there, and

$$g(z)^3 - [z(z-2)]^2 = \frac{4}{27} z(9z - 16) > 0.$$

This analytic reformulation is a written consequence; the proof of (RC3) above does not depend on differentiating g.

Scope and finite controls. A threshold cycle in the region $m^3 - m^{15/8} \le M < m^3$, $m \ge 7$, must therefore have at least one wrong-parity state. This does not exclude the lower part of the cubic band or the regime $M \ge m^3$, and it does not increase the period bound 780239. The integer ceiling at hypothetical minimum 9 improves the earlier minimum-only extrema ceiling from 674 to 482; at minimum 6569, it improves 283462537742 to 283388004962. These are conditional arithmetic evaluations, not assertions that cycles with those minima exist.

The threshold cycle

$$(9, 27, 140, 11, 36, 216, 14, 52, 374, 19, 82)$$

has $Y = \{9, 11, 14, 19\}$, $t = 19$, $z = 11$, and $w = 9$. It satisfies $19^3 = 6859 \le 9799 = [11(11-2)]^2 - 2$, yet has wrong parities, including the even retained base 14. Thus the new necessary extrema conditions are not a complete guard test. Appendix E records the exact carry formulas and the limitations of several proposed summaries of the missing guards.

3.12. Successive return gaps and larger height exclusions

The ordered return section supplies further restrictions because selected neighboring states remain distinct after a common return word. The floor errors must be bounded before this discreteness can be used. Successive left inductions preserve one boundary pair; only genuine right inductions transport it.

Retain the notation m, M, Y, t, w, z of Theorem 3.39, and use chronological words

$$A = OOE, \quad B = OE, \quad C = A^2B, \quad D = AC^2, \quad W = D^3C.$$

Write

$$\gamma = \frac{243}{256}, \qquad \delta = \frac{531441}{524288}, \qquad \rho = \delta^3\gamma = \frac{3^{41}}{2^{65}}, \qquad \sigma = \frac{13}{128} + 1 - \rho.$$

Here $7/64 < \sigma < 1/8$.

Theorem 3.40 (successive return height restrictions). Let an actual Juggler cycle have minimum m and maximum $M < m^3$. Then the following statements hold.

(i) If $m \ge 2^{24}$, two genuine transfers of the original boundary through C give

$$z - w \ge 6, \qquad z - w > \frac{26240}{59049} m^{13/128}, \qquad M < m^3 - \frac{1}{2} m^{253/128}. \tag{SR1}$$

(ii) If $m \ge 2^{128}$, a subsequent genuine transfer through W gives

$$z - w \ge 8, \qquad z - w > \frac{2}{5} m^{\sigma}, \qquad M < m^3 - \frac{1}{2} m^{381/128 - \rho}. \tag{SR2}$$

In particular,

$$M < m^3 - \frac{1}{2} m^{127/64}. \tag{SR3}$$

The exponent $381/128 - \rho$ is approximately 1.98795995227. The clean bounds (SR1) and (SR3) have the respective integer forms

$$m^{253} < \big[2(m^3 - M)\big]^{128}, \qquad m^{127} < \big[2(m^3 - M)\big]^{64}.$$

The natural-number gaps are positive under $M < m^3$.

Proof. The complete return on the sorted set $Y = \{y_0 < \cdots < y_{a+b-1}\}$ has lower word A, upper word B, and rank rotation $i \mapsto i + b \pmod{a+b}$. Appendix F proves the forced batch identities

$$a = 2b + r, \quad b = 2r + s, \quad 0 < s < r$$

on the first domain, and additionally

$$r = 3s + u, \qquad 0 < u < s$$

on the second. They place the successive boundary pairs at

$$(y_{b-1}, y_b), \quad (y_{b-r-1}, y_{b-r}), \quad (y_{s-1}, y_s), \quad (y_{s-u-1}, y_{s-u}).$$

The first two changes apply C; the third applies W. All displayed indices exist on their respective domains, and all traces belong to the same actual cycle.

The prescribed maps have absolute losses below 9/8 and 6/5, respectively, at the stated cutoffs. Their paired estimates strictly contract every integer gap at least two. Since each displayed pair has distinct odd endpoints, each genuine transfer decreases the positive even gap by at least two. The quantitative versions give the gap bounds in (SR1)–(SR2). Appendix F.4 combines them with the original exact cells

$$t^3 + 2 \le [w(w+2)]^2, \qquad M + 2 \le (t+1)^2, \qquad z^8 \le m^9,$$

and proves the displayed height restrictions. □

These domains are materially different. The first cutoff is $2^{24} = 16777216$; the second is approximately $3.4 \cdot 10^{38}$. The stronger exponent in (ii) does not replace the result in (i) below the larger cutoff. Both give wrong-parity obstructions for threshold cycles only in their stated excluded height strips. The remaining cubic region and every taller cycle are unexcluded, and the numerical period floor remains 780239.

3.13. The terminal mixed passage

The last two-point return must also be accounted for before gap contractions could imply no-cycle.

Proposition 3.41 (terminal prefix and suffix). At every Euclidean return stage with words U, V, there are chronological words P, Q such that

$$UV = P\,OE\,Q, \qquad VU = P\,EO\,Q. \tag{SR4}$$

Initially $P = O,\ Q = OE$. A left substitution changes P to UP; a right substitution changes Q to QV.

At a primitive actual cubic-band cycle's terminal two-base section $\{m, v\}$, the common prefix sends this adjacent pair to the largest odd and smallest even source pair (h, s). The mixed words OE, EO send it to $(t, q) = (E(M), O(m))$, and Q sends (t, q) back to (m, v). For $m \geq 3$,

$$0 < q - t < s - h. \tag{SR5}$$

Proof. The concatenation identities follow by induction from $U = OOE,\ V = OE$. Both full return traces follow their actual guards; their common prefix preserves adjacency until the labels differ. It therefore ends at the unique adjacent pair across the odd/even threshold. The middle and final images follow from $O(h) = M,\ E(s) = m,\ U(m) = v,\ V(v) = m$. Moreover,

$$h < m^2, \quad s \geq m^2 + 1, \quad t = \lfloor h^{3/4} \rfloor, \quad q = \lfloor (m^2)^{3/4} \rfloor.$$

Since the derivative of $x^{3/4}$ is below one for $x \geq h \geq 3$,

$$q - t < (m^2)^{3/4} - h^{3/4} + 1 < m^2 - h + 1 \leq s - h.$$

The cycle order gives $t < q$. Appendix F.5 records the complete word and gap accounting. $\square$

The mixed contraction supplies no bound on amplification through P, and current estimates control only the certified factors of Q. Their exact ideal exponent product is

$$p_P \frac{3}{4} p_Q = \frac{3^o}{2^L} > 1.$$

An additional estimate comparing the two parts is required. Appendix F.6 also shows why the present one-sided error certificate cannot stay uniform at a fixed minimum as the contracting word exponent approaches one. These observations leave the global no-cycle question open.

4. Cycle finance

A cycle itinerary is formally expanding (Theorem 3.2), yet the trajectory returns exactly. The multiplicative surplus $3^o - 2^L$ must be financed by the floor remainders, which are relatively $O(1/x)$ in logarithms. The resulting bound on the cycle minimum excludes every period that is not admissible for the inequality, once a uniform logarithmic floor-error bound is available above a verified descent floor.

The identity is unrolled along a circular word. After rotation to a cycle minimum that itinerary is a necklace of odd-run excursions. The geometry below records that itinerary. It introduces no new theorem: every named constraint is Theorem 3.2, Lemma 3.4, Lemma 3.21b, or the last-even one-step preimage.

The excursion necklace

Write n for a cycle minimum and

$$w = O^{a_1} E O^{a_2} E \cdots O^{a_e} E$$

for a minimum-based orientation (Lemma 3.21b), so $a_1 \geq 2$, $\sum_i a_i = o$, and $e = L - o$. Let v_i be the landing state at the start of block $i+1$, and p_i the even state just before its final E. If $a_{i+1} > 0$, then v_i is odd and the block rises to p_i. If $a_{i+1} = 0$, then $v_i = p_i$ is even; it is neither an odd valley nor a new peak. Thus the valley/peak terminology below applies to actual transitions between maximal odd and even runs. In general the block-boundary identities are

$$v_0 = n, \qquad p_i = J^{a_{i+1}}(v_i), \qquad v_{i+1} = J(p_i) = \lfloor \sqrt{p_i} \rfloor \quad (0 \leq i < e),$$

with $v_e = n$. The itinerary is

$$n \xrightarrow{OO} \text{first high region} \xrightarrow{E} v_1 \xrightarrow{O^{a_2} E} \cdots \xrightarrow{O^{a_{e-1}} E} v_{e-1} \xrightarrow{O^{a_e} E} n.$$

Two meanings of *entry* must not be conflated. *Cycle entry* is the distinguished cut that places the minimum at the start of the itinerary. *Dynamical entry* is the last even step into n. The second is a genuine boundary condition; the first is a choice of origin.

Cycle minimum and the forced lift

The minimum is odd, so the itinerary cannot begin with E or OE, and it cannot end with an odd letter (Theorem 3.2). The first two letters are therefore OO:

$$n \xrightarrow{O} y \xrightarrow{O} z.$$

For $n \geq 5$ one has $z = J^2(n) \geq (n+1)^2$ (Lemma 3.4(i)). In particular OOE cannot be a cycle itinerary (`no_cycle_itinerary_ooe`). On a cycle minimum the first even residual overshoots the entry one-step preimage: the first peak satisfies $p_0 \geq (n+1)^2$. That is the opposite of the last-peak condition below. Even $J^2(n)$ may continue with E; odd $J^2(n)$ continues with O. Either way the minimum-based prefix is still OO.

Excursions, valleys, and peaks

An ordinary excursion is one block $O^a E$. In the itinerary semantics that itinerary is `oddEvenBlock a 1`. Write

$$\mu(a) = \frac{3^a}{2^{a+1}}$$

for the ideal (floor-free) exponent of the block. The block is formally expanding if and only if $\mu(a) > 1$, equivalently $2^{a+1} < 3^a$. Thus OE contracts ($\mu(1) = 3/4$) and OOE expands ($\mu(2) = 9/8$). This is a reparameterization of the itinerary envelope of Theorem 2.2, not a transition law on pairs (a_i, a_{i+1}). Lemma 3.4(v) forbids $O^a E$ as a *cycle itinerary* for $a \geq 3$; it does not forbid an internal block of that shape.

The trajectory is then the wave

$$v_0 \to p_0 \to v_1 \to p_1 \to \cdots \to v_{e-1} \to p_{e-1} \to v_0.$$

On a cycle minimum every even state is already at least n^2 (Theorem 3.2(iii)). Valleys dominate a logarithmic defect sum because $1/(x \log x)$ is largest there; peaks are huge and cheap. That is why the run-type packing of Theorem 4.7 charges `OOE`-scale valleys, `OE`-scale valleys, and evens separately. The packed comparison is an extremal charge of the same sum, not a uniqueness theorem for the actual word. Expanding blocks can climb and a later OE can drop without crossing the anchor: four consecutive expanding blocks occur already at the certified start 1999 recorded in Section 6.

Closure and the entry one-step preimage

The valley sequence is circular. The last letter is E, so the last peak occupies the last-even one-step preimage of Lemma 3.4(iv):

$$n^2 \le p_{e-1} < (n+1)^2.$$

The minimum is odd, so $p_{e-1} \neq n^2$ (`cycle_last_even_ne_odd_sq`) and p_{e-1} is even. Thus

$$n^2 + 1 \le p_{e-1} < (n+1)^2, \qquad p_{e-1} \text{ even.}$$

An ordinary excursion needs only $v_i \ge n$. The last excursion must hit this one-step preimage and land on n:

$$v_{e-1} \xrightarrow{O^{a_e}} p_{e-1} \xrightarrow{E} n.$$

The first peak overshoots the same one-step preimage; the last peak lands in it. Those are different even states.

Once the trajectory returns to n, the prefix OO is forced again. A genuine cycle is a closed necklace of excursions whose first block starts OO, whose last peak lies in the entry one-step preimage, and whose entire trajectory stays at least n. The global constraints already proved are

$$\sum_i a_i = o, \qquad e = L - o, \qquad 2^L < 3^o,$$

together with periodicity $\prod_{i=0}^{e-1} v_{i+1}/v_i = 1$.

What remains

Finance (Theorem 4.4) sits around the necklace: the surplus $3^o - 2^L$ must be paid by floor losses, and at the verified floor $N_0 = 10^6$ this forces $L \ge 25781$. The run-type packing of Theorem 4.7 is a refinement of the same defect sum along the valleys and peaks just named. Subsequent refinements of one component of the necklace recovered Theorem 4.7 or closed. They are not claims of this note.

The pieces are understood separately: the forced lift at n, the blocks $O^{a_i}E$, and the last-even landing. What is not proved is a link strong enough to exclude the leftover lengths,

$$\text{minimum geometry} + \text{necklace of excursions} + \text{entry one-step preimage} \;\Rightarrow\; \perp.$$

That is a formulation of the remaining cycle problem, not a theorem. A genuine lower bound on the number of odd runs would feed Theorem 4.7; none is proved here.

Throughout this section write $L = |w|$ and $o = \#O(w)$ for a cycle itinerary w based at a cycle minimum $n \ge 2$. Natural logarithms are written log in this section and ln in Section

5; both denote the natural logarithm. The unique one-step fibres of Section 3 give, for every state $x \ge 1$ with image $y = J(x)$,

$$y^2 \le x^h < (y+1)^2, \qquad h = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}. \end{cases}$$

Lemma 4.1 (dyadic one-step-preimage logarithm). If $z, y \ge 1$ and $z < (y+1)^2$, then $\log z \le 2\log y + 2/y$.

Proof. The hypothesis gives $\log z \le 2\log(y+1)$. The inequality $\log(1+u) \le u$ for $u > 0$ yields $\log(y+1) = \log y + \log(1+1/y) \le \log y + 1/y$. □

Lemma 4.2 (one-step bounds). Let $x \ge 2$ and $y = J(x)$. If x is even, then $\log x \le 2\log y + 2/y$. If x is odd, then $3\log x \le 2\log y + 2/y$.

Proof. The even one-step preimage is $y^2 \le x < (y+1)^2$. The one-step-preimage logarithm lemma on $z = x$ is the first claim. The odd one-step preimage is $y^2 \le x^3 < (y+1)^2$. The same lemma on $z = x^3$ gives $\log(x^3) \le 2\log y + 2/y$. □

On a cycle minimum every proper prefix is non-contracting: a prefix with $3^{\#O} < 2^k$ would satisfy $J^k(n) < n$ by Corollary 2.3, contradicting minimality. Thus $2^k \le 3^{o_k}$ for every prefix of length k, where o_k is the odd count of that prefix.

Lemma 4.3 (unrolled envelope). Write $x_k = J^k(n)$ and o_k for the odd count of the length-k prefix. For every $0 \le k \le L$,

$$3^{o_k}\log n \le 2^k \log x_k + \frac{k\,3^{o_k}}{n}.$$

Proof. The case $k = 0$ is an equality. Suppose the claim holds at $k < L$, and write $x = x_k$ and $y = x_{k+1}$. Minimality gives $n \le y$, and the prefix law gives $2^{k+1} \le 3^{o_{k+1}}$, hence

$$\frac{2^{k+1}}{y} \le \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then $o_{k+1} = o_k$ and the one-step bound gives $\log x \le 2\log y + 2/y$. Multiply by 2^k and add the inductive remainder:

$$3^{o_k}\log n \le 2^{k+1}\log y + \frac{2^{k+1}}{y} + \frac{k\,3^{o_k}}{n} \le 2^{k+1}\log y + \frac{(k+1)\,3^{o_k}}{n}.$$

If the next letter is odd, then $o_{k+1} = o_k + 1$. Multiply the inductive bound by 3 and apply the one-step bound in the form $3\log x \le 2\log y + 2/y$:

$$3^{o_k+1}\log n \le 2^{k+1}\log y + \frac{2^{k+1}}{y} + \frac{k\,3^{o_k+1}}{n} \le 2^{k+1}\log y + \frac{(k+1)\,3^{o_k+1}}{n}.$$

This is the claim at $k+1$. □

Theorem 4.4 (finance). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \ge 2$. Then

$$n\log n \cdot (3^o - 2^L) \le L \cdot 3^o.$$

Proof. Apply Lemma 4.3 at $k = L$. Periodicity gives $x_L = n$ and $o_L = o$, so

$$3^o \log n \le 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The itinerary is formally expanding, so $3^o > 2^L$. Rearranging and multiplying by n is the claim. □

This is the conceptual centre of the note. Exact one-step preimages give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. Ideal dynamics expands and exact dynamics returns, so the floor errors finance the expansion. The inequality $\log(1+u) \le u$ is the only analytic input; the content is that interaction. The Lean form is exactly Theorem 4.4 (constant 1): `cycleMin_finance`.

Hierarchy of forms. These three layers must not be conflated.

1. *Theorem 4.4* is the conceptual sharp inequality. Constant 1:

   $$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o,$$

   equivalently $\theta \le L/(n \log n)$ with $\theta = 1 - 2^L/3^o$. It charges every one-step-preimage defect at the cycle minimum. Lean: `cycleMin_finance`.

2. *Corollary 4.4c* (below) is the same one-step-preimage-log unroll with remainders kept as $1/x_{i+1}$. It is the strongest proved form of those defects. Lean: `cycleMin_finance_inv_sum`.

3. *Corollary 4.5* is the convenient length-only statewise bound: a verified descent floor N_0 plus the three-class charge of the defect sum produces a per-length threshold $n_{\max}(L)$ and excludes every L with $n_{\max}(L) \le N_0$. The charge is

   $$\sum_i \frac{1}{x_i \log x_i} \;\le\; \frac{e}{n \log n} + \frac{o-e}{t \log t} + \frac{e}{2n^2 \log n}, \qquad t = \lfloor n^{3/2} \rfloor,$$

   charging valleys at n, internal odds at t, and evens at n^2. It is called the *parity* charge in the tables and in places below, which is historical and slightly misleading: it is **not** a two-class split of odds against evens, and the t-scale middle term is exactly what makes it sharper than one. Lean: `cycleMin_threeTerm` from `CycleMin` alone, with no hypothesis about `EE`.

4. *Theorem 4.6* is the numerical certification of Corollary 4.5 at $N_0 = 10^6$. The certified identity is the conservative relative-defect form

   $$1 - \frac{2^L}{3^o} \le \frac{6}{5} \sum_{i=1}^{L} \frac{1}{x_i \log x_i}.$$

   The factor $6/5$ is a convenient uniform majorant of $-\log(1-\delta)/\delta$ on $[0, 1/6]$. It is not Theorem 4.4, and the headline cutoff 25781 is not an artifact of that majorant. The identity itself is Lean for any cycle minimum $n \ge 400$ (`cycleMin_defect_finance`, `DefectFinance.lean`); the per-length numeric table stays a verified computation.

The relative-defect unroll is a parallel identity, not Theorem 4.4 multiplied by 6/5. Charging every state at the cycle minimum in that identity recovers the coarser comparison $\theta \le (6/5)L/(n\log n)$, which is Theorem 4.4 with coefficient 6/5. At this floor that uniform charge excludes only through length 1053. The computational table uses a stricter length-only three-class charge of the same identity.

Corollary 4.4c (inv-sum). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \ge 2$. Write $x_i = J^i(n)$. Then

$$(3^o - 2^L)\log n \le 3^o \sum_{i=1}^{L} \frac{1}{x_i},$$

equivalently $\theta \le \left(\sum_i 1/x_i\right)/\log n$.

Proof. The same induction as Lemma 4.3, keeping each one-step preimage defect as $2^{k+1}/x_{k+1}$ instead of replacing it by $3^{o_{k+1}}/n$. At $k = L$ one has $x_L = n$. Lean: `cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`. □

The computational table of Theorem 4.6 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state x with image $y = J(x) \ge 12$, the relative defect $\delta = (x^h - y^2)/x^h$ satisfies $\delta \le 2/y \le 1/6$. Writing $\varepsilon = -\frac{1}{2}\log(1-\delta)$ and using $-\log(1-\delta) \le \frac{6}{5}\delta$ on $[0, 1/6]$ gives $\varepsilon \le (6/5)/y$. Unrolling $t_{i+1} = (h_i/2)\, t_i - \varepsilon_i$ around the cycle then yields the conservative identity displayed in item 4 of the hierarchy. This chain is Lean end to end (`DefectFinance.lean`): the per-step losses in image form (`log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, from $-\log(1-\delta) \le \frac{6}{5}\delta$ on $[0, 1/6]$, `neg_log_one_sub_le_sixth`), the amplification priced by the upper invariant $\log x_k \le w_k \log n$ (`cycleMin_log_le_weight`), and the charged unroll (`cycleMin_charge_prefix`), closed at $x_L = n$.

On a minimum-based cycle of length L with o odd letters and $e = L - o$ even letters, the last letter is even, so $e \ge 1$ and the number of odd-run starts is at most e. Every even state satisfies $x \ge n^2$. Every odd state preceded by an odd state satisfies $x \ge t = \lfloor n^{3/2} \rfloor$. Therefore

$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \le \frac{e}{n\log n} + \frac{o-e}{t\log t} + \frac{e}{2n^2\log n},$$

and

$$n\log n\left(1 - \frac{2^L}{3^o}\right) \le \frac{6}{5}\left(e + (o-e)\frac{n\log n}{t\log t} + \frac{e}{2n}\right).$$

The optimal uniform coefficient on $[0, 1/6]$ is $6\log(6/5) \approx 1.093$. Replacing 6/5 by that constant, or even by 1, on the same parity charge does not change the first surviving length: one still has $n_{\max}(25780) \le 10^6 < n_{\max}(25781)$. The published table and the count 141 keep the proved coefficient 6/5. The cutoff 25781 is therefore not an artifact of an avoidable loss in the majorant.

Lemma 4.4b (odd-count monotonicity). Write $\theta(o) = 1 - 2^L/3^o$ and

$$R(o) = e + (o-e)\alpha + \frac{e}{2n}, \qquad e = L - o, \qquad \alpha = \frac{n\log n}{t\log t}.$$

The certified parity comparison is $n\log n\cdot\theta(o)\le(6/5)R(o)$. For every $n\ge 12$ one has $\alpha<1/2$, hence

$$R(o+1)-R(o)=2\alpha-1-\frac{1}{2n}<0.$$

Also $\theta(o+1)>\theta(o)$. Therefore if the comparison fails at some admissible o, it fails at every larger odd count. Equivalently, the largest n at which the comparison can hold occurs at the least admissible odd count $o_{\min}(L)=\min\{o:3^o>2^L\}$. The same monotonicity holds for any positive coefficient in place of 6/5, and for the constant-1 comparison of Theorem 4.4.

Proof. The difference $R(o+1)-R(o)$ is the displayed coefficient of o. For $n\ge 12$ one has $t=\lfloor n^{3/2}\rfloor\ge n^{3/2}-1>2n$, hence $\alpha<(n\log n)/(2n\log(2n))=(\log n)/(2\log(2n))<1/2$. The map $o\mapsto\theta(o)$ is strictly increasing on integers o with $3^o>2^L$. If $n\log n\cdot\theta(o)>(6/5)R(o)$, then $n\log n\cdot\theta(o+1)>(6/5)R(o)>(6/5)R(o+1)$. □

Define

$$\gamma(L)=\frac{3^{o_{\min}(L)}}{2^L}-1,$$

and write $n_{\max}(L)$ for the largest integer n at which the displayed parity inequality can still hold at $o=o_{\min}(L)$. The coarser comparison $n\log n\le(6/5)L\cdot 3^{o_{\min}}/(3^{o_{\min}}-2^L)$ is used only as a check; Theorem 4.6 uses $n_{\max}(L)$ from the parity form. Lemma 4.4b is why that table may be computed at $o_{\min}$ alone.

Proposition 4.4a (finance-survivor algorithm). Fix a verified descent floor N_0. For each integer $1\le L\le 10^5$, compute $o_{\min}(L)=\min\{o:3^o>2^L\}$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L)>N_0$. The resulting finance-survivor set is

$$\mathcal{E}(N_0)=\big\{L:1\le L\le 10^5,\ n_{\max}(L)>N_0\big\}.$$

The printed instance is $\mathcal{E}=\mathcal{E}(10^6)$, with $|\mathcal{E}|=141$.

Corollary 4.5. If every integer $2\le n\le N_0$ reaches 1, then no nontrivial cycle of length L exists whenever $n_{\max}(L)\le N_0$.

Proof. A periodic state never reaches 1, so every cycle state is at least N_0+1. The length-only three-class charge of the certified relative-defect identity then forces the minimum to satisfy the displayed inequality, hence $n\le n_{\max}(L)$. This is the convenient statewise bound in the hierarchy after Theorem 4.4; it is not a replacement for that theorem. □

Record values of the three-class $n_{\max}$ include $n_{\max}(19)=133$, $n_{\max}(84)=2323$, $n_{\max}(569)=23568$, $n_{\max}(1054)=788014$, and $n_{\max}(25781)=26254995$.

Theorem 4.6 (verified computation). Every integer $2\le n\le 10^6$ reaches 1. Consequently:

(A) there is no nontrivial Juggler cycle of length at most 25780;

(B) if a nontrivial cycle has period $L\le 10^5$, then $L\in\mathcal{E}$, where $|\mathcal{E}|=141$.

The bound closes continuously through 25780; 25781 is the first integer for which this particular inequality does not exclude a cycle. The coefficient 6/5 is the certification majorant of item 4 in the hierarchy after Theorem 4.4, not the source of the cutoff.

Proof. The descent floor is Proposition 1.3. The gap table computes $o_{\min}(L)$ and $n_{\max}(L)$ for every $1 \le L \le 10^5$ by Lemma 4.4b. Corollary 4.5 at $N_0 = 10^6$ excludes every L with $n_{\max}(L) \le 10^6$. The contiguous excluded prefix is $L \le 25780$. The complementary set in range is $\mathcal{E}$; checksums are Appendix B. □

Run packing

Theorem 4.7 refines the length-only charge by separating the unique minimum, `OOE`-scale valleys, `OE`-scale valleys, internal odds, and evens. It is a supporting comparison at the same floor; the leftover count 141 → 99 does not raise the cutoff. A cycle cannot put an `OE`-start at the minimum (Theorem 3.2), and an n-circuit of k odds and ℓ evens stays at least n only if $3^k \ge 2^{k+\ell}$.

Theorem 4.7 (run-type packing). Let w be a cycle itinerary of length L with $o = o_{\min}(L)$ odd letters and $e = L - o$ even letters, based at a cycle minimum $n \ge 12$. Assume further that w is primitive and that w contains no `EE`. Write v for the least odd integer with $v^3 \ge n^4$, and write $t = \lfloor n^{3/2} \rfloor$. Then $o - e < e$, the largest number of n-scale valleys compatible with the even cap is $o - e$ copies of `OOE`, and the remaining $2e - o$ circuits are `OE` from v. The cycle minimum occurs once, so

$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \le \frac{1}{n \log n} + \frac{o-e-1}{(n+2)\log(n+2)} + \frac{2e-o}{v \log v} + \frac{1}{t \log t} + \frac{o-e-1}{t_+ \log t_+} + \frac{e}{2n^2 \log n},$$

where $t_+ = J(n+2)$. Combined with the 6/5 unroll this is strictly smaller than the three-class sum of Corollary 4.5 whenever $2e - o > 0$. Sending the cycle maximum to infinity removes one even term and does not change the valley packing.

Proof. The statement is at $o = o_{\min}(L)$. Lemma 4.4b already excludes every larger odd count from the parity comparison used in Theorem 4.6; the packing is a refinement at that same odd count. Formal expansion at $o_{\min}$ forces $3o < 2L$ on every leftover length in range, equivalently $o - e < e$. The even-cap comparison $3^k \ge 2^{k+\ell}$ is the ideal power envelope; floors only help. An `OE`-start is followed by an even state, so Theorem 3.2 gives $J(v) \ge n^2$ and therefore $v^3 \ge n^4$. Unique visit of the cycle minimum is primitivity, which is why that hypothesis is stated: a cycle word concatenated with itself is again based at a cycle minimum, and the display charges only one valley at n. The no-`EE` hypothesis enters at the split of the e valleys into cheap and expensive. Counting on indices rather than on blocks, an odd letter is a *valley* when its cyclic predecessor is even; the valleys are then in bijection with the maximal even runs, and the counting lemma gives $\#\text{cheap} \le o - \#\text{blocks}$. Only when no `EE` occurs do the blocks number e, giving the $\#\text{cheap} \le o - e$ the packing uses; the subsection *The packing hypothesis and its price* below shows the hypothesis is not removable. The displayed sum charges one cheap valley at n, the remaining $o - e - 1$ cheap valleys at the next odd integer $n + 2$, the expensive valleys at v, one internal odd at t, the remaining internals at $J(n + 2)$, and every even at n^2. Any deeper odd run or any higher valley only decreases the sum. □

Theorem 4.8 (run-type table). At the verified descent floor $N_0 = 10^6$, the packed comparison of Theorem 4.7 excludes the 42 lengths $56347 + 1054k$ for $k = 0, \dots, 41$ and leaves an explicit set $\mathcal{E}_{\text{run}} = \mathcal{E}_{\text{run}}(10^6)$ of 99 lengths. The first survivor remains 25781. In particular, if a nontrivial cycle has period $L \le 10^5$ and its itinerary satisfies the hypotheses of Theorem 4.7, then $L \in \mathcal{E}_{\mathrm{run}}$.

The two halves of that progression do not carry the same weight. The 24 lengths with $k \ge 18$, from 75319 up, are excluded with no hypothesis about `EE` at all: at most half the odd letters can be cheap valleys whatever the word does (`two_mul_cheap_le_odd`), and charging $\lfloor o/2 \rfloor$ of them at $n+2$ still excludes exactly those lengths. The 18 with $k \le 17$, from 56347 to 74265, are excluded only under the no-`EE` hypothesis, and the subsection below exhibits admissible words that defeat each. Dropping that hypothesis therefore leaves 117 lengths in place of 99. It does not move the cutoff, which is 25781 either way and comes from Corollary 4.5.

Proof. The 141 lengths of Theorem 4.6(B) are tested at $n = 10^6 + 1$ against the packed right-hand side. The comparison is certified on each length: no entry is left uncertain. The 42 excluded lengths are exactly the arithmetic progression named in the statement. The complementary set in that range is $\mathcal{E}_{\mathrm{run}}$. Checksums are Appendix B. The period cutoff remains 25781. □

The packing hypothesis and its price

Theorem 4.7 assumes the itinerary contains no `EE`. The assumption is not cosmetic and it is not removable, and since 18 of Theorem 4.8's 42 exclusions rest on it, it is worth saying exactly what it buys and what it costs.

Why the packing needs it. The extremality argument counts *blocks* $\mathtt{O}^a\mathtt{E}$. Counting instead on indices, where nothing is assumed, call an odd letter a *valley* when its cyclic predecessor is even and an *internal* when that predecessor is odd. The valleys are then in bijection with the maximal even runs. If the itinerary carries m cyclic `EE` adjacencies, the e even letters form $e - m$ maximal runs, so

$$\#\text{valleys} = e - m, \qquad \#\text{internals} = o - e + m, \qquad \#\text{cheap} \le \min(o - e + m,\ e - m),$$

with the expensive valleys making up the rest. The bound $\#\text{cheap} \le o - e$ that Theorem 4.7 uses is the case $m = 0$; in general the counting lemma gives only $\#\text{cheap} \le o - \#\text{blocks}$, and with `EE` present the even letters outnumber the blocks. The paper's phrasing — $o - e$ copies of `OOE` and $2e - o$ circuits of `OE` — already presupposes one even letter per block.

What it is worth. At every one of the 42 lengths, at $n = 10^6 + 1$, the ratio of the Corollary 4.5 charge to the packed one is 1.4048, uniformly; to leading order it is $e/(o - e)$, the packing moving the n-scale charge from e valleys to $o - e$. That constant is the whole budget of the refinement, and each of the 42 dies with margin $\theta/\text{packed} \in [1.0033, 1.3535]$ — inside that budget of necessity, since each survives the unpacked charge.

The second cap, and the split. The quantity #cheap is bounded both by $o - e + m$ and by the valley count $e - m$ it is part of. The two cross at $m = e - o/2$, where both equal $o/2$; past the crossing the valleys themselves run out and the majorant falls again. So `EE` cannot inflate the n-scale charge beyond $(o/2)/(o - e) = 1.2047$, however much of it a word carries. Since θ/packed rises monotonically along the progression and crosses that ceiling between $L = 74265$ and $L = 75319$, the 42 split cleanly: the 24 lengths from 75319 up are excluded whatever the itinerary, and the 18 from 56347 to 74265 are not.

The 24 are a theorem, not an observation. The split above is read off a scan over m, but the surviving half needs no scan. Call an odd letter a *cheap valley* when its cyclic predecessor is even and its cyclic successor is odd — the start of an odd run of length at least two. A cheap valley is a valley; its successor is an odd letter with an odd predecessor, hence an internal; and the successor map is injective on the window. So the cheap valleys sit inside the valleys and inject into the internals, and those two classes partition the odd letters (`valley_add_internal`), giving

$$2 \cdot \#\text{cheap} \;\le\; o$$

with no reference to `EE` — Lean `two_mul_cheap_le_odd`, from `cheap_le_valley` and `cheap_le_internal`. Feeding $\lfloor o/2 \rfloor$ to `sixTerm_bound_packed` in place of the packing's $o-e$ therefore yields a comparison carrying none of Theorem 4.7's hypotheses, and at $N_0 = 10^6$ that comparison excludes exactly the 24. The bound is of course weaker than the packed one — it charges more valleys at the $n+2$ scale — and it remains below the charge of Corollary 4.5, as it must.

The 18 are genuinely lost without the hypothesis. For each there is a word satisfying every restriction this paper proves for a cycle-minimum itinerary — $3^{a_j} \ge 2^j$ at every prefix, the per-run cap of Theorem 3.31, $o = o_{\min}(L)$, and the `OO`…`E` shape — and carrying more `EE` than the comparison can absorb. The witness is the Beatty interleaving of `OOE` and `OE` blocks over $e-k$ even letters followed by a tail of k even letters, with k between 51 and 3926. Its odd runs have length at most two, which is the packing's own extremal shape, so no claim about run structure is violated; what fails is only the correspondence between blocks and even letters. Granting the extremality of the run packing in full therefore does not restore the counting, and no sharpening of the run analysis will: what would be needed is a restriction sensitive to the floors, whereas every restriction used above is exponent bookkeeping, exact for the multipliers and blind to the floors.

Remark (why the run cap cannot help). The least admissible odd count is defined by $3^o > 2^L$, that is $o \log(3/2) > e \log 2$, and the per-run cap of Theorem 3.31, which Theorem 3.29's run–suffix law supplies, is $\lfloor (e-i) \log 2 / \log(3/2) \rfloor$ with i the number of even letters already spent. The two use the same constant, so at every one of the 42 lengths

$$\left\lfloor e \cdot \frac{\log 2}{\log(3/2)} \right\rfloor = o_{\min}(L) - 1.$$

The single-run word $\mathtt{O}^o\mathtt{E}^e$ is therefore forbidden by exactly one letter, and two blocks are already admissible — carrying $e-2$ adjacencies. The cap and the odd count are the same inequality read twice, which is why the cap is never far from binding and never actually binds.

What this does not say. Theorem 4.7 is true as stated, with its hypotheses, and Theorem 4.8's 42 exclusions are correct under them. A defeated exclusion does not produce a cycle of that length; it means the comparison does not rule one out. And the count m is treated here as free, whereas a realised itinerary's `EE` count is fixed by the dynamics — the witnesses are admissible for the restrictions this paper proves, not exhibited as cycles.

Arithmetic structure of the finance survivors

The leftover lengths cluster around the continued-fraction approximants of $\log 2/\log 3$. That organizes the table; it does not constrain a hypothetical cycle.

Proposition 4.9 (finance-survivor lattice). Write $v_* = (25781, 16266)$ and $v_{1054} = (1054, 665)$. Then

$$25781 \cdot 665 - 1054 \cdot 16266 = 1,$$

so these two vectors are a unimodular basis of $\mathbb{Z}^2$. Every length in $\mathcal{E}_{\mathrm{run}}$, and every one of the 42 packing deaths of Theorem 4.8, is of the form $(L, o_{\min}) = a\,v_* + b\,v_{1054}$. With the table cap $L \le 10^5$ the 99 survivors fall in three affine slices of counts 29, 47, and 23. The identification with $\mathcal{E}_{\mathrm{run}}$ is Theorem 4.8. This is a change of coordinates for (L, o), not a relation between hypothetical cycles.

Proof. The determinant identity is integer arithmetic. The generator comparison $3^{665} > 2^{1054} > 3^{664}$ gives $o_{\min}(1054) = 665$. Direct evaluation of $o_{\min}$ on the 141 lengths of Theorem 4.6(B) places each pair $(L, o_{\min})$ on the displayed lattice, with the 42 packing deaths the F_1 continuation $b \ge 29$. □

The gap transfer and the short-cycle reduction

Everything above turns a verified descent floor into per-length exclusions. At the five upper-convergent denominators $19, 84, 1054, 50508, 176251$, the computed ratio $n_{\max}(q_k)\log n_{\max}(q_k)/(q_k q_{k+1})$ lies in $[0.41, 0.53]$. This finite comparison motivates the discussion in Section 6.2; it is not by itself a global scaling theorem.

This subsection records the one statement of the note that needs no floor. It transfers a lower bound on the linear form

$$\Lambda = \Lambda(L, o) = o\log 3 - L\log 2 = -\log\Big(1-\theta\Big), \qquad \theta = 1 - \frac{2^L}{3^o},$$

into a bound on the cycle minimum. The only new input is $\log\frac{1}{1-\theta} \le \frac{\theta}{1-\theta}$.

Theorem 4.10 (gap transfer). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \ge 2$. Then

$$n\log n \cdot \min(\Lambda, 1) \le 2L.$$

Proof. Write $A = 3^o$, $B = 2^L$, $P = n\log n \ge 0$. Theorem 4.4 reads $P(A-B) \le LA$. If $A \le B$ then $\Lambda \le 0$ and the left side is nonpositive. If $A \ge 2B$, then $A - B \ge A/2$, so $PA/2 \le LA$, hence $P\min(\Lambda, 1) \le P \le 2L$. If $B < A < 2B$, then $\Lambda = \log(A/B) \le A/B - 1 = (A-B)/B$, so $P\Lambda \le P(A-B)/B \le LA/B \le 2L$. □

The Lean form is `cycleMin_gap_transfer`; the abstract corollary "$\varepsilon \le \min(\Lambda, 1)$ implies $n\log n \cdot \varepsilon \le 2L$" is `cycleMin_length_of_gap`.

Corollary 4.11 (short cycles are excluded). Rhin's effective estimate [15, Proposition, p. 160, (7)] applies to the absolute value of an integer linear form in $1, \log 2, \log 3$. For a nontrivial cycle, $L \ge 11$, $0 < o < L$, and $\Lambda = o\log 3 - L\log 2 > 0$. Substituting $(u_0, u_1, u_2) = (0, -L, o)$ gives height $H = L$ and

$$\Lambda \ge L^{-13.3} > e^{-6.1256} L^{-13.3}.$$

We retain the weaker constant to state the following consequence:

$$n \log n \le 2e^{6.1256} L^{14.3} < 915 L^{14.3}, \qquad L > \Big(\frac{n \log n}{915}\Big)^{1/14.3}.$$

In particular, a cycle with $L^{14.3} \le n \log n/915$ is excluded without a descent-floor input.

Proof. Corollary 3.23 supplies $L \ge 11$, so Rhin's height condition is satisfied. Set $\varepsilon = e^{-6.1256} L^{-13.3}$. Then $0 < \varepsilon \le \min(\Lambda, 1)$, and Theorem 4.10 gives the first displayed upper bound. The inequality $2e^{6.1256} < 915$ completes the proof. This use of Rhin is an external theorem, not a Lean proof of the transcendence estimate. □

Remark (what the reduction does and does not do). Corollary 4.11 is a reduction of the no-cycle problem, not a kill. It says that the only cycles left to exclude are the *long* ones, $L > (n \log n/915)^{1/14.3}$; every shorter cycle is excluded by transcendence plus finance alone. At the certified floors of Section 5 this is toothless — at $N_0 = 3.5 \cdot 10^8$ it forces only $L \ge 4$, while the finance table forces $L \ge 780239$ — which is the floor-level statement that a Baker-type transfer cannot compete with the exact gap. The value of the corollary is that it is floor-free and identifies the frontier exactly: the finance survivors of Theorems 4.6 and 5.9 have $L \approx n^{0.59}$ — the measured exponent $\log L/\log n_{\max}(L)$ is 0.595 at $L = 25781$, 0.573 at 50508, 0.582 at 176251 and 0.611 at 780239 — far inside the long regime, and no refinement of the defect *upper* bound can move them into the short one, because along the convergents the required minimum $n_{\max}(q_k)$ grows quadratically in L. Excluding long cycles is a statement about the parity word of a specific orbit at depth L, which no estimate in this note or in the companion discrepancy manuscript [16] controls. The problem "no nontrivial Juggler cycle" is therefore exactly the problem "no long Juggler cycle," and it remains open.

5. The laboratory instance and the walk-charge envelope

Everything in Sections 2–4 is floor-generic: Corollary 4.5 accepts any verified descent floor. This section first records a second, laboratory-certified instance of the same architecture, then replaces the length-only charge by a coupled exponent-walk charge. The extremal walk is identified exactly as a rotation itinerary, and a Denjoy–Koksma bound over certified Ostrowski blocks produces an envelope valid for *every* length in an explicit window — no per-length census and no dynamic program is needed on that window. The kill table at the laboratory floor then yields the period bound 176251, and the second certified floor raises it to 478245 (Corollary 5.10) and the third to 780239 (Corollary 5.11). Throughout, survivors are finance-survivors in the sense of Section 1.

5.1 The laboratory floor

Proposition 5.1 (laboratory descent floor; certified computational input). Every integer $2 \le n \le 26254995$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n, a finite realized itinerary with image strictly below the start; strong induction on that image reaches 1. The run walks the 13127497 odd starts (even starts descend by E) over 106 contiguous chunk records. Three bit-cap seeds (7110201, 13184021, 13782577) are resolved exactly at a $512 \cdot 10^6$-bit cap with exact integer square roots, the largest intermediate having 298912128 bits (seed 7110201); the maximum first passage is 325 steps (seed 15909091). Certificate hashes are in Appendix B. The floor $26254995 = n_{\max}(25781)$ is the cheapest floor that moves the Theorem 4.6 cutoff.

Theorem 5.2 (raised cutoff; verified computation). At $N_0 = 26254995$ the parity table of Theorem 4.6 excludes every length $L \le 50507$: any nontrivial Juggler cycle has period at least 50508. The first finance survivor is $L = 50508$ with $n_{\max}(50508) = 162848324$; 19 parity survivors remain through $L = 2 \cdot 10^5$ (run packing leaves 11 at the same cutoff). This is the one row in the paper where the exact $n_{\max}$ of Section 4 and the value in the committed table differ, and the reason is worth recording: the crossing at $L = 50508$ is sharp to a relative $2.7 \cdot 10^{-10}$, which is *below* the 10^{-9} relative guard the table's generator adds to the right-hand side. That guard is deliberate and conservative — it can only make $n_{\max}$ larger, hence can only refuse to exclude a length — so the table prints 162848325. Both values give the same exclusion here, since the floor 162849448 exceeds either.

Proof. Proposition 5.1 supplies the floor; Corollary 4.5 and the gap table of Proposition 4.4a exclude every L with $n_{\max}(L) \le 26254995$. The contiguous excluded prefix is $L \le 50507$; checksums are Appendix B. □

5.2 Transport to a reduced base

The parity and run-pack tables price valleys independently. On a real minimum-based cycle every state is coupled through one closed exponent walk. With a_k the number of odd letters among the first k, set

$$u_k = (1+\mu)a_k - k, \qquad \mu = \log_2(3/2).$$

The defect-free floors give the upper envelope $x_k \le n^{2^{u_k}}$, and cycle minimality forces $u_k \ge 0$ at every step. The lower envelope requires controlling the accumulated floor losses; that is the transport lemma.

Theorem 5.3 (transport). On a minimum-based cycle with minimum $n \ge 400$, length L, o odd and e even letters, every state satisfies

$$x_k \;\ge\; \left(n\,e^{-D}\right)^{w_k}, \qquad w_k = 2^{u_k}, \qquad D = \frac{1.05\,e}{n} + \frac{0.7\,o}{n^{3/2}}.$$

Proof. Write $\ln x_k \ge w_k \ln n - E_k$. The floor losses give the recursion $E' \le \frac{3}{2}E + 1.05\,x^{-3/2}$ at an odd letter and $E' \le \frac{1}{2}E + 1.05\,x^{-1/2}$ at an even letter, using $-\ln(1-t) \le 1.05\,t$ on $t \le 0.05$. Unrolling, the amplification from injection j to state k is exactly w_k/w_{j+1}. Odd injections have $x_j \ge n$ and $w_{j+1} \ge \frac{3}{2}$, contributing at most $0.7\,n^{-3/2}$ each; even injections have $x_j \ge n^2$ (Theorem 3.2(iii)) and $w_{j+1} \ge 1$, contributing at most $1.05/n$ each. Hence $E_k \le w_k D$. □

The transport inequality is Lean end to end in log form: the walk weight $w_k = 2^{u_k} = 3^{a_k}/2^k$ is rational, so the per-step floor losses (`log_floorPower_odd_ge`, `log_floorPower_even_ge`), the exact weight recursion, and the full induction (`cycleMin_transport`, `WalkTransport.lean`) are elementary real arithmetic over the formalized cycle envelopes `cycleMin_iterate_ge`, `cycleMin_even_ge_sq`, and `cycleMin_prefix_pow_le`.

Consequently the cyclic defect sum $\sum_i 1/(x_i \ln x_i)$ is bounded above by the maximum of the walk charge $\sum_k g(u_k)$, $g(u) = 1/(n'^{2^u} 2^u \ln n')$, over all nonnegative exponent walks of length L, evaluated at the *reduced base* $n' = n\,e^{-D}$. No free parameter remains; the walk value feeds the 6/5 unroll of Theorem 4.4 exactly as the parity charge did. At the laboratory floor, for the original sub-window $L < 301994$, one has $D \le 4.6 \cdot 10^{-3}$ and $\ln n' \ge 17.07$. The extended window requires the bounds in Theorem 5.8.

This consequence is itself Lean end to end (`WalkChargeMax.lean`): writing the charge through the rational weight, $g = 1/(e^{W\nu} W\nu)$ with $W = 2^{u_k} = 3^{a_k}/2^k$ and $\nu = \ln n'$, exponentiating the transport inequality gives $\sum_k 1/(x_k \ln x_k) \le \sum_k g(w_k)$ (`cycleMin_defect_le_charge`), and the charge of the realized word is dominated by the hug charge of the same length (`cycleMin_defect_le_hug_charge`, using Theorem 5.4 below), all under the recorded hypothesis $\nu > 0$.

5.3 The adversary is the hug itinerary

Theorem 5.4 (hug exchange). Among all nonnegative exponent walks of length L, the *hug itinerary* — take E at every step where $u \ge 1$, else O — is prefix-minimal: writing a_k for the odd count of a length-k prefix,

$$a_k^{\mathrm{hug}} \le a_k \qquad \text{for every admissible walk and every } k,$$

equivalently $u_k^{\mathrm{hug}} \le u_k$. Consequently the hug itinerary maximises the walk charge: since g is strictly decreasing in u,

$$g\big(u_k^{\mathrm{hug}}\big) \ge g(u_k) \quad \text{for every } k, \qquad \text{hence} \qquad \sum_k g(u_k) \;\le\; \sum_k g\big(u_k^{\mathrm{hug}}\big).$$

Proof. At the first disagreement with any other admissible itinerary, the hug itinerary holds E where the other holds O, because hug takes O only when E is illegal and the two words share the same prefix state. The odd-count gap $\delta_k = a_k(\text{other}) - a_k(\text{hug})$ is a path with steps in $\{-1, 0, +1\}$ that cannot go negative, since $\delta = 0$ restores the same state. This is the displayed prefix-minimality $a_k^{\mathrm{hug}} \le a_k$; since $u_k = (1+\mu)a_k - k$ with the same k, it transfers verbatim to $u_k^{\mathrm{hug}} \le u_k$. The greedy word uses exactly $o_{\min}(L)$ odd letters (Lemma 5.6). It therefore belongs to the class with that prescribed odd count, while still dominating every admissible word with a larger odd count.

Applying the strict antitonicity of g termwise and summing over k is the charge comparison. The prefix-minimality core is Lean: `hugOdds_le_of_admissible`. □

Remark (uniqueness). The hug itinerary is in fact the *unique* prefix-minimal admissible path in the class $(L, o_{\min}(L))$, so it uniquely maximises the charge; at the first disagreement the competitor already carries a strictly larger prefix odd count, and strict monotonicity of g makes the total comparison strict. The profile equality for every charged prefix $k < L$ is also formalized by `hug_charge_unique`; the prescribed total odd count determines the last letter. Uniqueness is not needed for the exclusion criterion.

The analytic half is also Lean, in a strengthened form (`WalkChargeMax.lean`): the charge is antitone in the rational weight (`stateCharge_antitone`, elementary exp monotonicity — no charge integral), so the exact hug itinerary maximises the total charge over *all* admissible exponent walks, not just a fixed (L, o) class (`hug_charge_maximal`). The equality case for the charged profile is formalized by `hug_charge_unique`, using `stateCharge_strictAnti` and `stateCharge_inj`.

The statement is about the $u \ge 0$ relaxation, not about realized cycle itineraries. Word-order (Christoffel) prefix-dominance is *false* for this family — the greedy word $OOEO$ beats $OOOE$ at $(L, o) = (4, 3)$ — so the exchange argument above, not a dominance order, is the correct mechanism.

Realized itineraries do, however, dominate the hug itinerary at the level of odd counts: on any minimum-based cycle itinerary, every length-k prefix carries at least $o_{\min}(k)$ odd letters. This is Lean end to end (`cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`), composing the formalized cycle prefix envelope $2^k \le 3^{a_k}$ (`cycleMin_prefix_pow_le`) with the hug minimality `hugOdds_least`. It is exactly the sense in which the hug itinerary is the cheapest adversary any hypothetical cycle can present. The survivor-lattice generators of Proposition 4.9 lie on this hug diagonal: $(1054, 665)$, $(25781, 16266)$, and the seed $(50508, 31867)$ all satisfy $o = o_{\min}(L)$ (Lean: `hugOdds_1054`, `hugOdds_lattice_base`, `hugOdds_seed`).

Proposition 5.5 (rotation average). The infinite hug walk is the rotation by $\alpha = \log_2(3/2)$ on $\mathbb{R}/(1+\alpha)\mathbb{Z}$. Unique ergodicity of the irrational rotation — extended, in the standard way, from continuous observables to Riemann-integrable ones — gives the charge-per-letter

$$C_*(n') = \frac{1}{\ln 3}\int_1^3 {n'}^{1-t}\, t^{-2}\, dt \;<\; \frac{1}{\ln 3\, \ln n'}.$$

Proof. Substituting $s = (t-1)\ln n'$ gives $C_* = \frac{1}{\ln 3\, \ln n'}\int_0^{2\ln n'} e^{-s}(1 + s/\ln n')^{-2}\, ds$, and the integrand is at most e^{-s}. □

The display and its quantitative sharpening are Lean (`RotationAverage.lean`): the quadratic majorant $t^{-2} \le 1 - 2(t-1) + 3(t-1)^2$ on $[1, 3]$ — the product with t^2 is $1 + 4(t-1)^3 + 3(t-1)^4$ — turns the Laplace bound into an exact antiderivative evaluation, giving $C_*(n') \le \bigl(1 - \frac{2}{\nu} + \frac{6}{\nu^2}\bigr)/(\ln 3\, \nu)$ at $\nu = \ln n'$ with no quadrature (`rotation_average_lt`, `rotationAverage_le`, gap form `rotationAverage_gap`). Lean's `hugCharge_sub_circleMean_le` identifies the limiting hug average with the circle integral `circleMean n'`; the elementary change of variables equating that integral with the displayed $C_*(n')$ remains prose. The observable has one wrap discontinuity, so bare unique ergodicity — uniform Birkhoff convergence for *continuous* observables — is not invoked directly. But the observable is monotone with a single jump, hence of bounded variation, and for such an observable §5.5's Denjoy–Koksma bound `denjoy_koksma_blocks` already gives $|C_L - \mathrm{circleMean}| \le s(L)\,\mathrm{Var}(F)/L$ at every L. Unique ergodicity is not needed; that Mathlib has no `UniquelyErgodic` costs this paper nothing.

This is the infinite-itinerary average, not a finite-L inequality: on the certified survey the finite leftover charge exceeds C_* by up to $1.57 \cdot 10^{-5}$, so $C_L \le C_*$ is false. The next two subsections quantify the finite-L error.

5.4 Itinerary identity

Write C_L for the charge-per-letter of the budgeted hug word at $(L, o_{\min}(L))$.

Lemma 5.6 (itinerary identity). For every L, the budgeted hug itinerary at $(L, o_{\min}(L))$ equals the exact rotation L-prefix generated by the integer rule: E at step k if and only if $3^a \geq 2^{k+1}$, where a is the number of odd letters already used. In particular C_L is a Birkhoff average of the rotation.

Proof. The exact rule keeps $u \in [0, 1 + \alpha)$, so its L-prefix uses exactly $o_{\min} = \lceil L \log 2/ \log 3 \rceil$ odd letters. A first budget-forced divergence between the two words would make the exact prefix use more of one letter than its own total, which is impossible. Lean: `budgetedWord_eq_hugWord`, with the window invariant `hugOdds_pow_ge` / `hugOdds_pow_lt` and minimality `hugOdds_least` (`WalkChargeItineraries.lean`).

The identification with the rotation is Lean too, and it is a floor identity rather than analysis (`HugRotation.lean`). Minimality says `hugOdds` is the least a with $2^k \leq 3^a$, and a least element of that shape is a ceiling: `hugOdds_eq_ceil` reads it as $\lceil k \log 2/ \log 3 \rceil$, hence `hugOdds_eq_sub_floor` as $k - \lfloor k\theta \rfloor$ and `hugEvens_eq_floor` as $\lfloor k\theta \rfloor$. The letter rule is then the wrap (`hugLetter_iff_floor_step`: even exactly when $\lfloor (k+1)\theta \rfloor = \lfloor k\theta \rfloor + 1$), the exponent walk is the rotation orbit (`hugWalk_eq_fract`: $u_k = \log_2 3 \cdot \{k\theta\}$), and `periodicObservable_hugWalk` is the "in particular" of the statement — the k-th term of the rotation's ergodic sum at phase 0 is the observable at the walk's position after k steps. Neither the certified sandwich nor irrationality of θ is used. □

5.5 Denjoy–Koksma over certified Ostrowski blocks

Two classical ingredients are used here in a fixed coordinate, so we record both explicitly.

The coordinate change. The infinite hug walk is rotation by $\alpha = \log_2(3/2)$ on the circle $\mathbb{R}/(1+\alpha)\mathbb{Z}$ (Proposition 5.5). Rescaling by $1/(1+\alpha)$ conjugates it to rotation by

$$\theta = \frac{\alpha}{1+\alpha} = \frac{\log(3/2)}{\log 3}$$

on the standard circle $\mathbb{R}/\mathbb{Z}$: with $\alpha = \log(3/2)/\log 2$ one has $1 + \alpha = \log 3/ \log 2$, and the quotient is the display. The observable $F(u) = n'^{1-2^u}/2^u$ is transported by the same rescaling; a homeomorphic change of coordinate does not change its total variation. All Ostrowski data below — quotients, convergents, digits — refer to this θ.

The Denjoy–Koksma inequality (classical; see [18, Theorem 3.1, p. 73]). If $f : \mathbb{R}/\mathbb{Z} \to \mathbb{R}$ has bounded variation $\mathrm{Var}(f)$ and p_j/q_j is a continued-fraction convergent of the irrational θ, then for every x,

$$\Big|\sum_{k=0}^{q_j-1} f(x + k\theta) - q_j \int_0^1 f\Big| \leq \mathrm{Var}(f).$$

The inequality is invoked below only at convergent denominators of θ, never at an arbitrary good rational approximation: the certified pairs (p_j, q_j) are produced by the continued-fraction recurrence from the certified shared quotient prefix of θ (`theta_convergent_denominators`, `theta_convergent_numerators`, `cf_lower_prefix`, `cf_upper_prefix`), so they are genuine convergents of θ, unimodular and coprime (`theta_convergents_unimodular`, `theta_convergents_coprime`). Their approximation quality $|\theta - p_j/q_j| < 1/q_j^2$ and the *residue* permutation are additionally verified in Lean (`theta_convergent_quality`, `theta_block_permutations`): the latter says $i \mapsto p_j i$ is a bijection of $\mathbb{Z}/q_j\mathbb{Z}$.

The residue permutation must not be confused with a permutation of fixed half-open cells. For example, at $p/q = 3/8$, the points 0 and 3θ both lie in $[0, 1/8)$. The argument below instead anchors cells at the starting phase and chooses their endpoint convention according to the sign of $\theta - p/q$.

The variation-versus-integral inequality itself is classical, and its analytic half is now Lean (`Problems/Juggler/DenjoyKoksma.lean`), for arbitrary cut points rather than for a rotation: `value_sub_mean_le_variation` — a value of f on a cell differs from the mean of f there by at most the variation there — together with the n-fold additivity of variation (`sum_eVariationOn_Icc`, Mathlib having only the binary form) gives `denjoy_koksma_abstract`: one sample point per cell, and the sample sum is within the total variation of the integral. `denjoy_koksma_unit` is that on $[0, 1]$ with q uniform cells, the shape applied per block below.

The *geometric* step — that the orbit $x, x+\theta, \dots, x+(q-1)\theta$ visits each phase-anchored, sign-oriented cell exactly once, which is where $|\theta - p/q| \le 1/q^2$ is used — is now Lean as well (`Problems/Juggler/DenjoyKoksmaOrbit.lean`), and with it the whole inequality. `denjoy_koksma_rotation` and its mean form `denjoy_koksma_rotation_mean` are the display above, for a one-periodic f of bounded variation, uniformly in the starting phase x. Both modules are in this paper's barrel. Mathlib has neither Denjoy–Koksma nor unique ergodicity of the irrational rotation — there is no `UniquelyErgodic` in Mathlib at all — so this is the only Lean path to the display.

Phase anchoring. A fixed-grid argument would place the cells at $[i/q, (i+1)/q)$ and asked the blocks to permute them. That is false, and the counterexample is small: $q = 2$, $\theta = 0.7$, $p = 1$, $x = 0.49$ puts both orbit points in $[0, \frac{1}{2})$. Two changes make it true. The cells must be anchored at the starting phase x, and which way they are half-open must follow the sign of $\delta = \theta - p/q$. Writing $k\theta = \lfloor kp/q \rfloor + (kp \bmod q)/q + k\delta$, point k sits at the left endpoint of cell $kp \bmod q$ displaced by $k\delta$, and $|k\delta| \le (q-1)/q^2 < 1/q$ is under one cell width — so it stays in that cell when $\delta \ge 0$ and falls into the previous one when $\delta \le 0$. `orbitCell` is that assignment; `orbitCell_inj` is its injectivity, which needs only $\gcd(p, q) = 1$ — the residue permutation this paper already certified as `theta_block_permutations`; and `orbit_mem_cell` is the displacement bound. `denjoy_koksma_cellmap` then takes the assignment in the direction the orbit supplies it, point $\mapsto$ cell, so no permutation has to be inverted.

The cell argument therefore proves the required rotation estimate uniformly in the starting phase, with the endpoint convention just stated.

And blocks compose. The proof below cuts a length $L = \sum_j b_j q_j$ into consecutive blocks whose starting phases differ, and leans on the inequality being uniform in x so that each block may be treated independently, whatever phase it inherits from its predecessor. That step is `denjoy_koksma_blocks`: an induction over a list of certified pairs (p_j, q_j), repeating a pair for each of the b_j copies, giving

$$\Big|\sum_{k<L} f(x + k\theta) - L\int_0^1 f\Big| \;\le\; s(L)\,\mathrm{Var}(f), \qquad s(L) = \sum_j b_j,$$

where $\mathrm{Var}(f)$ bounds the variation over every window of length one. The displayed bound of Theorem 5.7 is this with $\mathrm{Var}(F) < 2$, divided by L.

And the observable's own variation. That $F(u) = n'^{1-2^u}/2^u$ has $\mathrm{Var}(F) < 2$ including the wrap jump is `observable_window_variation_lt_two` (`JumpVariation.lean`). The rescaling is not a separate step there: `periodicObservable` is defined as F of the fractional part times the period $1 + \alpha = \log_2 3$, so it lives on $\mathbb{R}/\mathbb{Z}$ from the start, and the rotation number $\alpha/(1 + \alpha)$ is `walkTheta`. The variation itself needed three general facts Mathlib lacks — that variation ignores a sign (`eVariationOn_neg`, giving the antitone counterpart of `MonotoneOn.eVariationOn_eq`), that it is subadditive in the *function* and not only in the set (`eVariationOn_add_le`), and the shape itself (`eVariationOn_le_of_jump`): a fall, plus a jump, and the proof is a decomposition $f = g + h$ into an antitone part and a monotone step rather than a computation of the supremum.

`periodic_window_variation_le` then reads that off for *every* window $[y, y + 1]$, which is what the uniformity in x requires, and `block_envelope` and `theta_block_envelope` are the display: for any list of certified convergents — a pair repeated once per Ostrowski digit — the ergodic sum is within $2\,s(L)$ of $L\,C_*$.

The Ostrowski assembly closes too. `ostroBlocks` gives level i one copy of the certified pair at level $12 - i$ per Ostrowski digit, matching the association in `theta_sum_eq`; `ostroBlocks_snd_sum` says the denominators sum to L and `ostroBlocks_length` that the length is $s(L)$, so `theta_block_envelope_of_length` is the display at an arbitrary length with no window hypothesis. Feeding the digit cap $s(L) \le 47$ through it, `theta_block_envelope_window` makes the bound the constant $94 = 2 \cdot 47$ for every $L < 301994$.

The reading of C_L as that ergodic sum — Lemma 5.6's rotation identification — is Lean as well (`HugRotation.lean`), so the chain from the integer rule to the envelope is formal throughout, and `HugChargeEnvelope.lean` states the theorem once rather than leaving it a chain to compose: `hugCharge_sub_circleMean_le` is $|C_L - C_*| \le 2s(L)/L$ at every $L > 0$, and `hugCharge_sub_circleMean_window` is $94/L$ on the certified window.

One thing there is a definition and not a theorem, and the file says so. `hugCharge` is *defined* as the average of this section's observable along the exponent walk. That the walk positions are the budgeted word's is `budgetedWord_eq_hugWord`; that the k-th term of the rotation's ergodic sum is the observable at the k-th walk position is `periodicObservable_hugWalk`. What no theorem can supply is that the phrase "charge per letter" denotes this normalisation rather than another — `stateCharge`, Theorem 5.4's envelope charge, is proportional to it but not equal. The two named theorems are what make this the defensible choice.

This settles the ergodic-convergence part of what Proposition 5.5 called classical. The observable is monotone with one jump, hence of bounded variation, and the display above forces $|C_L - \mathrm{circleMean}| \le s(L)\,\mathrm{Var}(F)/L$. The elementary change of variables from `circleMean n'` to the displayed explicit $C_*(n')$ remains human. Unique ergodicity is not needed once Denjoy–Koksma is available — which is why Mathlib's not having `UniquelyErgodic` costs this paper nothing.

Theorem 5.7 (block envelope). For the exact rotation prefix of length L at reduced base n',

$$\bigl|C_L - C_*(n')\bigr| \;\le\; \frac{2\,s(L)}{L}, \qquad s(L) = \sum_j b_j,$$

for any decomposition $L = \sum_j b_j q_j$ into convergent denominators q_j of $\theta = \log(3/2)/\log 3$.

Proof. The observable $F(u) = {n'}^{1-2^u}/2^u$ decreases on the circle from $F(0) = 1$ to $F\big((1+\alpha)^-\big) = n'^{-2}/3$, so its variation including the wrap jump is < 2; the rescaled observable on $\mathbb{R}/\mathbb{Z}$ has the same variation. The Denjoy–Koksma inequality stated above, applied per block with the certified convergent quality $|\theta - p_j/q_j| < 1/q_j^2$, bounds the ergodic sum of each block of length q_j within $\mathrm{Var}(F)$ of $q_j C_*$. The decomposition $L = \sum_j b_j q_j$ partitions the length-L trajectory segment into consecutive blocks whose starting phases differ; since Denjoy–Koksma holds uniformly in the starting phase x, it applies independently to each block, whatever phase that block inherits from its predecessor. Summing the $s(L)$ blocks gives the display. □

The denominator list

$$1,\ 2,\ 3,\ 8,\ 19,\ 65,\ 84,\ 485,\ 1054,\ 24727,\ 50508,\ 125743,\ 176251$$

is certified by an interval continued fraction on the big-integer sandwich $2^{17087915} > 3^{10781274}$ and $2^{16785921} < 3^{10590737}$. The sandwich, the resulting real bounds on θ, the shared quotient prefix of the two rational endpoints, and the convergent recurrence are Lean (`theta_sandwich_upper`, `theta_sandwich_lower`, `lower_lt_walkTheta`, `walkTheta_lt_upper`, `cf_lower_prefix`, `cf_upper_prefix`, `theta_convergent_denominators`, `OstrowskiSandwich.lean`); the two power inequalities are checked by the Lean kernel through `norm_num`, not by the compiled runtime, so the sandwich that carries this certification rests on the kernel alone. The cylinder-interval bridge identifying the displayed list as a literal continued-fraction prefix of θ remains classical prose. It is not needed for the formal block bound: the quantitative hypothesis Denjoy–Koksma needs per block is Lean directly. With the matching numerators $0, 1, 1, 3, 7, 24, 31, 179, 389, 9126, 18641, 46408, 65049$ (`theta_convergent_numerators`), consecutive pairs are unimodular, all pairs are coprime, every certified convergent satisfies $|\theta - p/q| < 1/q^2$ against the sandwich (`theta_convergents_unimodular`, `theta_convergents_coprime`, `theta_convergent_quality`), and each certified pair has the requisite residue permutation (`theta_block_permutations`). The variation-versus-integral inequality, the signed anchored orbit-cell argument, its mean form, and uniform block composition are Lean (`denjoy_koksma_abstract`, `denjoy_koksma_rotation`, `denjoy_koksma_rotation_mean`, `denjoy_koksma_blocks`). A Koksma-type bound with constant 1 (that is, $+1/L$) is *false* for this observable; the correct constant is $2s(L)$.

5.6 The window theorem

Theorem 5.8 (uniform window envelope). For every $L \in [50508,\ 16785921)$, evaluate the reduced base at $n = 26254996$ and $o = o_{\min}(L)$. Then

$$C_L \ \le\ C_*(n') + \frac{2\,s(L)}{L} \ <\ \frac{1}{\ln 3\,\ln n'}.$$

Proof. Greedy Ostrowski digits obey $b_j \le a_{j+1}$, so with the certified quotients $\theta = [0; 2, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, \ldots]$ the digit sum satisfies $s(L) \le \sum_{j \le 13} a_j = 47$ for $L < q_{13} = 301994$, and on the remaining range $L = b\,q_{13} + r$ with $1 \le b \le a_{14} = 55$ and $r < q_{13}$, so $s(L) \le b + 47$. The general digit cap, exact reconstruction $L = \sum_j b_j q_j$, and digit-sum bound are Lean for any denominator sequence satisfying the convergent recurrence (`ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, `OstrowskiNumeration.lean`). The instantiated Lean theorem `theta_digitSum_le` supplies $s(L) \le 47$ only for $L < q_{13} = 301994$, and `hugCharge_sub_circleMean_window` has the same range. The extension from q_{13} to q_{14} is the human arithmetic just displayed: identify the next quotient $a_{14} = 55$, write $L = bq_{13} + r$, and apply the general block envelope with $s(L) \le b + 47$. Lean does certify the endpoint power inequalities (`theta_sandwich_lower`, `theta_sandwich_upper`, $2^{16785921} < 3^{10590737}$ and $3^{10781274} < 2^{17087915}$); it does not contain the named extended-window instance. Put $\nu = \ln n'$. Since $n = 26254996$ and $0 \le D \le 1.05L/n$, the entire window satisfies

$$16.41 < \log n - \frac{1.05 \cdot 16785921}{n} \le \nu < 17.084.$$

The digit caps give, without a scan,

$$\frac{2s(L)}{L} \le \max\left(\frac{94}{50508}, \frac{96}{301994}\right) < 0.001862.$$

For the second range this follows from $2(b+47)/(bq_{13}) \le 96/q_{13}$, since $b \ge 1$. The quadratic integral majorant of Proposition 5.5 gives

$$\frac{1}{\ln 3\, \nu} - C_*(e^\nu) \ge h(\nu) := \frac{2\nu - 6}{\ln 3\, \nu^3}.$$

Because $h'(\nu) = (18 - 4\nu)/(\ln 3\, \nu^4) < 0$ on this interval, $h(\nu) > h(17.084) > 0.00514 > 0.001862$. Combining this with Theorem 5.7 proves the strict comparison. The digit reconstruction and block inequality are formalized as described above; these explicit extended-window bounds are human arithmetic. □

Why the window reaches q_{14}, and why that is the natural stop. The bound $2s(L)/L$ is worst at the *small* end, not the large one: a large Ostrowski digit forces a large L, since $b_j = c$ requires $L \ge c\, q_j$. On $[q_{13}, q_{14})$ the two effects give $2s(L)/L \le 2(b+47)/(b\, q_{13}) \le 3.18 \cdot 10^{-4}$, maximised at $b = 1$ and an order below the value at $L = 50508$; an exhaustive scan of $[50508, 2 \cdot 10^6)$ puts the true maximum at $9.3766 \cdot 10^{-4}$, attained at $L = 74654$ with $s = 35$. So the window costs nothing to extend across the nonendpoint members of the $a_{14} = 55$ fan, and it stops at $q_{14} = 16785921$ only because that is where the next partial quotient begins. The same sufficient comparison can be checked at another fixed evaluation floor by bounding ν for that floor and comparing $h(\nu)$ with the digit bound above. No assertion uniform over arbitrarily large floors follows from the present calculation.

Two consequences, and one thing that is *not* a consequence. First, $q_{14} = 16785921$ is exactly L_{55}, the last member of the semiconvergent fan of Proposition 5.12. Because the window is half-open, it covers precisely $L_0, \ldots, L_{54}$ and excludes L_{55}. Second, the window theorem is not what limits the walk charge; that is Remark 5.8a.

What does *not* follow is that the kill tables become census-free. The window theorem is a uniform upper bound on the **charge**, so no per-length dynamic program is needed to bound $B(L)$ anywhere on $[50508, 16785921)$. The kill decision is the comparison $\theta(L) > \frac{6}{5}B(L) \cdot \text{guard}$, and its left-hand side $\theta(L) = 1 - 2^L/3^{o_{\min}(L)}$ is a per-length Diophantine quantity that the envelope says nothing about. Corollaries 5.10 and 5.11 identify exactly that quantity as the obstruction at the surviving fan members. So the extension removes the per-length *dynamic program* from every nonendpoint fan member and leaves the per-length *arithmetic* in place. Eliminating the latter would need a lower bound on the Diophantine deficit that is uniform along the fan — a genuinely different statement, and one this paper does not prove.

Remark 5.8a (what the walk charge can ever buy). The charge of Theorem 5.3 prices a state at exponent u by $f(u) = 1/(x \ln x)$ with $x = (n')^{2^u}$, so $f(u)/f(0) = 2^{-u} \exp(-(2^u - 1) \ln n')$. Since $2^u - 1 = u \ln 2 + O(u^2)$, the charge decays on the scale $u \asymp 1/\ln n'$: it is a Laplace-type boundary layer at $u = 0$, not a hard cutoff, and $1/(\ln 3 \, \ln n')$ — the same quantity Theorem 5.8 carries — is the scale of its effective mass rather than the width of an interval. Integrating the layer against the walk gives an *empirical scaling law* for the advantage over the length-only parity charge,

$$\frac{\text{parity}}{\text{walk}} \approx c \ln n', \qquad c \approx 0.44.$$

The constant is fitted, not derived; only the $\ln n'$ dependence is explained by the boundary layer. Measured at $L = 50508$ over ten orders of magnitude in the floor, the ratio (improvement)$/\ln n'$ runs 0.461, 0.451, 0.446, 0.445, 0.439, 0.432, 0.427 at $n_0 = 10^6$, $2.6 \cdot 10^7$, $1.6 \cdot 10^8$, $3.5 \cdot 10^8$, 10^{10}, 10^{13}, 10^{16}: constant to 8% across the range.

So the walk charge is neither exhausted nor cheap to improve. It is not limited by certification depth, which the extension above showed has room to spare, and it is not limited by the envelope: substituting the census-free envelope of Theorem 5.8 for the exact lattice program changes the margin at $L = 50508$ from 1.1204 to 1.1196, a difference of 0.07%. It is limited by the shape of f. Since the advantage grows like $\ln n'$, **doubling the walk charge's efficiency requires squaring the descent floor**.

One caution about that cross-check, because it is easy to over-read. Both quantities compared there live on the *relaxed* class: the lattice program admits every binary word with o odds, e evens and $u_k \geq 0$, realizable or not. The agreement to 0.07% therefore says the envelope is tight against the relaxed optimum, and says nothing about how far that optimum sits above the true maximum over *realizable* words. That is a separate question, and it is the only place in this construction where a factor of the size in question could plausibly hide, so it is worth measuring rather than assuming.

Proposition 5.8b (the relaxation cannot hide a constant). For $14 \leq L \leq 24$ and $o = o_{\min}(L)$, let $\mathcal{A}(L)$ be the class the lattice program maximises over — all masks with o odds whose exponent walk stays nonnegative — and let $\mathcal{R}(L) \subseteq \mathcal{A}(L)$ be those realized as $\mathrm{word}_L(m)$ for some odd $m < 2 \cdot 10^7$. Then

1. the charge-maximising element of $\mathcal{A}(L)$ lies in $\mathcal{R}(L)$ at every one of those eleven lengths, so the two maxima agree exactly; and
2. that agreement does not depend on (1). Ordering $\mathcal{A}(L)$ by charge, the second element carries at least 0.9846 of the maximum and the tenth at least 0.9214, both ratios *increasing* in L (at $L = 24$: 0.9898 and 0.9481); and membership of $\mathcal{R}(L)$ is independent of charge to measurement accuracy — at $L = 24$ the realized fraction is $0.698, 0.702, 0.689, 0.694, 0.689, 0.691, 0.697, 0.698, 0.697, 0.695$ across the ten charge deciles. So the first realized word appears within the first few ranks whatever the argmax does, and the realized maximum is within about one percent of the relaxed maximum regardless.

Discussion. The relaxation is not vacuous: the realized fraction $|\mathcal{R}|/|\mathcal{A}|$ is 1.000 through $L = 21$ and then falls to 0.992, 0.906, 0.695, so by $L = 24$ the program maximises over half again as many words as can occur. But the words it adds are spread uniformly through the charge ordering rather than concentrated where the charge is small, and the ordering is flat at the top; those two facts together, not the coincidence in (1), are what bound the slack.

It is worth recording what this rules out. A relaxation that cost a factor of 8 would have to remove almost the entire top of the charge ordering, and $\mathcal{R}$ removes about 30% of it uniformly. Whatever explains the walk charge's advantage being $\approx 0.44 \ln n'$ rather than something larger, it is not the exponent-walk relaxation.

Proposition 5.8c (flatness at the kill-table lengths). Part (2) of Proposition 5.8b rests on the charge ordering being flat at the top, and enumeration establishes that only for $L \leq 24$. It can be established directly at the lengths the kill tables use, without enumerating, by running the lattice program of Theorem 5.3 with the K best partial sums per state in place of the best one: max-plus becomes top-K-plus, the rolling array grows by a factor K, and the pass stays linear in L. At $K = 16$, writing r_j for the j-th largest charge,

$$1 - \frac{r_{16}}{r_1} = \begin{cases} 9.44 \cdot 10^{-2}, & L = 18, \\ 6.24 \cdot 10^{-2}, & L = 24, \\ 5.42 \cdot 10^{-8}, & L = 50508,\ n' = 2.6 \cdot 10^{7}, \\ 6.80 \cdot 10^{-9}, & L = 176251,\ n' = 1.6 \cdot 10^{8}, \\ 1.08 \cdot 10^{-9}, & L = 780239,\ n' = 3.5 \cdot 10^{8}. \end{cases}$$

So the top does not merely stay flat at the operative lengths: it flattens by seven orders of magnitude between $L = 24$ and $L = 50508$, and continues to flatten with L.

Proof of the computation. The recursion and admissible class are those of Theorem 5.3; only the accumulator changes. The top-K program reproduces the exhaustive top ten at $L = 18$ to 10^{-18}, and its GPU form agrees with the host form bitwise, so the values above are the same objects the kill tables use. $\square$

The reason is visible in the charge. At the operative lengths the sum is carried by the $\asymp L/(\ln 3 \ln n')$ states with $u \approx 0$, and the closest a nonnegative walk can come to $u = 0$ other than exactly is $\asymp 1/o$; perturbing the extremal walk therefore moves one contribution by a relative $\asymp \ln n'/o$ out of a total of $\asymp L/\ln n'$ equal terms. At $L = 50508$ that predicts $\approx 8 \cdot 10^{-8}$ against the measured $5.4 \cdot 10^{-8}$.

Together with Proposition 5.8b this closes the question the relaxation raised. Whatever removes 30% of the admissible words, it must remove *all sixteen* of the leading walks before the realized maximum falls by as much as $5 \cdot 10^{-8}$; realizability is spread uniformly through the ordering, so that is not what is happening. The exponent-walk relaxation is not hiding a constant, at the lengths where the constant would matter.

One limit remains on how far this should be read. $\mathcal{R}(L)$ is a scanned lower bound on the realizable set, so a word missing from it can only raise the realized maximum: the measurement bounds the slack above, which is the direction that matters. An earlier and thinner scan ($m < 2 \cdot 10^{6}$) put the $L = 22$ argmax outside $\mathcal{R}$ and the realized fraction at 0.38; at $2 \cdot 10^{7}$ both reverse, which is the expected behaviour when a length-L word occurs with density $\approx 2^{-L}$, and a caution against reading one absence as an exclusion.

What the boundary layer does suggest is that improvement will not come from a better envelope for this charge. It raises a sharper question than "find a better charge": is *every* charge that depends only on the transported exponent u, under the same one-step defect budget, subject to the same $O(1/\ln n')$ concentration? A positive answer would turn the observation above into a limitation theorem for the whole class, and would be worth more than another constant.

5.7 Kill table and the period bound

The envelope controls the charge; kill decisions are the per-length finance comparison $\theta(L) > \frac{6}{5}\,B(L)\cdot$guard of the Theorem 4.6 architecture, which is Diophantine, not envelope-limited.

The comparison template is itself Lean (`cycleMin_hug_kill_criterion`, `DefectFinance.lean`): every minimum-based cycle at $n \geq 400$ with positive reduced log-base must satisfy $1 - 2^L/3^o \leq \frac{6}{5}\sum_k g(\text{hugWeight}_k)$ at the reduced base, chaining the Lean finance inequality (`cycleMin_defect_finance`, the certified identity of Theorem 4.6) with the defect-to-hug-charge envelope of §5.2. The kill decisions below verify the numeric *failure* of this inequality per length; only that evaluation, and the parity refinement of Corollary 4.5, remain verified computation.

Theorem 5.9 (verified computation; period bound at the laboratory floor). At $N_0 = 26254995$, the walk charge of Theorem 5.3 excludes 18 of the 19 parity survivors of Theorem 5.2 through $L = 2 \cdot 10^5$. Kill margins run from 1.1204 at the seed $L = 50508$ (required improvement over parity 6.87, walk supplies 7.70; deficit $D = 7.4566{\cdot}10^{-4}$), through 1.1195 and 1.1187 at its multiples 101016 and 151524, up to 7.69 at the parity-marginal lengths. The sole walk survivor is $L = 176251$ (margin 0.159; required improvement 48). The combined parity + walk contiguous excluded prefix is 176250: any nontrivial Juggler cycle has period at least 176251.

Proof. Theorem 5.2 leaves the 19 parity survivors. Each is priced by the exact (step, odd-count) lattice dynamic program at the reduced base (Theorem 5.3) and compared against $\theta(L)$ under the 6/5 unroll; substituting the census-free envelope of Theorem 5.8 instead recovers the same 18 kills (margin 1.1196 at $L = 50508$; $L = 176251$ survives at 0.1588). Checksums are Appendix B. ☐

Calibration. At the base instance $(L, N_0) = (25781, 10^6)$ the walk charge gives margin 0.196 — correctly no kill (required improvement 32.5, walk supplies 6.37). The mechanism did not change between the two floors; the target did.

Corollary 5.10 (verified computation; second laboratory floor). Every integer $2 \leq n \leq 162849448$ reaches 1 (first-passage certificate exactly as in Proposition 5.1: the extension segment walks 68297226 odd starts over 547 contiguous chunk records with exact integer square roots and no failures of any kind; the maximum first passage is 433 steps at seed 78641579, and the largest intermediate has 463362780 bits at seed 92502777; hashes in Appendix B). At $N_0 = 162849448$ the parity table excludes the previous survivor cluster $\{50508, 101016, 151524\}$ outright and leaves 25 survivors through $L = 6\cdot 10^5$; the walk charge of Theorem 5.3 kills the 15 below 478245 (margins 1.198 at 176251 and 352502, up to 8.44); lengths $L \leq 176250$ stay excluded because both exclusions are monotone in the floor. Every survivor here lies inside the window $[50508, 16785921)$ of Theorem 5.8, so the *charge* side of each kill is census-free; the per-length lattice program is run as a cross-check and agrees to

0.07%. The comparison against $\theta(L)$ remains per-length. The combined contiguous excluded prefix is 478244: any nontrivial Juggler cycle has period at least 478245.

The sole survivor at this floor is the semiconvergent fan member $478245 = 176251 + 301994$: required improvement over parity 19.46, direct walk margin 0.4334 — the obstruction is the Diophantine quality of $|3^o - 2^L|$ along the fan, not the envelope. The kill table recomputes on commodity hardware in minutes (Appendix B). Corollary 5.11 evaluates the same criterion at the next cheap floor.

Corollary 5.11 (verified computation; third laboratory floor). Every integer $2 \leq n \leq 350000000$ reaches 1 (first-passage certificate exactly as in Proposition 5.1: the extension segment from 162849449 walks 93575276 odd starts over 749 contiguous chunk records with exact integer square roots; two bit-cap seeds 172376627 and 240154767 are resolved at a $3{\cdot}10^9$-bit cap, the largest intermediate having 1493770145 bits at seed 172376627; the maximum first passage is 466 steps at seed 198424189; hashes in Appendix B). At $N_0 = 350000000$ the parity table leaves 17 survivors through $L = 8 \cdot 10^5$; the five leftovers below 478245 stay excluded because both exclusions are monotone in the floor, and the walk charge of Theorem 5.3 kills the 10 leftovers in $[478245, 755512]$ (margins 1.00555 at 478245, up to 7.824 at 504026). Every survivor here lies inside the window $[50508, 16785921)$ of Theorem 5.8, so the *charge* side of each kill is census-free and the per-length lattice program is only a cross-check; the comparison against $\theta(L)$ remains per-length. The combined contiguous excluded prefix is 780238: any nontrivial Juggler cycle has period at least 780239. This is the main numerical result of the paper.

The sole survivor is the semiconvergent fan member $780239 = 176251 + 2{\cdot}301994$: required improvement over parity 14.46, direct walk margin 0.6049 — the obstruction remains the Diophantine quality of $|3^o - 2^L|$ along the fan, not the envelope. The kill table is the GPU floating-point certified comparison of the same Theorem 5.9 criterion (Appendix B).

At and beyond $q_{14} = 16785921$, the stated half-open window would need a further block argument and certified quotient data. Within this walk-charge criterion, the remaining near-convergent comparisons depend on the small positive forms $o \log 3 - L \log 2$, starting with the fan member $L = 780239$, which is already inside the present window. The geometric restrictions of Sections 3.10–3.13 and 6.3 supply separate necessary conditions. The family leftover — the semiconvergent fans of $\log 2/\log 3$ reduced to dangerous-position partial quotients — is the working draft `juggler_near_convergent_diophantine_note.md`. The survivors are finance-survivors, not candidate cycles.

5.8 The price of the fan

The paragraph above says the obstruction is Diophantine and stops. It can be made quantitative, and the answer is short: the frontier lengths form one explicit arithmetic progression of 56 terms, ending on the next convergent, and the descent floor each term costs is computable in closed form. Nothing here is a new exclusion. It is a price list for the exclusions that remain, and it says exactly what the walk charge of this section bought.

Proposition 5.12 (fan law). Let $q_{12} = 176251$, $q_{13} = 301994$ be the consecutive convergent denominators of $\log 2/\log 3$ bracketing the frontier, with numerators $p_{12} = 111202$,

$p_{13} = 190537$. Put

$$L_k = q_{12} + k\, q_{13}, \qquad o_k = p_{12} + k\, p_{13}, \qquad 0 \le k \le 55.$$

Then $o_k = o_{\min}(L_k)$ for every such k; the linear form is exactly affine in k,

$$\Lambda_k = o_k \log 3 - L_k \log 2 = \Lambda_0 + k\Lambda', \qquad \Lambda_0 = 3.6002 \cdot 10^{-6}, \quad \Lambda' = -6.4508 \cdot 10^{-8};$$

and $k = 55$ is the last index with $\Lambda_k > 0$, because $\Lambda_0/|\Lambda'| = 55.81$. At that last index $L_{55} = 16785921 = q_{14}$ and $o_{55} = 10590737 = p_{14}$: the fan ends on the next convergent, which is where the partial quotient $a_{14} = 55$ is consumed.

Proof. Λ is linear in (o, L) and $(o_k, L_k) = (p_{12}, q_{12}) + k\,(p_{13}, q_{13})$, which is the affine formula; $\Lambda' = \Lambda(q_{13}) < 0$ because $3^{p_{13}} < 2^{q_{13}}$, consecutive convergents lying on opposite sides. Since $\Lambda_k \in (0, \log 3)$ for $0 \le k \le 55$, one has $3^{o_k} > 2^{L_k} > 3^{o_k - 1}$, which is $o_k = o_{\min}(L_k)$. The sign change at 55.81 is arithmetic, and $q_{14} = a_{14} q_{13} + q_{12}$ with $a_{14} = 55$ identifies the endpoint. $\square$

Because Λ_k *decreases* in k, so does $\theta(L_k)$, and $n_{\max}(L_k)$ increases: the fan grows strictly more expensive as it is climbed, and the cheapest member is always the one at the current frontier within this fan. The following transition has been checked computationally at $k = 1, \ldots, 12, 20, 31, 40, 52, 54$:

$$N_0 \;\ge\; n_{\max}(L_k) \qquad \Longrightarrow \qquad \text{period} \;\ge\; L_{k+1},$$

No all-index transition theorem is asserted: excluding a fan member also requires ruling out intervening non-fan lengths. (At $k = 0$ alone one extra length intervenes, the doubling $2q_{12} = 352502$, whose threshold 1044095006 sits 1793 above $n_{\max}(q_{12})$; the multiples mq_{12} cluster just above $n_{\max}(q_{12})$ in the same way and are all cleared by $n_{\max}(L_1)$.)

k	L_k	Λ_k	$n_{\max}(L_k)$ = floor that passes L_k
0	176251	$3.600 \cdot 10^{-6}$	$1.044 \cdot 10^{9}$
1	478245	$3.536 \cdot 10^{-6}$	$2.756 \cdot 10^{9}$
2	780239	$3.471 \cdot 10^{-6}$	$4.480 \cdot 10^{9}$
3	1082233	$3.407 \cdot 10^{-6}$	$6.238 \cdot 10^{9}$
6	1988215	$3.213 \cdot 10^{-6}$	$1.182 \cdot 10^{10}$
31	9538065	$1.600 \cdot 10^{-6}$	$1.040 \cdot 10^{11}$
52	15879939	$0.246 \cdot 10^{-6}$	$1.034 \cdot 10^{12}$
54	16483927	$0.117 \cdot 10^{-6}$	$2.199 \cdot 10^{12}$
55	16785921	$0.052 \cdot 10^{-6}$	$4.866 \cdot 10^{12}$

Three readings of this table.

What the walk charge is worth. Corollary 5.11 reaches $L_2 = 780239$ from $N_0 = 3.5 \cdot 10^8$. Finance and parity alone reach it only from $n_{\max}(L_1) = 2.756 \cdot 10^9$. The walk charge is therefore worth a factor 7.9 in descent floor at the present frontier — and a factor 6.4 at the previous one, where Corollary 5.10 reached L_1 from $1.63 \cdot 10^8$ against a finance requirement of $1.044 \cdot 10^9$.

What the next step costs. A purely computational route to $L_3 = 1082233$ — no walk charge, no new idea, only a longer first-passage run — needs $N_0 \geq n_{\max}(780239) = 4479642886$, a floor 12.8 times the present one. The walk charge does very much better, and by a factor that can be measured rather than guessed.

Observation 5.13 (finite margin scaling). For two lengths evaluated at two floors each, fitting the kill margin $\theta(L)/(\frac{6}{5}B(L, N_0))$ to a power of $N_0 \log N_0$ gives effective exponents 1.0491 and 1.0458. Their mean, approximately 1.047, is an empirical interpolation; it is not a proved asymptotic law.

Evidence. The committed kill records contain two lengths priced at two floors each. At $L = 176251$ the margin runs 0.158796 at $N_0 = 26254995$ to 1.198309 at 162849448; at $L = 478245$ it runs 0.433383 at 162849448 to 1.005552 at 350000000. The implied exponents are 1.0491 and 1.0458, agreeing to 0.31%. The excess over $\beta = 1$ is the secondary dependence of the reduced base on the floor; it is not needed for the conclusion below, which is unchanged at $\beta = 1$.

Applied to the present survivor, $L = 780239$ at margin 0.604888, this predicts a kill at $N_0 = 5.53 \cdot 10^8$. Evaluating the criterion directly gives

$$\text{the walk charge kills } L = 780239 \text{ from } N_0 = 553906250,$$

so the prediction is accurate to 0.2%, and the walk charge is worth a factor $4479642886/553906250 = 8.09$ in descent floor here — against 7.9 at the previous frontier and 6.4 at the one before. These three comparisons suggest a useful local estimate. Other fan members require direct evaluation; a uniform factor of eight over the entire fan is not established.

Corollary 5.14 (conditional; the next period bound). If every integer $2 \leq n \leq 554000000$ reaches 1, then any nontrivial Juggler cycle has period at least 1082233.

Proof, modulo the floor. At $N_0 = 554000000$ the parity table of Corollary 4.5 leaves twenty survivors below $L_3 = 1082233$, and the walk charge of Theorem 5.3 kills all of them. Everything at or below 780239 is excluded because both exclusions are monotone in the floor and $554000000 > 350000000$, except 780239 itself, which is killed here at margin 1.000884. The nine survivors above it are killed with margins

806020	8.076	830747	2.908	881255	4.596
931763	6.101	956490	1.663	982271	7.451
1006998	3.203	1032779	8.669	1057506	4.595

all by the certified evaluation of the Theorem 5.9 comparison at the reduced base. The next length, $L_3 = 1082233$, survives at margin 0.70815. Hence the contiguous excluded prefix is 1082232. $\square$

The kill table is therefore *already done*; the only missing input is the descent floor. Extending the certified floor from $3.5 \cdot 10^8$ to $5.54 \cdot 10^8$ is a first-passage run over roughly $1.0 \cdot 10^8$ further odd starts — comparable in size to the extension Corollary 5.11 already performed, and a factor 1.58 in floor rather than the 12.8 that finance alone would demand. We do not carry out that run here; Corollary 5.14 is stated conditionally so that the computation and the criterion are separable, which is the architecture of Corollaries 5.10 and 5.11 as well.

Further computational costs. The fan table reports parity-charge thresholds through L_{55}. Extrapolating the locally measured walk improvement to every member would be a heuristic; the conditional next-period claim in Corollary 5.14 uses direct comparisons instead. The limitation proved in Proposition 6.2a concerns fixed-floor charges with a positive length-uniform anchor contribution.

6. Limitations and future directions

Section 3.10 isolates a second unresolved question: whether absolute floor-cell alignment forces every threshold cycle to contain a wrong-parity state. Theorems 3.39–3.40 exclude successively wider top strips at their explicit minimum thresholds. The remaining region and the taller-cycle case are still unresolved. Appendix E's family and guard results do not establish the uniform conclusion.

Proposition 3.41 identifies a paired-gap issue: amplification through the terminal common prefix must be compared with the mixed passage and suffix contractions using the shared absolute cells. The present estimates cover only certified suffix factors. Their one-sided error certificate also requires increasingly large minima as the contracting exponent approaches one (Appendix F.6). Neither observation excludes the terminal mixed-word return.

The result does not imply termination. The remaining finance-survivor lengths are uncontrolled, and existence of a cycle at such a length is open.

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite itinerary w with $J^{|w|}(n) < n$. Even starts realize E; an odd start with even image realizes OE. The complement of those two words is the odd-to-odd class. If every start above 1 has some descent certificate, strong induction yields arrival at 1. That hypothesis is not proved, and a uniform run bound on expanding blocks is unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471.$$

The number p of odd runs on a minimum-based cycle — equivalently, the number of excursions on the necklace of Section 4 — is another natural statistic to test. The run form gives $p \leq e$ and, because the first odd run has length at least two, $p \leq o - 1$, hence $p \leq \min(e, o-1) < 0.3691\,L$ on an expanding itinerary; that is only the trivial ceiling, and a genuine lower bound on p, or a peak-height / peak-count tradeoff, would feed Theorem 4.7.

The following comparison records a limitation of this statistic. A bound on p is only useful if it constrains the *adversary*, and for the walk charge the adversary is the extremal walk of Theorem 5.3, which can be recovered from the lattice program by storing its decisions. That walk turns out to saturate both ceilings at once. Writing p for its odd-run count:

L	84	1054	25781	50508
p	31	389	9515	18641
$\min(e, o-1)$	31	389	9515	18641
longest odd run	2	2	2	2

So $p = e$ exactly, at every length tested: every even run has length one, and the walk is a word in the two blocks `OE` and `OOE` alone, mixed at the density $\log 2/\log 3$ — the hug itinerary of Theorem 5.4, in its Sturmian form. At $L = 84$ it is $\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{OE}\,\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{OE}\cdots$.

These relaxed extremal words attain the displayed run-count ceiling at the tested lengths. Thus a lower bound on run count alone would not exclude those particular relaxed words. A constraint coupling peak height, parity realization, and exact return could still be stronger.

The repository also measures realization of short hug prefixes among odd $m < 4 \cdot 10^6$: relative to $2^{-(l-1)}$, the observed frequency ratio is about 1.00 at $l = 8, 9, 10$ and 1.38 at $l = 18$. This finite experiment establishes realization of those short prefixes. It does not establish realization at lengths such as 780239, realization by cycle minima, or exact closure of the full word. These remain possible sources of further restrictions.

Corollary 4.11 excludes a short-period regime in terms of the cycle minimum. None of the finite realization or charge comparisons supplies a lower bound on that minimum growing with the period. The general cycle and termination problems remain open.

6.1 Relation to the companion manuscripts

The companion manuscripts [16,17] use the power envelope and descent floor proved or recorded here. Their analytic and counting results are separate inputs, and the period bounds of Sections 2–5 do not depend on them.

The envelope supplies a bounded target. If the exponent walk of a start $n \in (y, 2y]$ reaches

$$u_t \le -\log_2\left(\frac{\log(2y)}{\log N_0}\right),$$

Theorem 2.2 gives $J^t(n) \le N_0$, after which the verified descent floor gives arrival at 1. Paper C [17] combines a quantitative almost-all version of this event with a lower bound for backward-closed sets. Its parity or stopped-pressure hypotheses remain unproved; a larger finite floor alone does not establish them.

For numerical scale comparisons at $N_0 = 350000000$,

$$N_0^{4/3} \approx 2.47 \cdot 10^{11}, \qquad N_0^{3/2} \approx 6.55 \cdot 10^{12}, \qquad N_0^2 \approx 1.23 \cdot 10^{17}.$$

These are conversions of the supplied floor, not additional cycle exclusions or a proof of an almost-all hypothesis.

A cycle's basin has a separate lower bound. Paper C's stated contagion theorem gives $\sum_{m \in B(C),\, m \le x} 1/m \gg (\log x)^\lambda$ for every $\lambda < \lambda^{**} = 0.4926\ldots$, if the basin $B(C)$ of a nontrivial cycle exists. This is a result of the companion manuscript, not a new theorem here. For a primitive cycle C with minimum n and period L, the present paper instead gives

$$\theta \log n \le \sum_{x \in C} \frac{1}{x} \le \frac{L}{n}, \qquad \theta = 1 - 2^L/3^o.$$

The first inequality is Corollary 4.4c and the second follows from minimality. These constraints concern the cycle states; they do not give an upper bound on the basin that could contradict contagion.

Odd share and the finite frontier. Every nontrivial cycle has $o/L > \log 2/\log 3$. If $n \log n > 2L$, Theorem 4.10 also implies

$$0 < \frac{o}{L} - \frac{\log 2}{\log 3} \leq \frac{2}{n \log n \log 3}.$$

The frontier lengths $176251, 478245, 780239$ are members of the finite fan in Proposition 5.12. The auxiliary denominator 301994 is the adjacent lower-convergent denominator used to generate that fan; it is not itself one of these surviving lengths. A constraint on the odd share of live starts averaged over many starts is a different statement from a restriction on one periodic orbit.

Current scope of the companion hypotheses. Paper B's depth-five certificate class has stated density 7/8; the proposed length-seven and length-eight extensions remain outside its established conclusions. Paper C's revised cylinder hypotheses concern bad words or bad prefixes, and its stopped-pressure formulations concern live starts. The unrestricted all-word versions are not assumed: terminating cylinders already obstruct them. The scale-average stopped-pressure bound remains an open arithmetic input. These corrections do not change Paper A's finite cycle bounds, and no companion result is used here to exclude the surviving lengths.

6.2 Scope of the method's limitations

At the tabulated upper-convergent denominators, the finance threshold is well approximated by a constant multiple of $q_k q_{k+1}$ after multiplication by $\log n_{\max}$. The exact denominator recurrence is

$$q_{k+1} = a_{k+1} q_k + q_{k-1}.$$

The numbers $3.4, 5.8, 23.5, 2.5, 1.7$ in the numerical comparison are approximate ratios q_{k+1}/q_k, not the integer partial quotients a_{k+1}. A square-root scaling in the floor can be a useful local heuristic, but a global asymptotic for the first surviving period would require control of continued-fraction quotients and the intervening lengths. No such global theorem is claimed here.

The following proposition gives a precise limitation under an explicit normalization assumption.

Proposition 6.2a (what an anchor-normalized charge can exclude). Call Φ a *charge* if every cycle minimum n with word of length L satisfies $\sum_{i<L} 1/(x_i \log x_i) \leq \Phi(n, L)$, with Φ nonincreasing in its first argument. Fix $N_0 \geq 2$, and call Φ *anchor-normalized at* N_0 if there is a constant $c_{N_0} > 0$, independent of L, such that $\Phi(N_0, L) \geq c_{N_0}$ for every L. Say Φ *excludes* L at floor N_0 when $\theta(L) > \frac{6}{5}\Phi(N_0, L)$. Then an anchor-normalized charge excludes only finitely many denominators q_k of the upper convergents $p_k/q_k > \log 2/\log 3$.

Proof. Anchor normalization makes exclusion at N_0 require $\theta(L) > \frac{6}{5} c_{N_0}$, a positive threshold fixed once N_0 is. Along an upper convergent $p_k/q_k > \alpha := \log 2/\log 3$, one has $p_k = o_{\min}(q_k)$, and therefore

$$0 < -\log\bigl(1 - \theta(q_k)\bigr) = p_k \log 3 - q_k \log 2 = q_k \log 3 \left(\tfrac{p_k}{q_k} - \alpha\right) < \frac{\log 3}{q_{k+1}}.$$

The last bound is the standard convergent estimate. Since $q_{k+1} \to \infty$, it gives $\theta(q_k) \to 0$, so only finitely many upper-convergent denominators clear the fixed threshold. $\square$

The explicit finance, run-packing and walk-charge majorants in Sections 4 and 5 are anchor-normalized: their displayed fixed-floor formulas retain a positive first-state contribution uniformly in L, the last entering `cycleMin_hug_kill_criterion`. Proposition 6.2a therefore applies to refinements that preserve this feature. It does not rule out a different charge formulation whose value at N_0 has no length-uniform positive anchor, nor does it turn the present finite period bounds into a universal no-go theorem.

The run–suffix inequalities require separate care. Theorem 3.26 compares the finite-state lower envelope $4(n/4)^{(3/2)^a}$ with the backward envelope $B(u)$. A leading-exponent comparison can suggest which shapes to test, but does not remove the constants or finite-n margin. In particular, it does not prove that every proper tail beginning with O is formally non-expanding. Equality of two finite shape counts likewise does not prove that the run–suffix inequalities are ineffective at every length surviving finance.

The finite shape enumerations and charge comparisons describe the instances computed in this paper. They do not rule out stronger enumeration methods, improved charges without the normalization of Proposition 6.2a, or restrictions valid specifically on realized cycles. Primitivity makes orbit states distinct; it does not make every block-boundary landing odd when adjacent even letters are permitted. Consequently, any refinement using distinct odd valleys must count actual maximal odd runs or impose the no-EE hypothesis.

A proof excluding all nontrivial cycles would need a further argument beyond the finite period bounds and the scoped limitation above.

6.3 Absolute upper cells and the remaining signed-loss question

The cubic rank grid also strengthens the total floor-loss bound. This section records that consequence, a local integer refinement, and the precision still needed to convert these restrictions into an exclusion.

Lemma 6.3a (odd-to-odd upper-square gap). If both x and $J(x)$ are odd, then

$$x^3 + 3 \leq (J(x) + 1)^2. \tag{UC1}$$

Proof. More generally, let y be odd. The equality $x^3+1 = (y+1)^2$ would give $x^3 = y(y+2)$. Since $\gcd(y, y+2) = 1$, unique factorization would make both positive factors cubes, say $y = a^3$ and $y+2 = b^3$. But $a \geq 1$, $b \geq a+1$, and $b^3 - a^3 \geq 3a^2 + 3a + 1 \geq 7$, a contradiction. For $y = J(x)$, the exact upper cell makes $(y+1)^2 - x^3$ positive; it is odd when x, y are odd. Having excluded one, it is at least three. $\square$

The integer argument is formalized by `cube_add_one_ne_odd_succ_sq` and `floorPower_odd_image_upper_gap`. It is a local floor-cell restriction and requires no cycle assumption.

Proposition 6.3b (upper-cell charge on the sorted grid). Let C be a primitive exact threshold cycle, or a primitive actual Juggler cycle with minimum $m > 1$ and maximum $M < m^3$. Write $m = \min C$, let $L = |C|$ be its least period, and let o be its lower-branch count, equivalently its odd count in the actual case. Set

$$T = \log 3, \qquad \Lambda = o \log 3 - L \log 2, \qquad A = (\log m)e^{-(1-1/L)\Lambda}.$$

Then $A > 0$, $\Lambda > 0$, and

$$\Lambda < \frac{e^{-A}}{A} \sum_{i=0}^{L-1} e^{-iT(A+1)/L} < \frac{e^{-A}}{A} \left(1 + \frac{L}{T(A+1)}\right). \tag{UC2}$$

Proof. Retain the sorted states and defects of Proposition 3.36. For an edge with target y and real branch exponent $p \in \{1/2, 3/2\}$, the upper cell gives $x^p < y + 1$. Consequently

$$0 \le \delta = \log \frac{p \log x}{\log y} < \eta(y) := \log \frac{\log(y+1)}{\log y} < \frac{1}{y \log y}.$$

The last inequality uses $\log(1 + t) < t$ twice. Since targets permute the states and the defects sum to Λ, $\Lambda < \sum_i 1/(c_i \log c_i)$. The grid bound (CB5) gives $\log c_i \ge Ae^{iT/L}$. For $t = iT/L \ge 0$, monotonicity and $e^t \ge 1 + t$ yield

$$c_i \log c_i \ge Ae^{Ae^t+t} \ge Ae^A e^{(A+1)t}.$$

This proves the first inequality in (UC2). With $z = T(A + 1)/L > 0$, the geometric sum is less than $1/(1 - e^{-z}) = 1 + 1/(e^z - 1) < 1 + 1/z$, proving the second. ☐

For example, along a sequence with $\log m \to \infty$, $\log m = o(L)$, and $\Lambda \log m \to 0$, this gives

$$m(\log m)^2 \Lambda \le (1 + o(1)) \frac{L}{\log 3}. \tag{UC3}$$

Indeed $A/\log m \to 1$ and $e^A/m \to 1$. If additionally $\Lambda \sim c/L$ with fixed $c > 0$ and m grows polynomially in L, then $m(\log m)^2 \le (1 + o(1))L^2/(c \log 3)$. These are conditional scale restrictions, not an actual-cycle scaling law or a uniform lower bound for Λ. The bound also applies to threshold cycles with wrong parity, so it does not itself detect that defect.

Corollary 6.3c (minimum range at the first surviving count pair). An actual primitive cubic-band cycle with

$$(L, o, e) = (780239, 492276, 287963)$$

must satisfy

$$350000000 < m < 520000000. \tag{UC4}$$

Proof and numerical boundary. The lower bound is the certified descent input of Corollary 5.11. At these fixed counts,

$$3.4711981668 \cdot 10^{-6} < \Lambda < 3.4711981670 \cdot 10^{-6}.$$

At $m = 520000000$, outward interval evaluation makes the last right-hand side of (UC2) less than $3.230293215 \cdot 10^{-6}$. For fixed counts, A increases with m, while both positive factors in that bound decrease. Thus (UC2) fails at every larger minimum as well. ☐

The scalar comparison and its rational outward endpoints are reproduced in the repository's cycle-rank-curvature control report, distributed with the software. The finite upper-cell and geometric bounds in (UC2) are now verified in Lean, including the scaled scalar consequence. The monotonicity argument and interval evaluation for this numerical cutoff remain written proof and computation, respectively. The asymptotic consequence (UC3) also remains written. The conclusion concerns these exact counts and the cubic-height hypothesis. It does not exclude period 780239 or raise the descent floor.

The unresolved signed comparison. Return to an actual primitive cubic-band cycle. Suppose $o \geq 3$, let $k = e^{-1} \pmod{L}$ lie in $(L/2, L)$, and put $\ell = L - k$, $h = 2k - L$. Start the chronological orbit at m. Its states at times h, k, L have ranks $2, 1, 0$, respectively. Let $Q = [h, k)$ and $R = [k, L)$ denote their source-time arcs. Adding ℓ pairs every Q edge with an R edge whose source and target ranks are both one lower. No source crosses a branch cut, so the complete two arcs have the same branch word.

For an actual edge $x \to y$, put $N = x^3$ on O and $N = x$ on E, and define the positive unused upper-cell capacity

$$s = Z((y+1)^2) - Z(N), \qquad Z(t) = \log\log t.$$

Write s_Q, s_R for its arc sums and C_Q, C_R for the sums of $\eta(y)$. As η decreases and the target ranks are adjacent pairs,

$$0 < \varepsilon := C_R - C_Q \leq \eta(m) - \eta(M).$$

Telescoping the actual losses along the two arcs gives

$$s_R - s_Q = \varepsilon + \chi, \qquad \chi = \log \frac{\log c_2 \, \log m}{(\log c_1)^2}. \tag{UC5}$$

For $t = 0, 2$, define

$$H_t(y) = \log \frac{\log(2y - m + t) \, \log m}{(\log y)^2}.$$

The condition

$$H_0(c_1) + \varepsilon < s_R - s_Q < H_2(c_1) + \varepsilon \tag{UC6}$$

would be equivalent to $0 < c_2 - 2c_1 + m < 2$, impossible for three odd integers. This is an exact reformulation of the missing estimate, not an inequality that has been established.

The complete universal rank-envelope cap sums at the counts of Corollary 6.3c leave a relaxed signed-loss interval of approximately $[-1.307, 1.307] \cdot 10^{-6}$. At the boundary-scale first-triple control, the forbidden window has width of order 10^{-10}. That control uses $m_0 = 350000001$ only as a diagnostic scale: $O(m_0) = 6547900454916$ is even, so it is not an actual cycle minimum. No simultaneous realization of all integer cells is asserted.

Lemma 6.3a changes the odd-to-odd cap from $V_1(y)$ to $V_3(y)$, where $V_a(y) = Z((y+1)^2 - a) - Z(y^2)$. For $y > 1$,

$$0 < V_1(y) - V_3(y) = \int_{(y+1)^2-3}^{(y+1)^2-1} \frac{du}{u \log u} \leq \frac{1}{y^2 \log y}. \tag{UC7}$$

Its total additional effect on those envelopes is at most $L/(m_0^2 \log m_0) < 3.238 \cdot 10^{-13}$. It cannot repair this cap test's missing precision. Multiplying the actual arc identities likewise only exponentiates (UC5). Balanced rotation counts do not bound the variation of the actual unused capacities. What remains is an arithmetic restriction on their joint distribution at the shared integer states, strong enough to prove (UC6).

Appendix A. Lean names and trust boundary

The proof object is the import closure of `formal/Problems/JugglerPaper.lean`. From `formal/`, run `lake build Problems.JugglerPaper`, followed by `lake env lean AxiomCheckPaperA.lean` to inspect the cited declarations' dependencies. The exact toolchain is pinned in `formal/lean-toolchain`. The optional formalization map contains the detailed declaration-level explanations. Every name cited in this manuscript must be reachable from the paper barrel; the associated trust-boundary tests enforce that condition.

The core financing, transport, hug domination, and bounded-variation rotation estimates are Lean theorems. The displayed change of variables identifying the circle mean with C_*, the explicit extension of Theorem 5.8 beyond q_{13}, and the use of Rhin are human arguments. The descent floors, per-length comparisons, and Theorem 3.31's enumeration are computational inputs. The native digit scan is recorded separately. None of these distinctions is removed by compiling the barrel.

Text	Lean declaration or evidence boundary
itinerary semantics	`follows_iff_itinerary`, `image_eq_iterate`, `image_append`
Theorem 2.1	`image_monotone_of_follows`
Theorem 2.2	`power_bound_word`
Corollary 2.3	`power_bound_contracts`
Theorem 2.4	`global_defect_identity`
Theorem 2.5	`global_defect_eq_zero_iff_localsTight`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`
Theorem 2.6	`global_defect_append`
Corollary 2.7	`image_eq_start_defectRatio`
per-step slack	`one_plus_eta_lt_succ_sq`
Lemma 3.1	`odd_preimage_unique`
CycleMin	`CycleMin`
Theorem 3.2	`cycle_itinerary_formally_expanding`, `cycleMin_start_odd`, `cycleMax_start_even`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`
Lemma 3.3	`lower_growth_word`
Lemma 3.4	`oo_suffix_threshold`, `ooo_suffix_threshold`, `threshold_inherits_odd_append`, `cycle_last_even_interval`, `cycle_last_even_ne_odd_sq`, `no_cycle_odd_run_append_even`, `no_cycle_itinerary_ooe`
odd-run block	`oddEvenBlock`

Text	Lean declaration or evidence boundary
Lemma 3.5	`no_cycle_itinerary_oooeoe`, `no_cycle_itinerary_ooooee`
Theorem 3.6	`no_cycle_itinerary_length_le_six`
Lemma 3.7	`no_cycle_itinerary_ooooeoe`, `no_cycle_itinerary_oooooee`
Theorem 3.8	`no_cycle_itinerary_length_le_seven`, with `no_cycle_itinerary_ooeoooe`, `no_cycle_itinerary_oooeooe`
Lemma 3.9	`cycle_trailing_evens_lt`
Lemma 3.10	`lowerDenom_replicate_odd`, `odd_run_lower_growth`
Lemma 3.11	`no_follows_seven_odds_of_lt256`
Theorem 3.12	`no_cycle_itinerary_two_even_ee`, `no_cycle_itinerary_two_even_eoe`
Theorem 3.13	`no_cycleMin_gapped_three_even_ee`, `no_cycleMin_gapped_three_even_eoe`
Theorem 3.14	`no_cycle_itinerary_three_even_eee`, with `no_cycle_itinerary_oooooooeee`
Theorem 3.15	`no_cycle_itinerary_three_even_eoee`
Theorem 3.16	`no_cycle_itinerary_three_even_eooee`
Theorem 3.17	`no_cycle_itinerary_three_even_eoooee`
Theorem 3.18	`no_cycle_itinerary_three_even_eeoe`
Theorem 3.19	`no_cycle_itinerary_three_even_eoeoe`
Theorem 3.20	`no_cycle_itinerary_three_even_eooeoe`
Theorem 3.21	`no_cycle_itinerary_gapped_three_even_ee`, `no_cycle_itinerary_gapped_three_even_eoe`
Theorem 3.22	`no_cycle_itinerary_even_count_le_three`
Corollary 3.23	`cycle_itinerary_length_ge_eleven`
Lemma 3.24	closed form of Lemma 3.10; no separate Lean name
Lemma 3.25	generalizes `cycle_trailing_evens_lt` past pure even runs
Theorem 3.26	Corollary 3.27's ten rows are Theorems 3.12–3.21 above
Lemma 3.28	`absorb_odd_step`, `cross_mul_pow`, `odd_run_ge` (`O7EEEEGap.lean`, now imported by `Problems.JugglerPaper`); X_7, Y_7 are that module's 6177 and 3990
Theorem 3.29	`no_cycle_itinerary_oooooooeeee` is the row $u = EEEE$, $a = 7$

Text	Lean declaration or evidence boundary
Theorem 3.31	arithmetic core Lean (`EvenCountEight.lean`): per-run cap `runCapConst_gt`, `runCapConst_lt`; the table is the floor, `runCap_eq_floor`, `runCap_antitone`, tuples `runCap_tuple_three`–`runCap_tuple_seven`; period floor `period_ge_22_of_even_count`, sharp by `expansion_holds_at_22` and `expansion_fails_below_22`. The enumeration of 353044 canonical forms (`run_suffix_law.closure`, Python) is verified computation, not Lean
Theorem 3.33	Exact parity and sorted order: `cubicBand_sorted_rotation`; coprimality and actual odd prefixes: `cubicBand_mechanical_itinerary`
Proposition 3.34	`thresholdMap_invariant`, `thresholdMap_ne_self`, `thresholdMap_exists_primitive`, `threshold_sorted_rotation`, `thresholdMap_eq_floorPower_iff`
Theorem 3.35	`threshold_all_periodic_rank_rotation`, `threshold_periods_equal`; `rankRotation_orbit_iff_residue`, `rankRotation_orbit_representatives`, `rankResidue_parameterization`, `rankRotation_orbit_upper_card`, `rankResidue_interlaces`; prefixes: `rankRotation_mechanical_prefix`
Proposition 3.36	Exact-cycle realization: `threshold_cycle_grid`, `cubicBand_cycle_grid` with conclusion `RealizedGridBounds`; the illustrative asymptotic width comparison remains written analysis
Proposition 3.37	`cubicParityProject_greatest`, `cubicRounding_eq_or_pred`, `cubicRounding_exists_primitive`, `cubicRounding_real_loss`, `cubicRounding_finite_invariant_rotation`; modular consequences as in Theorem 3.33

Text	Lean declaration or evidence boundary
Proposition 3.38	`branchOffset_same_branch_difference`, `branchOffset_same_branch_smooth_gap`, `branchOffset_nearest_even_gap`; kernel-checked witness: `branchOffsetCycle_cells`, `branchOffsetCycle_primitive`, `branchOffsetCycle_bounds`, `branchOffsetCycle_counts`, `branchOffsetCycle_rank_rotation`, `branchOffsetCycle_mechanical_prefix`
Theorem 3.39	Exact actual-cycle seam: `cycleMin_exact_return_seam`; integer height gap: `cubic_return_height_algebra`, `cycleMin_height_strip`, `cycleMin_all_states_height_strip`; top-strip obstruction: `threshold_cycle_wrong_parity`. The maximum-odd-integer restatement and the sharper fractional bound for t are written consequences
Appendix E.1	OE cell: `oe_eq_iff`; OOE floor and odd endpoint: `ooe_one_integer`, `ooe_odd_maximal`; hidden-parity family: `oe_perfect_power_hidden_odd`. General root iteration, OOEOE projection and fixed-word asymptotics remain written
Appendix E.2	Exact one-step rank returns: `left_subtractive_first_return`, `right_subtractive_first_return`; word statistics: `left_word_statistics`, `right_word_statistics`; guards: `follows_append`. Full accelerated tower partition remains written
Appendix E.3	Square carry: `baseline_bounds`, `baseline_zero_iff`, `baseline_one_iff`, `baseline_two_iff`, `square_guard_iff`; genuine family block: `ooeFamily_juggler_block`. The unbounded real quotient-substitution error remains written
Theorem E.4	`ooeFamilyReturn_mod_fortyeight`, `ooeFamilyReturn_valuation_drop`, `ooeFamily_juggler_chain_bound`, `ooeFamily_no_infinite_juggler_chain`. The additional 3-adic identity is written; six terminating traces are finite computations

Text	Lean declaration or evidence boundary
Appendix E.5	Initialized record: `record_initializes`, `endpoint_validation`; correction and signed floors: `exact_remainder_correction`, `exact_quotient_gap`, `corrected_quotient_integer`; executable recovery and hidden guards: `recoverPeak_eq`, `recoverPeak_guard_iff`. The further OOEOE composition is written
Appendix E.6	Every-modulus witness: `guardResidueFamily_every_modulus`; cells and parities: `guardResidue_ooe_traces`, `guardResidue_nat_parities`; exact remainder and valuation: `guardResidue_first_remainders_zero`, `guardResidue_aggregate_valuation`; record collision and classifier obstruction: `guardResidueFamily_record_collision`, `guardResidueFamily_no_record_classifier`; common domain and threshold edges: `guardResidue_common_band_and_section`, `guardResidue_threshold_blocks`. The general positive-b construction and supplementary bookkeeping remain written
Theorem 3.40	Actual-cycle gap and height bounds: `dc_cycle_gap`, `lr_cycle_gap`, `dc_cycle_height`, `lr_cycle_height`. The last two include the displayed real bounds and exact integer endpoints, from the stated minimum and cubic-band hypotheses
Theorem 3.40, cycle and threshold consequences	Ordinary cycle predicates: `cycleMin_dc_height`, `cycleMin_lr_height`. Wrong parity in the excluded threshold strips: `threshold_cycle_dc_wrong_parity`, `threshold_cycle_lr_wrong_parity`
Appendix F.1	Exact transported loss: `loss_exact`; concave-tail bound: `loss_lt_budget`; common-floor paired estimate: `paired_bound`; exponent/count identity: `exponent_eq_counts`
Appendix F, finite even-gap bookkeeping	`even_transfers_sum`, `even_transfers_positive`: each strict even-gap step consumes at least two, with every transfer retained as a hypothesis

Text	Lean declaration or evidence boundary
Appendix F.5, induced word counts	`count_determinant`, `count_coprime`, `expanded_count_gcd`: the induced count matrix has determinant one and preserves the gcd of the expanded totals
Appendix F.2–F.3	Unconditional word bounds: `c_loss`, `w_loss`; gap contraction: `c_contract`, `w_contract`. Required growth: `ac_grows`, `d_grows`, `v_grows`; the sharper written helper errors and the earlier 2^{15} growth cutoff are not needed in these formal proofs
Appendix F.2–F.4	Actual section: `periodicExtrema_return_model`; forced batches: `dc_rank_stages`, `lr_rank_stage`; guarded pair placement: `periodicExtrema_dc_transfers`, `periodicExtrema_lr_transfers`; height transport: `height_of_power_gap`
Proposition 3.41 and Appendix F.5	Guarded factorization: `induced_terminal_actual_factorization`. Primitive termination: `primitive_terminal`. Full actual-orbit construction, global adjacency, cut/extrema identification and strict mixed gap for $m \geq 3$: `periodicExtrema_terminal_cut`, `periodicOrbit_terminal_cut`. The periodic-set interface requires connectedness; the orbit interface derives it. Local cell inequality: `mixed_gap`
Appendix F.6	Exact necessary threshold for the stated sufficient certificate: `certificate_requires_large_minimum`. Its near-unit asymptotic interpretation remains written; this is not a counterexample to actual paired contraction
Lemma 3.21b	canonical run form; Theorem 3.2
Lemma 3.21a	the case split of Theorem 3.22
Lemma 4.1	`log_le_two_log_add`
Lemma 4.2	`log_step_even`, `log_step_odd`
Lemma 4.3	`cycleMin_log_envelope`
Theorem 4.4	`cycleMin_finance`
Corollary 4.4c	`cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`

Text	Lean declaration or evidence boundary
Lemma 4.4b	`packingR_step` (the step in o is the constant $2\alpha - 1 - 1/2n$), `alpha_lt_half` ($\alpha < 1/2$ once $t \geq 2n$), `two_n_add_one_lt_rpow_three_halves` (which $t = \lfloor n^{3/2} \rfloor$ gives for $n \geq 12$), `packingR_step_neg`, `theta_strictMono`, and `comparison_fails_upward` for an arbitrary positive coefficient — 6/5 here and 1 in Theorem 4.4 (`OddCountMonotone.lean`)
Corollary 4.5	See the accompanying formalization map.
Theorem 4.6	certified identity `cycleMin_defect_finance`, per-step losses `log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, invariants `cycleMin_log_le_weight`, `cycleMin_charge_prefix` (`DefectFinance.lean`); the numeric table is verified computation
Theorem 4.7	See the accompanying formalization map.
Theorem 4.8	run-type table; verified computation, not Lean
Proposition 4.9	`run_survivor_unimodular`, `run_survivor_seed_F2`, `run_survivor_seed_F3`, `three_pow_step_gt_two_pow_step`, `runSurvivors_length`
Theorem 4.10	`cycleMin_gap_transfer`; abstract length bound `cycleMin_length_of_gap` (`GapTransfer.lean`)
Corollary 4.11	`cycleMin_length_of_gap` with Rhin's measure [15] as hypothesis; the transcendence input is classical, not Lean
Proposition 5.1	laboratory floor; certified computation, not Lean
Theorem 5.2	raised cutoff; verified computation, not Lean
Theorem 5.3	transport inequality `cycleMin_transport`, per-step losses `log_floorPower_even_ge`, `log_floorPower_odd_ge` (`WalkTransport.lean`); §5.2 consequence `cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge` (`WalkChargeMax.lean`)

Text	Lean declaration or evidence boundary
Theorem 5.4	combinatorial core `hugOdds_le_of_admissible`; cycle-itinerary domination `cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`; charge maximisation `stateCharge_antitone`, `hug_charge_maximal` (`WalkChargeMax.lean`); strict uniqueness `stateCharge_strictAnti`, `stateCharge_inj`, `hug_charge_unique` — an admissible profile attaining the hug charge *is* the hug profile
Proposition 5.5	Lean proves convergence of the finite hug average to `circleMean` by `denjoy_koksma_blocks` and the Laplace bounds `inv_sq_le_quad`, `rotation_average_le`, `rotation_average_lt`, `rotationAverage_le`, `rotationAverage_lt`, `rotationAverage_gap`; the elementary change of variables identifying `circleMean n'` with the displayed `rotationAverage (log n')` remains prose
Lemma 5.6	itinerary identity `budgetedWord_eq_hugWord`, `hugOdds_pow_ge`, `hugOdds_pow_lt`, `hugOdds_pow_gt`, `hugOdds_least`; rotation identification (`HugRotation.lean`) `hugOdds_eq_ceil`, `hugOdds_eq_sub_floor`, `hugEvens_eq_floor`, `hugLetter_iff_floor_step`, `hugWalk_eq_fract`, and the Birkhoff reading `periodicObservable_hugWalk`
Theorem 5.7	`value_sub_mean_le_variation`, `sum_eVariationOn_Icc`, `denjoy_koksma_abstract`, `orbitCell_inj`, `orbit_mem_cell`; detailed scope in the formalization map
Theorem 5.8	`ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, `theta_digitSum_le`, `greedyDigitSum_le`; detailed scope in the formalization map
Theorem 5.9	kill template `cycleMin_hug_kill_criterion` (`DefectFinance.lean`); the per-length kill table is verified computation

Text	Lean declaration or evidence boundary
Proposition 5.12	`fanLength`, `fanOdd`, `fanLambda`, affine step `fanLambda_affine`, negativity `fan_step_pow`, `fanLambda_step_neg`, monotonicity `fanLambda_strictAnti`, endpoints `fanLambda_55_pos`, `fanLambda_56_neg` (these *are* `theta_sandwich_lower` and `theta_sandwich_upper`), length `fan_positive_iff`, and `fan_frontiers`, `fan_endpoint`, `fan_past_endpoint` (`FanLaw.lean`)
Propositions 5.8b, 5.8c	the two forced walk letters `walk_first_letter_odd`, `walk_second_letter_odd`, `step_lt_two` (`FanLaw.lean`); the relaxation and flatness measurements are verified computation
Observation 5.13, Corollary 5.14	finite fitted exponents and the separately evaluated conditional bound; computation, not Lean
Corollary 5.10	second floor and kill table; verified computation, not Lean
Corollary 5.11	third floor and kill table; verified computation, not Lean
Lemma 6.3a	`cube_add_one_ne_odd_succ_sq`, `floorPower_odd_image_upper_gap` (`UpperSquareGap.lean`); the integer upper-square gap only
Proposition 6.3b	Exact finite and closed charges: `power_cells_grid_charge`; scaled consequence: `power_cells_scaled_charge`; actual and threshold interfaces: `cubicBand_cycle_upper_charge`, `threshold_cycle_upper_charge`. Ordinary-orbit and minimum-based itinerary extraction at the true least period: `periodicOrbit_upper_charge`, `cycleMin_upper_charge`. The abstract statements retain both cell faces, rank translation, and transitivity; the interfaces derive these from actual connected cycles. The asymptotic consequence (UC3) remains written
Corollary 6.3c	Written monotonicity proof and outward interval computation at the stated counts; this numerical cutoff is not a Lean-certified comparison

Text	Lean declaration or evidence boundary
Section 6.3 signed comparison	Written transport identities and interval controls; no inequality forcing the forbidden window is established
Propositions E.7–E.8	Written residue and inverse-polynomial finite-difference proofs; no uniform theorem is inferred from finite trajectory checks
short certificates (Section 6)	`even_finiteProgress`, `odd_even_finiteProgress`
no certificate ⇒ odd-to-odd	`no_finiteProgress_implies_odd_odd`
induction to 1	`reachesOne_of_all_finiteProgress`
four-block chain	`four_block_pe_1999`

Appendix B. Admissible lengths

The record lengths of $\gamma(L)$ through 10^5, with the parity 6/5 bound $n_{\max}$ of Section 4, are

L	$o_{\min}$	$n_{\max}$
1	1	2
3	2	7
11	7	25
19	12	133
84	53	2323
569	359	23568
1054	665	788014
25781	16266	26254995
50508	31867	162848324

At the verified descent floor $N_0 = 10^6$ the first seven rows are excluded. The set $\mathcal{E} = \mathcal{E}(10^6)$ is defined by Proposition 4.4a: for each $1 \le L \le 10^5$, compute $o_{\min}(L)$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > 10^6$. This produces 141 lengths. The first few are

$$25781,\ 26835,\ 27889,\ 28943,\ 29997,$$

and a typical combination is $26835 = 25781 + 1054$. The last is 99561. The complete list is the `lengths` array of `data/research/juggler/cycle_finance/exceptions_parity.json`. The SHA-256 of that array, serialized as a JSON list of integers with no spaces, is `dd71aa1527656ba51cb031bafa5497f7bfdbbc43151ffba2c595793326bf7944`. The SHA-256 of the whole file `exceptions_parity.json` is `6b4eec79295b70cdeb9f7db677b7fd57bcb9bd1b51177e1e74aad7bd6e2262ff`. The 10^6 longest first-passage figures of Proposition 1.3 (253 steps at seed 78901) are the opening chunk of the laboratory-floor certificate, `data`

`/research/juggler/cycle_finance/floor_verify/N26254995/chunks/3_250002.json` ($3 \le n \le 250002$; SHA-256 `6303b62c9b1819deaf9715338f84899c1d75eb50dcab850a7b8fb28874ec19bc`). The file `floor.json` in the same directory as `exceptions_parity.json` is a later $n_{\mathrm{top}} = 2 \cdot 10^6$ companion and is not the 10^6 certificate. The parity table is written by `research.juggler_sequence.cycle_finance.write_parity_artifacts`. The table gives exact crossings of the stated 6/5 comparison. The historical guarded generator returned the conservative upper values 3 and 162848325 at $L = 1$ and $L = 50508$, respectively; these must not be confused with the exact crossings 2 and 162848324. Neither correction changes the stated period bounds.

The run-type table of Theorem 4.8 is `data/research/juggler/cycle_finance/budget_opt.json`. The 42 excluded lengths are `killed_by_budget`; their SHA-256, serialized as a JSON list of integers with no spaces, is `9d108776d6dc5dc1ae2594058850463cd2d3995cc57b9811471f08e3b818b90a`. The complementary 99 lengths form the table $\mathcal{E}_{\mathrm{run}}$ under Theorem 4.7's primitive and no-EE hypotheses. Their SHA-256, in the same serialization, is `9e2098923ccb39933630b116133a3fc2ddaf98ace4eb76dbab9b5ab9f6e604e6`. The first few survivors remain

$$25781,\ 26835,\ 27889,\ 28943,\ 29997;$$

the last survivor is 99477. The three families and the unimodular basis are Proposition 4.9. The finite leftover tables of Section 3 are the Lean `decide +kernel` evaluations named in Appendix A (`LeftoverEval.lean`, `LeftoverShort.lean`, `LeftoverFamilies.lean`). The lattice arithmetic of Proposition 4.9 is `RunSurvivorLattice.lean`.

The laboratory instance of Section 5 has its own artifacts. The first-passage certificate of Proposition 5.1 is `data/research/juggler/cycle_finance/floor_verify/N26254995/certificate.json`; the SHA-256 of its chunk records is `cbcbb540dd860b775d2a3b4351f7cf779609104d3aaf30c342aed6b87f36c9dc`. The parity survivor list of Theorem 5.2 has SHA-256 `d5946efdfac95c715ea46c81979a0eeaf40ad8a1dc893d161bab84ffa7f2afd9`. The walk-charge kill table of Theorem 5.9 is `data/research/juggler/cycle_walk_charge/survey.json`; the SHA-256 of the walk-alive list is `225d76ad12802a690934d01e2d37b3418865441a6825a015dd883c989d8942ec`. The second-floor certificate of Corollary 5.10 is `data/research/juggler/cycle_finance/floor_verify/N162849448/certificate.json`; the SHA-256 of its chunk records is `35d9755ce52a225a161b009d0f3674b24a5d0add16035f09ea83cf3880414ae4`. Its parity survivor scan is `data/research/juggler/cycle_walk_charge/new_floor_parity_leftovers.json`, and the 15 kill records under `data/research/juggler/cycle_walk_charge/new_floor_kills/` have SHA-256 `148180cbbfba93985b7a1be455fee16db97816f1534b1feca5a4740d475aeda0` (concatenated in length order); the direct non-kill record for the survivor 478245 (margin 0.4334) is stored alongside. The third-floor certificate of Corollary 5.11 is `data/research/juggler/cycle_finance/floor_verify/N350000000/certificate.json`; the SHA-256 of its chunk records is `c57f5ccc9bce980f478dafec00c449050a5e217a4f0f3e100916c41dacbe7472`. Its parity survivor scan is `data/research/juggler/cycle_walk_charge/N350000000_parity_leftovers.json`, and the 10 kill records under `data/research/juggler/cycle_walk_charge/N350000000_kills/` have SHA-256 `d16ccfed52757d4a44368a6549a8149ccbc926472737276c577912346db854ab` (concatenated in length order); the direct non-kill record for the survivor 780239 (margin 0.6049) is stored alongside. The exact-integer

CPU computation with guarded comparisons is the authoritative record for the first- and second-floor claims in this note. The third-floor kill table is the GPU floating-point certified comparison (`research.juggler_sequence.cycle_walk_charge_gpu`), the same IEEE-double kernel previously checked against stored CPU records to relative error below 10^{-13}; its performance figures live in the repository documentation. The probes are `research.juggler_sequence.cycle_walk_charge` (transport and kill table), `cycle_walk_ostrowski` (certified quotient sandwich and block envelope), and `cycle_walk_window` (window scan); artifacts live under `data/research/juggler/cycle_walk_*`.

Reproducibility of the second-floor kill table. From a fresh checkout of the repository, `pip install -e .` followed by, for a survivor length L,

```
python -m research.juggler_sequence.paper_a_audit
```

runs the Paper A arithmetic audit when executed from the repository root. It recomputes selected exact parity thresholds, compares stored walk margins, checks the finite fan identities, and checks the constants used in Corollary 4.11. It does not replay all descent trajectories. For a direct per-length CPU report, save the following as a script and run it from the repository root after installing the project:

```
import argparse
from research.juggler_sequence.cycle_walk_charge import certified_report

parser = argparse.ArgumentParser()
parser.add_argument("length", type=int)
parser.add_argument("floor", type=int)
args = parser.parse_args()
print(certified_report(args.length, args.floor))
```

For example, `python report_length.py 478245 162849448` evaluates the non-excluded endpoint of Corollary 5.10. The committed third-floor records use `cycle_walk_charge_gpu.gpu_certified_report` with floor 350000000, and therefore require the corresponding GPU runtime. Do not use its `--write` option to regenerate a different floor: that option selects a fixed output directory. The precise software version, dependency environment, and archived data must accompany a release.

Appendix C. Exact floor defect

This appendix records the exact identity behind Theorem 2.2. The present note uses only the nonnegativity $\Delta_w(n) \geq 0$. A nontrivial itinerary-dependent lower bound on Δ_w is left for future work.

For a single branch,

$$x^h = J(x)^2 + \rho(x), \qquad h = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write $\operatorname{gap}(a, \rho, h) = (a + \rho)^h - a^h$. The *global defect* $\Delta_w(n)$ is the terminal value of the resulting power-gap recurrence.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. Induct on w. The empty itinerary is $n = n + 0$. An even letter substitutes $x = J(x)^2 + \rho(x)$ into the inductive identity and lifts the new remainder through 2^ℓ. An odd letter cubes the identity and then substitutes $x^3 = J(x)^2 + \rho(x)$. Each step adds a nonnegative power-gap. □

For a one-letter illustration, $n = 3$ and $w = O$ give $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$.

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent: $\Delta_w(n) = 0$; every local remainder along w vanishes; and $\left(J^{|w|}(n)\right)^{2^{|w|}} = n^{3^{\#O(w)}}$. In that case w is monochrome: either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed itinerary therefore has $\Delta_w(n) > 0$.

Proof. Theorem 2.4 gives the first and third items. A power-gap vanishes if and only if its addend vanishes, so zero defect forces every remainder to vanish, and conversely. Vanishing remainders preserve parity, hence monochrome itineraries, and unique factorization produces the two power towers. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then $\Delta_{uv}(n)$ is the sum of two power-gaps, one lifting $\Delta_u(n)$ through $3^{\#O(v)}$ and one lifting $\Delta_v(m)$ through $2^{|u|}$. In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$, and every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Apply Theorem 2.4 to u, to v, and to uv, and expand the two power-gaps. □

Composition is polynomial, not additive. The Lean form is `global_defect_append`.

Corollary 2.7 (cycle surplus). If w is a cycle itinerary at n, then $\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$ exactly.

Proof. Theorem 2.4 with $m = n$. □

A mixed itinerary has strict total defect, but the relative slack of a single letter tends to 0 with the state, so no uniform local tax exists. Theorem 4.4 does not use these identities.

Appendix D. Family exclusions

This appendix records the small-cycle censuses (Lemma 3.5, Theorem 3.6, Lemma 3.7, Theorem 3.8) and the family-by-family exhaustion used by Lemma 3.21a and Theorem 3.22. Each geometry is killed by a next-square obstruction, by long odd-run growth against a last-even one-step preimage, or by a finite exceptional window.

Lemma 3.5 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle itinerary at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ because $(1 + 1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n+1) \leq 257\,n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $2 \le n < 256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_oooeoe` and `no_cycle_itinerary_ooooee` (Appendix A).

Now suppose $OOOOEE$ is a cycle itinerary at $n \ge 256$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on $OOOO$ gives $n^{81} \le 2^{130} z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $OOOEOE$ is a cycle itinerary at $n \ge 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on OOO gives $n^{27} \le 2^{38} z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-preimage bound $y^3 < (n+1)^4$. Write $A = n+1 \ge 257$. We claim $(y+1)^3 < 2A^4$. If $y \le A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \ge 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \le yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

Theorem 3.6 (small-cycle census). No itinerary of length at most six is a cycle itinerary at any $n \ge 2$.

Proof. Rotating a cycle itinerary by one letter moves the base point one step along the trajectory and yields another cycle itinerary; every state of the cycle is at least 2, because a trajectory that reaches 1 stays at 1 and cannot return to a start $n \ge 2$.

If every letter is odd, the start is odd, hence $n \ge 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \ge 3$, since $x^3 \ge (x+1)^2$. The trajectory ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle itinerary vE of the same length based at a cycle state $m \ge 2$. It therefore suffices to exclude even-terminating cycle itineraries of length at most six.

By Theorem 3.2(i) a cycle itinerary is formally expanding. No even-terminating itinerary of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE. If $m \ge 5$ realizes OO, Lemma 3.4(i) gives $J^2(m) \ge (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E, excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E, excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E. The itinerary $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this itinerary be a cycle itinerary at some start, and rotate to a cycle minimum $m \ge 2$. The three alignments of the two even letters are $OOEOOE$, $OEOOEO$, and $EOOEOO$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE: the first image is even and strictly below m^2, whereas every even state after a cycle minimum is at least m^2, as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \ge 3$. The prefix

OOE must be realized. If $m = 3$, then $3 \to 5 \to 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE. Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y, and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. □

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle itinerary at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128} n^{128}$. The finite comparison $2^{422} 15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115} n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_ooooeoe` and `no_cycle_itinerary_oooooee` (Appendix A).

Now suppose $OOOOOEE$ is a cycle itinerary at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix $OOOOO$. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on $OOOOO$ gives $n^{243} \leq 2^{422} z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose $OOOOEOE$ is a cycle itinerary at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130} z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters OE give the odd-preimage bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. □

Theorem 3.8 (small-cycle census through length seven). No itinerary of length at most seven is a cycle itinerary at any $n \geq 2$.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd itinerary cannot return, and every mixed cycle itinerary rotates to an even-terminating cycle itinerary based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle itineraries of length seven.

By Theorem 3.2(i) a cycle itinerary is formally expanding. The even-terminating expanding length-seven words are exactly O^6E, EO^5E, OEO^4E, $OOEO^3E$, O^3EO^2E, O^4EOE, and O^5EE. Lemma 3.4(v) excludes O^6E. The itinerary EO^5E rotates one step onto O^5EE, and OEO^4E starts OE, so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE; both leftovers are excluded by Lemma 3.7.

It remains to exclude the cyclic class containing $OOEO^3E$ and O^3EO^2E. At a cycle minimum m, Theorem 3.2 forces the orientation to start OO and end E. The two possible orientations in this class are exactly those two words; a rotation can interchange them. In the first orientation, the internal even state is followed by OOO, so Lemma 3.4(ii), valid at $m \geq 3$, contradicts the last-even preimage interval. In the second orientation, the corresponding OO bootstrap applies at $m \geq 5$. The remaining odd minimum $m = 3$ follows $3 \to 5 \to 11 \to 36 \to 6$: the E at 36 is realized, but the next prescribed O is impossible at the even state 6. Both minimum-based orientations are excluded. □

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle itinerary at n.

Proof. First, if $n \geq 256$, then

$$n^{3^{k-2}} > 2^{G_{k-2}}(n+1)^{2^k}.$$

The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \geq 6$, cubing both sides produces

$$n^{3^{k-1}} > 2^{3G_{k-2}}(n+1)^{3\cdot 2^k}.$$

The recurrence of Lemma 3.10 gives $G_{k-1} = 3G_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \geq 256$.

Now suppose $O^{k-2}EE$ is a cycle itinerary at such an n. Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \leq 2^{G_{k-2}} z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{G_{k-2}}(n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle itinerary at such an n. Write $z = J^{k-3}(n)$ and $y = \lfloor \sqrt{z} \rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3G_{k-3}}(y+1)^{3\cdot 2^{k-2}}$. The last letters OE give the odd-preimage bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $G_{k-2} = 3G_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{G_{k-2}}(n+1)^{2^k}$.

For $2 \leq n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE, O^5EOE, and O^6EOE fail to return on the same 254-start window; this is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_two_even_ee` and `no_cycle_itinerary_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \geq n$. In the first family the remainder after that letter is O^bEE with $b+2 \geq 6$; in the second it is O^bEOE with $b+3 \geq 6$. The trailing-even and last-odd one-step preimages of those remainders are measured against the cycle start n. Combined with Lemma 3.10 at the remainder start y, the same algebra as in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{G_{\ell-2}}(n+1)^{2^\ell} \leq 2^{G_{\ell-2}}(y+1)^{2^\ell},$$

where ℓ is the remainder length. If $y \geq 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y.

If $y < 256$, then $n \leq y < 256$. A gapped leftover of total length at least 17 has $a \geq 7$ or $b \geq 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \leq a \leq 6$ and $b \leq 6$ fail to be minimum-based cycle itineraries on the window $2 \leq n < 256$; this is the Lean `decide +kernel` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). □

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The itinerary O^aEEE is not a cycle word at n.

Proof. Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{G_a}(n+1)^{2^{a+3}}$ on any such cycle itinerary. For $n \geq 128$ the opposite comparison

$$n^{3^a} > 2^{G_a}(n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330}(n+1)^{512}$. Indeed, for $n \geq 128$ one has $(n+1)^{512} < (129/128)^{512} n^{512} < e^4 n^{512} < 64\, n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \geq 128 = 2^7$, the left side is at least 2^{1519}. Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \geq 128$.

For $2 \leq n < 128$, the case $a = 6$ is a table of evaluations of $OOOOOOEEE$: at every such start the itinerary fails to return. The same finite check is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_oooooeee` (Appendix A). For $a \geq 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEOEE is not a cycle word at n.

Proof. First let $n \geq 4$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \leq 4p^2 < 4(n+1)^8$. For $n \geq 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even one-step preimage at z then gives $z < (y+1)^2 \leq (n+1)^6$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{G_a}(n+1)^{6 \cdot 2^a}.$$

For $a = 5$ and $n \geq 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \geq 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the itinerary fails to return; these are the Lean `decide +kernel` evaluations behind `no_cycle_itinerary_three_even_eoee` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. □

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOOEE is not a cycle itinerary at n.

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix O^aEOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \leq 2^{10}p^4 < 2^{10}(n+1)^{16}$. For $n \geq 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even one-step preimage at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{G_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$.

For $2 \le n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_three_even_eooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.17 (mixed bunched family $EOOOEE$**).** Let $a \ge 3$ and $n \ge 2$. The itinerary $O^a EOOOEE$ is not a cycle itinerary at n.

Proof. First let $a \ge 4$ and $n \ge 3$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix $O^a EOOO$. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \le 2^{38} p^8 < 2^{38}(n+1)^{32}$. For $n \ge 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even one-step preimage at z then gives $z < (y+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{G_a}(n+1)^{2^{a+2}}.$$

For $n \ge 256$ and $a \ge 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \ge 256$. The same two-even one-step preimage gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38}(y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38}(n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38} \cdot 39^{16}$. The numerical comparison $2^{38} \cdot 39^{16} < 24^{27}$ contradicts $n \ge 24$.

If $y \ge 39$, the numerical comparison $40^{27} < 2 \cdot 39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39}(n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026}(y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624}(n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650}(n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \le n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_three_even_eoooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.18 (mixed bunched family $EEOE$**).** Let $a \ge 5$ and $n \ge 2$. The itinerary $O^a EEOE$ is not a cycle itinerary at n.

Proof. First let $n \ge 4$, and write $z = J^a(n)$ and y for the last odd letter of $O^a EEOE$. The suffix EOE is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are even, so $z < (y+1)^4$. For $n \ge 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle itinerary would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{G_a}(n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \ge 314$, and already for $a = 6$ and $n \ge 16$.

For $2 \le n < 314$ and $a = 5$, and for $2 \le n < 16$ and $a = 6$, the itinerary fails to return; these are the Lean `decide +kernel` evaluations behind `no_cycle_itinerary_three_even_eeoe` (Appendix A). For $a \ge 7$ and $n < 256$, Lemma 3.11 applies. For $a \ge 6$ and $n \ge 16$, the tail of the previous paragraph applies. □

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOEOE is not a cycle itinerary at n.

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO. The last-odd one-step preimage and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{G_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_three_even_eoeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. □

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOEOE$ is not a cycle itinerary at n.

Proof. First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10 at length two yields $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y, and the same small/large split applies.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `decide +kernel` evaluation behind `no_cycle_itinerary_three_even_eooeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. □

Theorem 3.21 (gapped leftovers as cycle itineraries). Let $n \geq 2$. No cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Every cycle itinerary has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a + b + 3$. The rotation by $k = 0$ is the original itinerary, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a + 1$ is the bootstrap itinerary O^bEEO^aE. That itinerary has an internal even letter and last gap at least 2. If $a \geq 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \geq 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \geq 4$. The rotation by $k = a + b + 2$ begins with the last even letter of w, which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a + b + 4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a + 1$ is O^bEOEO^aE, excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting an odd letter; the rotation by $k = a + b + 2$ begins OE, forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. □

Appendix E. Absolute return cells and parity obstructions

This appendix consolidates the short-return calculations used in Section 3.11 and the precise limitations found when trying to extend them. A prescribed word acts by the displayed branch functions even when an intermediate parity is wrong. Its *guard* is the conjunction of the required source parities at every letter. Statements about prescribed endpoints must therefore be distinguished from statements about actual Juggler trajectories.

E.1. Exact short-return endpoints

Proposition E.1 (root collapse and odd endpoints). For positive integers N, r, s,

$$\left\lfloor \left\lfloor N^{1/r} \right\rfloor^{1/s} \right\rfloor = \left\lfloor N^{1/(rs)} \right\rfloor.$$

Consequently

$$F_{E^k}(x) = \lfloor x^{1/2^k} \rfloor, \qquad F_{OE^k}(x) = \lfloor x^{3/2^{k+1}} \rfloor.$$

If $H = \lfloor x^{9/8} \rfloor$, then

$$F_{OOE}(x) \in \{H - 1, H\}.$$

In particular an odd OOE endpoint equals $Q(x^{9/8})$. For positive odd x, an odd $OOEOE$ endpoint similarly equals $Q(x^{27/32})$.

Proof. For each integer $j \geq 0$, the nested root is at least j if and only if $N \geq j^{rs}$, proving root collapse. Put $u = \lfloor x^{3/2} \rfloor$. Then $F_{OOE}(x) = \lfloor u^{3/4} \rfloor$. Concavity, or subadditivity of the 3/4 power, gives

$$0 \leq x^{9/8} - u^{3/4} < 1.$$

The two possible floors follow; exactly one of two consecutive integers is odd.

For $u = F_{OOE}(x)$ and $x \geq 5$, we have $u \geq 6$ and $0 \leq x^{9/8} - u < 2$. Concavity now gives

$$0 \leq x^{27/32} - u^{3/4} < \frac{3}{2} u^{-1/4} < 1,$$

since $u \geq 6 > 81/16$. Taking the final floor again leaves fewer than two units of total error. The odd-output conclusion follows. The remaining odd sources are 1, which is fixed, and 3, whose $OOEOE$ endpoint is the even integer 2. □

Endpoint compression does not determine the guard. For every odd $s \geq 3$, the prescribed block

$$s^4 \overset{O}{\longmapsto} s^6 \overset{E}{\longmapsto} s^3$$

has odd endpoints, but the source of its E step is odd. It lies in the threshold band with $b = s^3$. Likewise $13 \mapsto 46 \mapsto 311 \mapsto 17$ has odd endpoints and the stated OOE endpoint identity, but both hidden guards fail.

For completeness, a fixed prescribed word also has an exact floor-loss identity. Let its branch exponents be α_j, its successive states $n_j = \lfloor n_{j-1}^{\alpha_j} \rfloor$, its prefix products $p_j = \prod_{i \le j} \alpha_i$, and

$$q_j = p_k/p_j, \qquad \epsilon_j = n_{j-1}^{\alpha_j} - n_j.$$

Since $n_{j-1}^{p_k/p_{j-1}} = (n_j + \epsilon_j)^{q_j}$, telescoping gives

$$x^{p_k} - n_k = \sum_{j=1}^{k} \big((n_j + \epsilon_j)^{q_j} - n_j^{q_j} \big).$$

If every nonempty proper prefix has $p_j > p_k$, concavity gives the strict upper bound

$$1 + \sum_{j<k} q_j n_j^{q_j - 1}.$$

When the sum is at most 1, an odd final output is the odd projection of x^{p_k}. Each such fixed word eventually has this property as x grows, because its finitely many negative powers tend to zero. This supplies no uniform threshold for words whose length grows, and does not certify internal parities. This general fixed-word asymptotic statement remains a written proof.

E.2. Euclidean return words retain the complete guard

For the rank rotation

$$T(i) = \begin{cases} i + b, & 0 \le i < a, \\ i - a, & a \le i < a + b, \end{cases}$$

suppose the two branches execute chronological words A, B. Inducing on the prefix of length a, when $a > b$, gives

$$(a, b; A, B) \longmapsto (a - b, b; A, AB).$$

The first branch returns in one step; the second first leaves the prefix and then returns. For $b > a$, the prefix of length b instead gives

$$(a, b; A, B) \longmapsto (a, b - a; AB, B).$$

When $a = b$, every rank in the prefix of length a returns by AB and is fixed by the induced rank map. Iterating yields the accelerated formulas, for $q \ge 1$,

$$(qb + r, b; A, B) \longmapsto (r, b; A, A^q B), \qquad 0 \le r < b,$$

$$(a, qa + r; A, B) \longmapsto (a, r; AB^q, B), \qquad 0 \le r < a.$$

For example, in the first formula the ranks $i \ge r$ traverse $i + b, \dots, i + qb$ outside the retained prefix before returning to $i - r$. This also proves that the displayed words are first returns.

At each stage the length and letter-count identities are

$$a|A| + b|B| = L, \qquad av(A) + bv(B) = (o, e),$$

where v records the two original letter counts. Column addition preserves the determinant 1 of the word-count matrix. The return towers partition the original finite set: on each permutation cycle choose the most recent retained rank while traversing backwards. Every cycle meets the prefix, since rank cycles are residue classes modulo $d = \gcd(o, e)$ and each retained prefix has at least d ranks. At termination there are d towers, each of length L/d; a primitive cycle has a single tower.

For the full guard P_W, chronological composition is exactly

$$P_{AB}(x) = P_A(x) \wedge P_B(F_A(x)).$$

The tower partition therefore transfers every original parity condition. Fully expanding this representation still gives L tests across all retained bases. This is exact symbolic closure; it does not provide a uniformly shorter arithmetic test for those guards. The two subtractive rank returns and their word statistics are formalized. The complete accelerated tower partition remains written; the existing formal itinerary concatenation theorem `follows_append` supplies the semantic composition law.

E.3. An exact quotient form and its failed power substitution

Proposition E.2 (three-candidate square carry). Suppose $y \geq 1$ and $y^4 \leq N < (y+1)^4$. Set

$$u = \operatorname{isqrt}(N), \quad c = u - y^2, \quad d = \left\lfloor \frac{N - y^4}{2y^2} \right\rfloor, \quad h = \min(d, 2y).$$

Then $0 \leq c \leq 2y$ and

$$c = h - \kappa, \qquad \kappa \in \{0, 1, 2\}.$$

Writing $a = y^2 + h$, choose $\kappa = 0$ if $a^2 \leq N$, $\kappa = 1$ if $(a-1)^2 \leq N < a^2$, and $\kappa = 2$ otherwise. For odd OE endpoints x, y, taking $N = x^3$ makes the hidden E-source guard equivalent to c being odd.

Proof. Write $N = (y^2 + c)^2 + \epsilon$, where $0 \leq \epsilon \leq 2(y^2 + c)$. Then

$$d = c + \left\lfloor \frac{c^2 + \epsilon}{2y^2} \right\rfloor \geq c.$$

If $c \leq 2y - 1$, then $c^2 + \epsilon \leq 6y^2 - 1$, so $d - c \leq 2$. If $c = 2y$, clipping gives $h = c$. The adjacent square tests identify the unique correction. Finally $u = y^2 + c$ is even exactly when c is odd. □

The genuine block $93 \to 896 \to 29$ attains the correction 2: $d = h = 57$, whereas $c = 55$.

Proposition E.3 (unbounded error of a particular substitution). For each odd integer $s \geq 3$, define

$$\begin{aligned} x &= s^8 + 8, & u &= s^{12} + 12s^4, \\ v &= s^{18} + 18s^{10} + 54s^2 - 1, & z &= s^9 + 9s - 1. \end{aligned}$$

These form an actual OOE block inside the threshold band $b = s^8$, with x, u, z odd and v even. Its suffix displacement is

$$c_2 = v - z^2 = 2z - 27s^2.$$

Define the actual and substituted quotients

$$d_2 = \left\lfloor \frac{u^3 - z^4}{2z^2} \right\rfloor, \qquad D = \left\lfloor \frac{x^{9/2} - z^4}{2z^2} \right\rfloor.$$

Then

$$D - d_2 \in \{36s^2, 36s^2 + 1\}, \qquad \min(D, 2z) - c_2 = 27s^2.$$

Thus replacing the eliminated integer u by its continuous power cannot preserve Proposition E.2 with a uniformly bounded additive correction.

Proof. The three adjacent-square certificates follow from

$$x^3 - u^2 = 48s^8 + 512, \qquad (v+1)^2 - u^3 = 216s^{12} + 2916s^4,$$

$$(z+1)^2 - v = 27s^2 + 1, \qquad v - z^2 = 2s^9 + 18s - 27s^2 - 2.$$

For $s \geq 3$, the first remainder lies strictly between 0 and u; the second lies strictly between 0 and $v + 1$. The last two expressions are positive. These inequalities give all three floors; the parities and $b \leq x, u, z < b^2 \leq v < b^3$ follow directly.

Put $R = 48s^8 + 512$. The mean-value theorem bounds

$$\Delta = \frac{x^{9/2} - u^3}{2z^2} \quad \text{strictly between} \quad \frac{3uR}{4z^2} \text{ and } \frac{3(u+1)R}{4z^2}.$$

Subtracting $72s^2z^2$ from $3uR/2$ gives

$$P = 336s^{12} + 144s^{11} + 3384s^4 + 1296s^3 - 72s^2 > 0.$$

The corresponding upper excess is $P + 72s^8 + 768$, and

$$P + 72s^8 + 768 < 388s^{12} < 2s^{18} < 2z^2.$$

For the first bound use $144s^{11} \leq 48s^{12}$ and bound each of the four remaining positive terms by s^{12}. It follows that $36s^2 < \Delta < 36s^2 + 1$, proving the assertion about the two floors. Since $d_2 \geq c_2$, it also follows that $D > 2z$, giving the clipped difference. □

The ideal endpoint is nevertheless within one integer: $\lfloor x^{9/8} \rfloor = z + 1$. Taylor's formula gives

$$0 < x^{9/8} - (s^9 + 9s) < \frac{9}{2s^7} < 1.$$

The ideal radicand lies outside the original suffix endpoint cell. Thus Proposition E.3 tests the proposed substitution, rather than misapplying Proposition E.2 to a radicand in that cell. It refutes this bounded additive formula; it does not refute all fixed-register rules or all rules for parity alone.

E.4. The polynomial growth family has no infinite exact chain

Theorem E.4 (finite consecutive family chains). Let $X(r) = r^8 + 8$, for odd integers $r \geq 3$. If there are k consecutive transitions $J^3(X(r_i)) = X(r_{i+1})$ within this family, starting at $r_0 = r$, then

$$k \leq \max\left(0, \left\lfloor \frac{\nu_2(r-1) - 2}{2} \right\rfloor\right).$$

Every immediate transition requires $r \equiv 1 \pmod{48}$. In particular, no infinite chain can remain in this exact three-step family description.

Proof. Proposition E.3 makes a transition equivalent to

$$s^8 = r^9 + 9r - 9.$$

For odd r, s, reducing modulo 32 gives $10(r-1) \equiv 0 \pmod{32}$, hence $r \equiv 1 \pmod{16}$. The residue $r \equiv 2 \pmod 3$ is impossible. If $3 \mid r$, the right side has 3-adic valuation 2, whereas an eighth power has valuation divisible by 8. Thus $r \equiv 1 \pmod 3$.

If the destination s itself continues, both r, s are $1 \pmod{16}$. Factoring the transition after subtracting 1 gives

$$(s-1)(s+1)(s^2+1)(s^4+1) = (r-1)(r^8 + r^7 + \cdots + r + 10).$$

The last factor on the right is $2 \pmod{16}$; the last three factors on the left each have 2-adic valuation 1. Therefore

$$\nu_2(s-1) = \nu_2(r-1) - 2.$$

In a chain of $k \geq 1$ transitions the k source parameters all have valuation at least 4. The first $k-1$ valuation drops give $\nu_2(r-1) - 2(k-1) \geq 4$, which is the asserted bound. The final destination need not continue, and no additional drop has been assumed there. ☐

For continuing destinations one also has $\nu_3(s-1) = \nu_3(r-1) + 2$. If $r = 1 + 3j$, its right-hand factor is $18 + 108j + 756j^2 \equiv 18 \pmod{27}$; the other three left-hand factors are prime to 3, since $s \equiv 1 \pmod 3$. This additional identity is a written observation and is not needed for the finite bound.

This excludes escape by endless exact concatenation of this family. It says nothing about later re-entry, other growth patterns, or general escape to infinity. As separate finite computations, the six previously examined parameters

$$3,\ 5,\ 11,\ 101,\ 10^6+1,\ 10^{20}+1$$

reach 1 after respectively $28, 13, 64, 24, 60, 54$ steps. Their peak decimal lengths are $111, 13, 129, 41, 468, 1480$. The stored integer traces certify every edge by adjacent squares. These six computations are not a proof about all family members.

E.5. Retaining the exact first remainder repairs the short quotient

Proposition E.5 (exact remainder transport). Consider a prescribed *OOE* block $x \mapsto u \mapsto v \mapsto z$, with $u \geq 3$. Set

$$X = x^3, \quad R = X - u^2, \quad B = X - R = u^2, \quad t = \sqrt{X}, \quad \eta = t - u.$$

The remainder record and proposed endpoint must satisfy

$$B > 0, \quad B \text{ is a square}, \quad R \geq 0, \quad R^2 \leq 4B, \qquad z^8 \leq B^3 < (z+1)^8.$$

These conditions certify $u = \operatorname{isqrt}(X)$ and the suffix endpoint. Define

$$\Delta = \frac{t^3 - u^3}{2z^2}, \qquad C = \frac{3tR}{4z^2}.$$

Then

$$C - \Delta = \frac{\eta^2(t + 2u)}{4z^2}, \qquad 0 \leq C - \Delta < \frac{5}{6}.$$

Consequently, for the integer floors

$$d = \left\lfloor \frac{u^3 - z^4}{2z^2} \right\rfloor, \qquad q = \left\lfloor \frac{t^3 - z^4}{2z^2} - C \right\rfloor,$$

we have $d \in \{q, q+1\}$.

Proof. The initialized record gives $0 \leq \eta < 1$ and $R = (t-u)(t+u)$. Expansion proves the exact identity. The endpoint cell gives $z \geq 2$ and $z^2 \geq u$: otherwise $u \geq z^2+1$, contradicting $(z^2+1)^3 > (z+1)^4$ for $z \geq 2$. For that comparison, the difference is

$$z^3(z^3 - 4) + z(2z^3 - 3z - 4) > 0.$$

It follows that

$$0 \leq C - \Delta < \frac{3u+1}{4z^2} \leq \frac{3u+1}{4u} \leq \frac{5}{6}.$$

The two possible floors are immediate. □

The corrected q is signed and may a priori be -1; the relation with $d \geq 0$ guarantees $q + 1 \geq 0$. There is an exact integer implementation. With $K = 2X - 3R > 0$,

$$q = \left\lfloor \frac{\operatorname{isqrt}(K^2X) - 2z^4}{4z^2} \right\rfloor. \tag{ER1}$$

This uses $\lfloor(\alpha - a)/b\rfloor = \lfloor(\lfloor\alpha\rfloor - a)/b\rfloor$ for integers a, b, $b > 0$, and the identity $\lfloor K\sqrt{X}\rfloor = \operatorname{isqrt}(K^2X)$. Negative numerators in (ER1) require floor division.

Put $h = \min(q+1, 2z)$ and $N = z^2 + h$. Proposition E.2 now gives

$$v = N - \kappa, \qquad \kappa \in \{0, 1, 2, 3\}.$$

Indeed, for $c = v - z^2 < 2z$, we have $c \leq h \leq d + 1 \leq c + 3$; if $c = 2z$, clipping gives $h = c$. The correction is the smallest $j \in \{0, 1, 2, 3\}$ for which

$$(N - j)^4 \leq B^3.$$

All candidates are positive, since $N \geq z^2 \geq 4$. For odd endpoints x, z, the complete *OOE* guard is exactly

$$R \text{ even}, \qquad N - \kappa \text{ even}.$$

Thus the absolute record (x, z, R), a new exact square root in (ER1), and at most four fourth-power comparisons recover both hidden parities. This does not claim a computational speedup.

For the next word $OOEOE$, the candidate intermediate endpoint $z_* = Q(x^{9/8})$ must first pass the eighth-power cell validation. Then apply this prefix guard and the OE cell and guard at z_*. This yields one further exact composition. It does not prove uniform arithmetic closure for arbitrarily long induced words. Retaining the full remainder retains absolute information sufficient to reconstruct u; it is not a finite-valued summary.

E.6. Fixed residues do not determine the hidden guard

Proposition E.6 (a counterfamily for every fixed residue modulus). Even retaining the exact first remainder $R = 0$, no fixed finite collection of residue classes of x, u, z and $\mathcal{E} = x^9 - z^8$ determines the full OOE guard on its geometric first-return domain. The two blocks below also share the same full threshold and first-return section. This statement concerns those specified summaries, not unrestricted arithmetic in full absolute values and not a periodic-set invariant.

Proof. Let $Q \geq 2$ be any even integer, set $c = Q - 1$, and choose a positive integer $b \equiv 3 \pmod 4$ with $b^2 > 504c^5$. For $d = 1, -c$, set

$$t_d = b^4 + 4d, \quad x_d = t_d^2, \quad u_d = t_d^3,$$

$$V_d = b^{18} + 18db^{14} + 126d^2b^{10} + 420d^3b^6 + 630d^4b^2,$$

$$Z_d = b^9 + 9db^5 + \frac{45d^2b - 1}{2}.$$

All these expressions are integers. The exact branches are

$$O(x_d) = u_d, \quad O(u_1) = V_1, \quad O(u_{-c}) = V_{-c} - 1, \quad E(O(u_d)) = Z_d.$$

The first remainder is exactly zero in both cases.

To verify the floors, put $A = b^4$, so $A > 8c + 4$. Taylor expansion through degree four gives

$$(A + 4d)^{9/2} = V_d + 252d^5\xi^{-1/2},$$

for ξ between A and $A + 4d$. The remainder has the sign of d, and its magnitude is less than $504c^5/b^2 < 1$, giving the two second images. Expansion through degree two similarly gives

$$(A + 4d)^{9/4} = b^9 + 9db^5 + \frac{45d^2b}{2} + \frac{15}{2}d^3\zeta^{-3/4}.$$

Its remainder has magnitude less than $15c^3/b^3 < 15/64 < 1/2$, whereas the polynomial part is a half-integer. Its floor is therefore Z_d. Root collapse identifies this with $E(O(u_d))$.

The states x_d, u_d, Z_d are odd. The polynomial V_d is odd, so the negative-d block has an even final E source and is guard-valid, whereas the positive-d block fails that guard. For Z_d, oddness follows from $45d^2b - 1 \equiv 2 \pmod 4$.

These blocks have the same geometric domain. Set $T = t_{-c} > 8$ and $B = T^2$. Then $t_1 < 2T$, and

$$B \le x_d, u_d, Z_d < B^2 \le O(u_d) < B^3.$$

For example $u_1 < 8T^3 < T^4$, and $O(u_1)^2 \le t_1^9 < 512T^9 < T^{12}$. Also $Z_1 < T^3 = u_{-c}$, so both blocks are first returns to

$$D = [B, Z_1 + 1) \cap \mathbb{N} :$$

their initial and final points belong to D, and both intermediate states lie outside it.

Finally

$$t_1 - t_{-c} = 4Q, \qquad Z_1 - Z_{-c} = Q\left(9b^5 + 45b - \frac{45Qb}{2}\right).$$

Thus $x_d, u_d, Z_d, \mathcal{E}_d$ agree modulo Q. They also have the same exact valuation $\nu_2(\mathcal{E}_d) = 3$: since $t_d \equiv 5 \pmod 8$, $x_d \equiv 9 \pmod{16}$, while $Z_d^8 \equiv 1 \pmod{16}$. For any fixed finite list of moduli, choose an even common multiple as Q. The common data then give opposite guards, proving the claim. □

The formal witness specializes this construction, for any requested modulus $q > 0$, to $c = 512q - 1$ and $b = c^3$. It verifies the colliding record, including the exact aggregate valuation 3, and both blocks' common threshold and first-return section. Even supplying that exact threshold and section boundary to a classifier of the stated record cannot distinguish the final source parities. The general free-b argument above remains a written proof.

The two source values are different, and neither is asserted to belong to a cycle. The absolute quotient and cell comparisons in Section E.5 distinguish them. Adding a later intermediate parity or remainder would also distinguish them.

The dyadic endpoint summary has a related limitation. For a word of length $L \ge 1$, odd count o, and odd endpoint y,

$$x^{3^o} - y^{2^L} \equiv x^{3^o} - 1 \pmod{2^{L+2}}.$$

For an odd expanding closed return $y = x > 1$, its nonzero aggregate has valuation $\nu_2(x-1)$, because $3^o - 2^L$ is odd. This is endpoint information, not the hidden source guards. Likewise, an energy formed from squared differences of neighboring wrong-parity indicators counts changes in those indicators. It can vanish when every indicator is 1; it does not initialize their correct value.

The analogous aggregate calculation on an actual return permutation $y_i = x_{\sigma(i)}$, with peaks v_i and displacements $c_i = v_i - y_i^2$, is only

$$\sum_i c_i = \sum_i v_i - \sum_i x_i^2.$$

It is not a zero-sum identity. Even bases restricted to distinct odd integers admit all even-step cells by setting $v_i = x_{\sigma(i)}^2 + 1$; the omitted odd-prefix equations are essential. Products of the complete cells similarly return to the existing floor-defect identities unless a further estimate is supplied. Theorem 3.39 uses the coupled periodic order and actual odd-prefix cells to obtain a restriction beyond these bookkeeping identities. The uniform wrong-parity conclusion remains open.

E.7. Full arithmetic domains cannot preserve every OOE return

Write $A = E \circ O \circ O$ for the prescribed three-step map, whether or not its intermediate states realize the indicated parities. Proposition E.1 and the growth argument used in Theorem 3.39 give

$$0 \le x^{9/8} - A(x) < 2, \qquad A(x) > x \quad (x \ge 5). \tag{E7a}$$

Consequently a nonempty set S of odd integers at least five would give an infinite actual all-OOE trajectory if every $x \in S$ had $O(x)$ odd, $O(O(x))$ even, and $A(x) \in S$. The following restrictions concern this proposed invariant domain, rather than an arbitrary trajectory.

Proposition E.7 (eventually periodic domains). Every odd residue class modulo an even positive integer contains arbitrarily large x for which $O(x)$ is even. Hence no nonempty eventually periodic set of odd integers has the displayed invariant-domain property.

Proof. Fix an even integer $Q \ge 2$, an odd residue $1 \le a < Q$, and an integer $h \ge 1$. Put

$$t = 2Qh, \qquad x = t^8 + a, \qquad u = t^{12} + \frac{3a}{2} t^4.$$

Then x is odd and congruent to a modulo Q, while u is an even integer. Direct expansion gives

$$4(x^3 - u^2) = 3a^2 t^8 + 4a^3 > 0,$$
$$4((u+1)^2 - x^3) = 8t^{12} - 3a^2 t^8 - 4a^3 + 12at^4 + 4 > 0.$$

For the last sign, $a \le t^2$ implies $a^2 t^8 \le t^{12}$ and $a^3 \le t^6 \le t^{12}$, leaving at least $t^{12} + 12at^4 + 4$. Thus $O(x) = u$, which prevents the second O step. These sources grow without bound with h. An infinite eventually periodic odd domain contains a tail of one odd residue class after refining its period to an even modulus. A nonempty finite domain above four cannot be invariant because $A(x) > x$. □

Proposition E.8 (one full polynomial value family). Let P be a polynomial of degree $d \ge 2$, with positive leading coefficient, such that $P(n)$ is a positive integer for every sufficiently large integer n. Fix an eventual value set $S_P = \{P(n) : n \ge n_0\}$. At least one of every three sufficiently large consecutive integer parameters n has $A(P(n)) \notin S_P$.

Proof. Let $\alpha = 9/8$, let a be the leading coefficient of P, and use the increasing inverse branch of P near infinity to define

$$w(t) = P^{-1}(P(t)^\alpha), \qquad C = a^{(\alpha-1)/d} > 0.$$

Leading terms give $w(t) \sim Ct^\alpha$. Differentiating the exact identity $P(w(t)) = P(t)^\alpha$ first gives $w'(t) \sim C\alpha t^{\alpha-1}$, and a second differentiation gives

$$w''(t) = \frac{\alpha(\alpha-1)P(t)^{\alpha-2}P'(t)^2 + \alpha P(t)^{\alpha-1}P''(t) - P''(w(t))w'(t)^2}{P'(w(t))} \sim \frac{9C}{64} t^{-7/8}. \tag{E7b}$$

Thus the derivative estimate follows from an exact differentiated identity, without differentiating an unspecified error term.

If $A(P(n)) = P(s_n)$ with integer $s_n \geq n_0$, then $s_n \sim Cn^\alpha$: sufficiently large outputs force the target parameter onto the increasing tail. The bound (E7a), the mean value theorem, and $P'(u) \sim adu^{d-1}$ yield

$$0 \leq w(n) - s_n = O(n^{-\alpha(d-1)}) = o(n^{-7/8}). \tag{E7c}$$

If the three parameters $n, n+1, n+2$ all return into the family, integrating (E7b) over the unit square and applying (E7c) gives

$$s_{n+2} - 2s_{n+1} + s_n = \frac{9C}{64} n^{-7/8}(1 + o(1)).$$

For sufficiently large n this integer is strictly between zero and one, a contradiction. □

This theorem allows arbitrary integer target parameters s_n; it assumes no polynomial update rule. Its hypothesis is that the entire eventual source value family is retained. It does not exclude an orbit using only a sparse subsequence of parameters. Finite changes to the value set have no effect, and a fixed arithmetic progression of parameters is covered by replacing $P(n)$ with $P(qn + b)$, for integers $q > 0, b$. A degree-one polynomial that is integer valued on a full integer tail has integer slope and constant; if its value tail is odd, Proposition E.7 applies instead.

The valuation in Theorem E.4 is specific to consecutive transitions inside that family. For example the actual return

$$199 \xrightarrow{O} 2807 \xrightarrow{O} 148718 \xrightarrow{E} 385$$

raises $\nu_2(x - 1)$ from one to seven. Neither the family valuation nor Propositions E.7–E.8 establish a decreasing rank for every actual OOE return. Existence of an infinite all-OOE trajectory remains open.

Appendix F. Successive gaps and terminal return words

This appendix supplies the complete written proof of Theorem 3.40 and the terminal accounting of Proposition 3.41. Every application retains the actual cycle states and all intermediate parity guards.

F.1. A general floor-loss and paired-gap certificate

For a prescribed word of length $L \geq 1$ with positive integer source $x \geq 1$, let $a_j = 3$ for an O letter and $a_j = 1$ for an E letter. Let p_j be its ideal prefix exponents, $p = p_L$, n_j its actual integer states, and $\varepsilon_j = n_{j-1}^{a_j/2} - n_j \in [0, 1)$. The exact square remainder is

$$R_j = n_{j-1}^{a_j} - n_j^2, \quad 0 \leq R_j \leq 2n_j, \quad \varepsilon_j = \frac{R_j}{\sqrt{n_j^2 + R_j} + n_j}.$$

Thus no remainder or intermediate parity has been replaced by an independent random choice.

Assume every proper tail $q_j = p/p_j$, $j < L$, is below one. Telescoping gives the exact identity

$$e_W(x) := x^p - F_W(x) = \sum_{j=1}^{L} \big[(n_j + \varepsilon_j)^{p/p_j} - n_j^{p/p_j}\big].$$

If the proper states are at least $m \geq 1$, concavity implies

$$0 \leq e_W(x) < K_W(m), \qquad K_W(m) = 1 + \sum_{j<L} q_j m^{q_j - 1}. \tag{F1}$$

This remains an absolute-loss estimate when $p > 1$, provided the proper tails are below one. For the paired conclusion assume $0 < p < 1$ and two traces with sources $z > w \geq m$ and proper states at least m. Then, with $\kappa = pm^{p-1}$,

$$0 \leq F_W(z) - F_W(w) < \kappa(z - w) + K_W(m). \tag{F2}$$

Consequently the sufficient certificate

$$2\kappa + K_W(m) \leq 2 \tag{F3}$$

forces strict contraction for every input gap at least two. If the inputs and distinct outputs are odd, the even gap drops by at least two. The trace hypotheses are supplied by actual cycle membership in every application below.

F.2. Two transfers through the first contracting word

The floor-loss bound.

For $C = OOEOOEOE$, all seven proper prefix states satisfy $n_j \geq x$ when $x \geq 3$: O does not decrease a positive integer and $A(x) \geq x$, by $O(O(x)) \geq x^2$. The seven proper tail exponents are

$$\frac{81}{128}, \frac{27}{64}, \frac{27}{32}, \frac{9}{16}, \frac{3}{8}, \frac{3}{4}, \frac{1}{2}.$$

They are all below one. The concavity estimate in Section F.1 therefore yields

$$0 \leq e_C(x) := x^\gamma - C(x) < 1 + U(x), \qquad U(x) = \sum_j q_j x^{q_j - 1}.$$

At $x \geq 2^{24}$, the seven summands are bounded above respectively by

$$\frac{81}{32768}, \frac{27}{524288}, \frac{27}{256}, \frac{9}{16384}, \frac{3}{262144}, \frac{3}{256}, \frac{1}{8192}.$$

Their sum is $63121/524288 < 1/8$. Hence

$$0 \leq e_C(x) < \frac{9}{8}. \tag{F4}$$

This is unconditional for the prescribed map; it does not discard any guard in its subsequent application to actual cycle returns.

For $z > w \geq 2^{24}$, concavity now gives

$$C(z) - C(w) = z^\gamma - w^\gamma + e_C(w) - e_C(z) < \gamma w^{-13/256}(z - w) + \frac{9}{8}.$$

Also

$$\gamma w^{-13/256} < \frac{27}{64}.$$

Indeed $w^{13/256} \geq 2^{39/32} > 2^{6/5} > 9/4$; the last comparison follows by fifth powers from $2^{16} > 3^{10}$. For $d \geq 2$,

$$\frac{27}{64}d + \frac{9}{8} < d,$$

since $(37/64)d \geq 37/32 > 9/8$. Monotonicity of the prescribed branch maps supplies the lower bound in the paired estimate above. This proves the strict gap contraction.

Growing comparison words.

For $x \geq 16$,

$$A(x) > \frac{7}{8}x^{9/8}, \qquad B(x) \geq \frac{7}{8}x^{3/4}.$$

Here $A(x) > x^{9/8} - 2$ is the exact two-cell bound and $B(x) = \lfloor x^{3/4} \rfloor$. All intermediate A-iterates stay at least x. Consequently

$$A^3B(x) \geq \left(\frac{7}{8}\right)^{907/256} x^{2187/2048} > \frac{1}{2}x^{2187/2048} > x \quad (x \geq 2^{15}). \tag{F5}$$

The coefficient exceeds $1/2$ because $907/256 < 4$ and $(7/8)^4 > 1/2$. The final comparison follows from $15 \cdot 139 = 2085 > 2048$.

Now set $\alpha = 9/8$, $u = A(x)$, $v = C(u) = A^3B(x)$. For $x \geq 2^{24}$, both $u, v \geq x$. Concavity and (F4) give

$$0 \leq x^{\alpha\gamma^2} - D(x) < 2\gamma^2 u^{\gamma^2 - 1} + \frac{9}{8}\gamma v^{\gamma - 1} + \frac{9}{8} < \frac{17}{4}.$$

The exponent is $\alpha\gamma^2 = 531441/524288$. Since $24 \cdot 7153/524288 > 1/4$ and $2^{1/4} > 9/8$,

$$D(x) > \frac{9}{8}x - \frac{17}{4} > x \quad (x \geq 2^{24}). \tag{F6}$$

The forced first two batches.

Order the retained set as $Y = \{y_0 < \cdots < y_{a+b-1}\}$. Its lower a bases execute A; its upper b bases execute B. The return sends rank i to $i + b$ below a, and to $i - a$ above it. Both counts are positive: $A(x) > x$ and $B(x) < x$ throughout the present domain.

We have $a > 2b$. Otherwise the full return word has ideal exponent at most

$$(9/8)^{2b}(3/4)^b = (243/256)^b < 1,$$

and its prescribed floor composition cannot return a start > 1 to itself.

Write $a = qb + r$, $0 \leq r < b$. Thus $q \geq 2$. If $r > 0$, the maximal left batch has words A, A^qB, counts r, b, and its upper branch strictly decreases rank. If $r = 0$, its terminal return

A^qB fixes the retained prefix. For $q \geq 3$, (F5) makes $A^qB(x) > x$, contradicting either case. Therefore

$$a = 2b + r, \qquad 0 < r < b,$$

and the new pair is A, C.

Write $b = pr + s$, $0 \leq s < r$. If $p \geq 3$, the next lower word AC^p has ideal exponent at most

$$(9/8)(243/256)^3 = 129140163/134217728 < 1.$$

It cannot increase rank in a nonterminal return, or fix a terminal prefix. If $p = 1$, the next strict left step makes $(AC)C = D$ an upper branch that must decrease rank, contrary to (F6). Finally $p = 2, s = 0$ makes D the terminal fixed return, also contrary to (F6). Hence

$$b = 2r + s, \qquad 0 < s < r. \tag{F7}$$

Both right steps are genuine, with at least two retained bases. This proof applies to normalized threshold cycles on this domain before actual parity is imposed.

The two genuine transfers.

Initially

$$w = y_{b-1} = B(t), \qquad z = y_b = A(m).$$

A left induction step retains the prefix of length a. Its new largest base $t_1 = y_{a-1}$ satisfies $A(t_1) = t$, so its upper return $AB(t_1) = w$. Its lower minimum image is still z. The next left step again gives the same pair. Neither step supplies an additional independent gap.

After the maximal left batch the counts are r, b. At the first right step the boundary pair becomes

$$(w_1, z_1) = (C(w), C(z)) = (y_{b-r-1}, y_{b-r}).$$

At the second right step it becomes

$$(w_2, z_2) = (C(w_1), C(z_1)) = (y_{s-1}, y_s).$$

The indices exist by (F7). Every source and intermediate state is part of a genuine return tower in the original cycle. In an actual Juggler cycle, all four transferred endpoints and w, z are odd, and both transferred gaps are positive even integers.

Put $d_j = z_j - w_j$ and $\kappa = \gamma m^{-13/256} < 27/64$. Since every transferred source is at least m, the paired estimate above gives

$$d_1 < \kappa d_0 + \frac{9}{8}, \qquad d_2 < \kappa d_1 + \frac{9}{8}. \tag{F8}$$

Each even positive gap drops by at least two. Thus $d_0 \geq 6$. Also

$$2 \leq d_2 < \kappa^2 d_0 + \frac{9}{8}(1 + \kappa),$$

and therefore

$$d_0 > \frac{2 - \frac{9}{8}(1 + \kappa)}{\kappa^2} \geq \frac{205}{512\kappa^2} = \frac{26240}{59049} m^{13/128}.$$

This proves the gap bound in (SR1). Retaining κ gives the sharper expression

$$d_0 > \frac{57344}{59049} m^{13/128} - \frac{32}{27} m^{13/256}.$$

F.3. The next batch and the 65-letter return

The next word $W = D^3C$ has ideal exponent $\rho = 3^{41}/2^{65}$. For every integer $x \geq 2^{128}$, we prove $0 \leq x^\rho - W(x) < 6/5$; for integers $z > w \geq 2^{128}$ with $d = z - w \geq 2$, we prove $0 \leq W(z) - W(w) < \rho w^{\rho-1} d + 6/5 < d$.

The next forced batch.

Section F.2 leaves the pair D, C with counts r, s, $r > s > 0$. The ideal exponent of a complete return is $\delta^r \gamma^s > 1$, since all floors lie below their ideal powers and a cycle returns a start greater than one to itself. Since $\delta^3\gamma = \rho < 1$, we must have $r > 3s$.

Write $r = ks + u$, $0 \leq u < s$. Thus $k \geq 3$. The next upper word is D^kC when $u > 0$; when $u = 0$, the terminal concatenation D^kC fixes the remaining ranks.

The comparison word $V = D^4C$, with exponent $\eta = \delta^4\gamma = 3^{53}/2^{84}$, strictly grows on $x \geq 2^{128}$. Here is a direct finite-word proof. Each nonempty proper prefix of C has exponent at least α. Each nonempty proper prefix of $D = AC^2$ has exponent greater than δ. It follows that every nonempty proper prefix of V has exponent at least $\delta > \eta$. Therefore all its 83 proper tails are below one. Every actual prescribed state is at least one, so (F1) gives

$$0 \leq x^\eta - V(x) < 84.$$

The exact rational comparisons

$$\eta > 1 + \frac{1}{512}, \qquad (9/8)^4 < 2$$

give, for $x \geq 2^{128}$,

$$V(x) > x^\eta - 84 > \frac{9}{8}x - 84 > x.$$

Also $D(x) > x$ on this range by Section F.2. Monotonicity therefore gives $D^kC(x) \geq D^4C(x) > x$ for $k \geq 4$, contrary to either an upper branch decreasing rank or a terminal fixed return. Thus $k = 3$, and $r > 3s$ excludes $u = 0$. This proves $r = 3s + u,\ 0 < u < s$.

The exact error certificate.

For $W = D^3C$, every proper prefix exponent is at least $\delta > \rho$, by the same block calculation. All proper actual states are at least the source $x \geq 2^{128}$: D grows, its proper states stay above its source by the A, AC growth statements in Section F.2, and the proper C states stay above their source.

Let p_j run through the 64 proper prefix exponents and put $q_j = \rho/p_j$. The exact dyadic majorant at $m = 2^{128}$ is

$$\begin{aligned} U_W(m) &:= \sum_{j=1}^{64} q_j m^{q_j - 1} \\ &\leq \sum_{j=1}^{64} \frac{q_j}{2^{\lfloor 128(1-q_j)\rfloor}} \\ &= \frac{277910493483851358096413315698577150111783}{2^{140}} < \frac{1}{5}. \end{aligned} \tag{F9}$$

This is a finite rational identity obtained by expanding the prescribed word D^3C; the verifier records every prefix, tail and summand. Equations (F1) and (F9) prove the claimed absolute-loss bound.

For the slope, the exact comparisons

$$128(1-\rho) > \frac{10}{7}, \qquad 2^{10}3^7 > 8^7$$

imply $m^{1-\rho} > 2^{10/7} > 8/3$, hence

$$\kappa_W := \rho m^{\rho-1} < \frac{3}{8}.$$

Since $2(3/8) + 6/5 = 39/20 < 2$, (F2) proves the claimed paired contraction.

A third genuine transfer.

The preceding two C transfers end at

$$(w_2, z_2) = (y_{s-1}, y_s).$$

The next three left steps keep this exact boundary unchanged. After them the retained counts are u, s, with words D, W. Both seam points belong to the upper domain, because $0 < u < s$. Its first right step is therefore genuine and gives

$$(w_3, z_3) = (W(w_2), W(z_2)) = (y_{s-u-1}, y_{s-u}). \tag{F10}$$

All indices exist. Every intermediate is in a first-return tower of the same original cycle; the sources and endpoints are at least m. All four seams are pairs of distinct odd cycle states. Nothing in this step supplies a fourth or a terminal pair.

Write $d_j = z_j - w_j$ and $\kappa_C = \gamma m^{-13/256} < 1/64$. The three valid inequalities are

$$d_1 < \kappa_C d_0 + \frac{9}{8}, \quad d_2 < \kappa_C d_1 + \frac{9}{8}, \quad d_3 < \kappa_W d_2 + \frac{6}{5}.$$

All four gaps are positive even integers, so $d_0 \geq 8$. More generally, n strict transfers between even natural gaps give $d_0 \geq d_n + 2n$, and hence $d_0 \geq 2(n+1)$ if the final gap is positive. This finite statement does not supply any additional valid transfer. Moreover,

$$2 \leq d_3 < \kappa_W \kappa_C^2 d_0 + \frac{9}{8}\kappa_W(1+\kappa_C) + \frac{6}{5}.$$

The remaining numerator has the exact lower bound

$$\frac{4}{5} - \frac{9}{8}\frac{3}{8}\frac{65}{64} = \frac{7609}{20480}.$$

Therefore

$$\begin{aligned} d_0 &> \frac{7609}{20480\,\rho\gamma^2} m^\sigma \\ &> \frac{121744}{295245} m^\sigma > \frac{2}{5} m^\sigma. \end{aligned}$$

The rational comparisons $125/128 < \rho < 127/128$ give $7/64 < \sigma < 1/8$. This proves the gap bounds in (SR2).

F.4. Transporting a seam gap to the maximum

The original exact cells from Theorem 3.39 are

$$t^3+2 \le [w(w+2)]^2, \qquad M+2 \le (t+1)^2, \qquad z^8 \le m^9.$$

The gaps at least six and eight give the respective integer restrictions

$$t^3+2 \le [(z-6)(z-4)]^2, \qquad t^3+2 \le [(z-8)(z-6)]^2.$$

All factors are positive because $w \ge m > 0$.

For a common smooth consequence, suppose that $m \ge 2^{24}$, $0 < \nu \le 1/8$, and

$$H := z-w-1 > \frac{1}{3}m^{\nu}.$$

Set

$$X = m^{9/8}, \qquad a_* = m^{3/2}, \qquad R = \frac{1}{3}m^{3/8+\nu}.$$

We have $0 < w+1 = z-H \le X-H$. The elementary inequality $(1-u)^{4/3} \le 1-u$, $0 \le u \le 1$, gives

$$t < (w+1)^{4/3} \le (X-H)^{4/3} \le X^{4/3} - X^{1/3}H < a_* - R.$$

Here $R \ge m^{3/8}/3 \ge 512/3 > 8$, and $R/a_* \le 1/(3m) < 1/4$. Hence

$$R^2 \le a_*R/4, \qquad 2a_* \le a_*R/4.$$

All quantities being squared are positive, and

$$\begin{aligned} M &< (a_*-R+1)^2-2 \\ &= a_*^2 - 2a_*R + R^2 + 2a_* - 2R - 1 \\ &< a_*^2 - \frac{3}{2}a_*R = m^3 - \frac{1}{2}m^{15/8+\nu}. \end{aligned} \tag{F11}$$

For the two-transfer case $d_0 = z-w \ge 6$, so

$$H = d_0 - 1 \ge \frac{5}{6}d_0 > \frac{65600}{177147}m^{13/128} > \frac{1}{3}m^{13/128}.$$

Taking $\nu = 13/128$ proves (SR1). For three transfers $d_0 \ge 8$, and

$$H \ge \frac{7}{8}d_0 > \frac{7}{20}m^{\sigma} > \frac{1}{3}m^{\sigma}.$$

Taking $\nu = \sigma \in (7/64, 1/8)$ proves (SR2) and (SR3).

An integer ceiling can be evaluated without real powers. In the first case take the least even $G \ge 6$ with $(59049G)^{128} > 26240^{128}m^{13}$; in the second, take the least even $G \ge 8$ with $(5G)^{64} > 2^{64}m^7$. Then $d_0 \ge G$. Let z_* be the greatest odd integer with $z_*^8 \le m^9$, and T the greatest positive odd integer satisfying

$$T^3+2 \le [(z_*-G)(z_*-G+2)]^2.$$

The necessary ceiling is $M \le (T+1)^2-2$. The latter recipe uses the clean $7/64$ gap exponent, rather than the sharper σ. At the single illustrative input $m = 2^{128}+1$, it gives $G = 6554$. This is a conditional arithmetic evaluation, not a periodic orbit.

F.5. The terminal mixed-word passage

At a retained stage with words U, V, exponents λ, μ, and counts a, b, the exact exponent invariant is

$$\lambda^a \mu^b = \frac{3^o}{2^L} > 1.$$

Left and right substitutions preserve it:

$$\lambda^{a-b}(\lambda\mu)^b = \lambda^a \mu^b = (\lambda\mu)^a \mu^{b-a}.$$

For a primitive cycle the final two-base stage is $\{m, v\}$, with

$$U(m) = v, \qquad V(v) = m.$$

The terminal word UV fixes m, has ideal exponent $p = \lambda\mu = 3^o/2^L > 1$, and has exact loss

$$e_{UV}(m) = m^p - m. \tag{F12}$$

Applying V also to m would be an off-domain prescribed evaluation: periodicity supplies neither its cycle membership nor its guard pattern. The upper point disappears when the section becomes $\{m\}$. Even strict contraction for every earlier selected upper map would only decrease finitely many positive gaps; it would not manufacture a surviving terminal pair.

To exclude this terminal return one needs an absolute-loss estimate strictly below the positive excess $m^p - m$, or a new incompatible shared-cell constraint. Bounds on the difference of two losses do not by themselves supply that absolute calibration. Equation (F12) alone is a restatement of closure, not a new no-cycle result.

There is a more informative way to account for the terminal pair without inventing an extra same-word return. At every induced stage there are chronological words P, Q such that

$$UV = P\,OE\,Q, \qquad VU = P\,EO\,Q. \tag{F13}$$

Initially $U = A,\ V = B,\ P = O,\ Q = OE$, and direct concatenation verifies both identities. A left substitution $(U, V) \mapsto (U, UV)$ changes P to UP and keeps Q. A right substitution $(U, V) \mapsto (UV, V)$ keeps P and changes Q to QV. These rules prove (F13) by induction.

The same substitutions give an exact count invariant. Write $o(W), e(W)$ for the odd and even letter counts of a word. Then

$$o(U)e(V) - o(V)e(U) = 1.$$

The initial vectors are $(2, 1)$ and $(1, 1)$; either substitution adds one vector to the other and preserves the determinant. Thus both count vectors are primitive. If $L = a|U| + b|V|$ and $o = a\,o(U) + b\,o(V)$, the same integer change of coordinates gives $\gcd(a, b) = \gcd(L, o)$. Positive coprime section counts reach $(1, 1)$ by strict subtraction, since $a + b$ decreases until equality. The guard-preserving induction must accompany these count identities; the identities alone do not construct an actual orbit.

At the final primitive two-base stage write the bases as $\{m, v\}$, so $U(m) = v,\ V(v) = m$. They are the two smallest original cycle states and hence adjacent. Their full-cycle words

UV, VU first apply the same P. Common letters preserve adjacency inside the ordered cubic-band cycle until their labels diverge. Thus P sends (m, v) to (h, s), the largest odd source and smallest even source. The differing words OE, EO then send that pair to

$$(t, q) = (E(M), O(m)),$$

and the common suffix Q sends (t, q) back to (m, v). All these statements concern the actual two cycle traces, with every guard and square remainder retained.

The mixed two-step passage itself strictly contracts for an actual cubic-band cycle with odd $m \geq 3$. Indeed

$$h < m^2, \qquad s \geq m^2 + 1, \qquad t = \lfloor h^{3/4} \rfloor, \qquad q = \lfloor (m^2)^{3/4} \rfloor.$$

The extra unit in the lower bound for s uses its even parity and the odd parity of m^2. The normalized cycle order gives $t < q$. Since the derivative of $x^{3/4}$ is below one for $x \geq h \geq 3$,

$$0 < q - t < (m^2)^{3/4} - h^{3/4} + 1 < m^2 - h + 1 \leq s - h. \tag{F14}$$

This closes the pair passage geometrically through the mixed block. It does not give a net contraction of the complete passage: the common prefix P, which grows during left inductions, has not been controlled by the previous right-transfer estimates. Right-transfer words assemble Q; the existing bounds control only the factors already certified.

The exact ideal exponents make the missing comparison explicit:

$$p_P \frac{3}{4} p_Q = \frac{3^o}{2^L} > 1. \tag{F15}$$

Therefore simply multiplying ideal powers cannot turn (F14) into a contradiction. A successful gap proof needs an exact bound that compares amplification through P with loss through the mixed block and Q. The local inequality (F14) is an elementary floor-cell consequence; the new bookkeeping explains precisely which part of the terminal argument it does and does not control.

F.6. The limitation of a fixed-minimum certificate

Even if every inner loss were omitted optimistically, (F3) would require $pm^{p-1} \leq 1/2$. For a word with a proper prefix, $K_W(m) > 1$, so it actually requires the strict inequality

$$m > (2p)^{1/(1-p)} \qquad (1/2 < p < 1). \tag{F16}$$

As $p \to 1^-$, the logarithm of this threshold grows like $(\log 2)/(1-p)$. At any fixed m, the slope $pm^{p-1} \to 1$, so (F3) eventually fails even before its positive inner-loss terms are added.

This is a rigorous limitation of the specified sufficient certificate. It is not a counterexample to the true paired map inequality. Correlations between the two traces could conceivably improve the bound on $e_W(w) - e_W(z)$.

The qualification matters: not every possible later cycle-selected upper word has been proved to have exponent below one, or all its proper tails below one. Close to termination, a word of ideal exponent slightly above one can still decrease because of floors. One may not silently impose the infinite ideal Euclidean itinerary on an actual finite cycle.

7. Acknowledgments and use of AI

The author used large language models throughout the development of this work, including drafting and revising the prose, proposing and developing proof arguments, writing Lean formalizations, and designing and implementing computations. The September 2026 review with OpenAI Codex checked the mathematical claims and references, corrected the finite-window estimate, revised the rotation-cell argument and several hypotheses, distinguished formal proofs from numerical evidence, and prepared the document build and independent numerical audit. AI assistance is not independent mathematical validation. The explicit evidence boundaries in Section 1.2 and Appendix A apply to these contributions. The author is responsible for the statements, proofs, code, and final verification of this preprint.

References

1. C. A. Pickover, *Computers and the Imagination: Visual Adventures Beyond the Edge*, St. Martin's Press, New York, 1991, ch. 40, p. 232.

2. C. A. Pickover, *The Mathematics of Oz: Mental Gymnastics from Beyond the Edge*, Cambridge University Press, Cambridge, 2002, ch. 45, pp. 102–106.

3. OEIS Foundation Inc., "Juggler sequence: if n even then $\lfloor\sqrt{n}\rfloor$ else $\lfloor n^{3/2}\rfloor$," Sequence A094683 in *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A094683` (accessed 28 August 2026).

4. OEIS Foundation Inc., "Number of steps needed for n to reach 1 in the juggler sequence," Sequence A007320 in *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A007320` (accessed 29 August 2026).

5. E. W. Weisstein, "Juggler Sequence," *MathWorld*, `https://mathworld.wolfram.com/JugglerSequence.html` (accessed 29 August 2026).

6. OEIS Foundation Inc., "Largest value in trajectory of n under the juggler map of A094683," Sequence A094716 in *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A094716` (accessed 29 August 2026).

7. V. Prasad and M. A. Prasad, "Estimates of the maximum excursion constant and stopping constant of juggler-like sequences," ResearchGate preprint, 2025. `https://doi.org/10.13140/RG.2.2.14110.04168`.

8. J. C. Lagarias, "The $3x + 1$ problem and its generalizations," *Amer. Math. Monthly* 92 (1985), 3–23. doi:10.1080/00029890.1985.11971528.

9. J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x + 1$ *Problem*, American Mathematical Society, Providence, RI, 2010.

10. R. E. Crandall, "On the "$3x + 1$'' problem," *Math. Comp.* 32 (1978), 1281–1292. doi:10.1090/S0025-5718-1978-0480321-3.

11. K. R. Matthews and A. M. Watts, "A generalization of Hasse's generalization of the Syracuse algorithm," *Acta Arith.* 43 (1984), 167–175. doi:10.4064/aa-43-2-167-175.

12. J. L. Simons and B. M. M. de Weger, "Theoretical and computational bounds for m-cycles of the $3n + 1$-problem," *Acta Arith.* 117 (2005), 51–70. doi:10.4064/aa117-1-3.

13. S. Eliahou, "The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths," *Discrete Math.* 118 (1993), 45–56. doi:10.1016/0012-365X(93)90052-U.

14. L. Kirby and J. Paris, "Accessible independence results for Peano arithmetic," *Bull. London Math. Soc.* 14 (1982), 285–293. doi:10.1112/blms/14.4.285.

15. G. Rhin, "Approximants de Padé et mesures effectives d'irrationalité," in *Séminaire de Théorie des Nombres, Paris 1985–86*, Progress in Mathematics 71, Birkhäuser, Boston, 1987, 155–164. doi:10.1007/978-1-4757-4267-1_11.

16. P. Cochin, "Five-Step Descent Certificates for the Juggler Map: Parity Statistics of Nested Floor Powers," companion manuscript (Paper B), revision of 10 September 2026. Source manuscript.

17. P. Cochin, "Fate Contagion and Termination Criteria for the Juggler Map," companion manuscript (Paper C), revision of 9 September 2026. Source manuscript.

18. M. R. Herman, "Sur la conjugaison différentiable des difféomorphismes du cercle à des rotations," *Publ. Math. IHÉS* 49 (1979), 5–233, Theorem 3.1, p. 73. doi:10.1007/BF02684798.